

College of Technology

Instrumentation and Control

Module # 9 Level Measurement

Document Intent:

The intent of this document is to provide an example of how a subject matter expert might teach Level Measurement. This approach is what Idaho State University College of Technology is using to teach its Energy Systems Instrumentation and Control curriculum for Level Measurement. The approach is based on a Systematic Approach to Training where training is developed and delivered in a two step process. This document depicts the two step approach with knowledge objectives being presented first followed by skill objectives. Step one teaches essential knowledge objectives to prepare students for the application of that knowledge. Step two is to let students apply what they have learned with actual hands on experiences in a controlled laboratory setting.

Examples used are equivalent to equipment and resources available to instructional staff members at Idaho State University.

Level Measurement Introduction:

This module covers aspects of level measurement as used in process instrumentation and control. Level measurement addresses essential knowledge and skill elements associated with measuring level. Students will be taught the fundamentals of level measurement using classroom instruction, demonstration, and laboratory exercises to demonstrate knowledge and skill mastery of level measurement. Completion of this module will allow students to demonstrate mastery of knowledge and skill objectives by completing a series of tasks using calibration/test equipment, level indicating, and level transmitting devices.

References

This document includes knowledge and skill sections with objectives, information, and examples of how pressure measurement could be taught in a vocational or industry setting. This document has been developed by Idaho State University's College of Technology. Reference material used includes information from:

American Technical Publication – Instrumentation, Fourth Edition, by Franklyn W. Kirk, Thomas A Weedon, and Philip Kirk, ISBN 979-0-8269-3423-9 (Chapter 4)

Department of Energy Fundamentals Handbook, Instrumentation and Control, DOE-HDBK-1013/1-92 JUNE 1992, Re-Distributed by http://www.tpub.com

STEP ONE

Level Measurement Course Knowledge Objectives

Knowledge Terminal Objective (KTO)

KTO 3. Given examples, **EVALUATE** level measurement fundamentals as they apply to measuring level process pressure variables to determine advantages and disadvantages associated with different types of devices used to indicate, measure, and transmit level.

Knowledge Enabling Objectives (KEO)

KEO 3.1. **DEFINE LEVEL** and its importance as a process variable.

KEO 3.2. **DESCRIBE** how **LEVEL** is used to measure the VOLUME of material in a tank or vessel.

KEO 3.3. **DESCRIBE** what a **POINT LEVEL MEASUREMENT** is and how it is accomplished.

KEO 3.4. **DESCRIBE** what a **CONTINUOUS LEVEL MEASRURMENT** is and how it is accomplished.

KEO 3.5. **DESCRIBE** how **GAUGE GLASSES** are used to measure liquid level.

KEO 3.6. **DESCRIBE** how **REFLEX GAUGE CLASSES** are used to measure liquid level.

KEO 3.7. **DESCRIBE** how **MAGNETIC-COUPLED** level gauges measure liquid level.

KEO 3.8. **DESCRIBE** how **CABLE and WEIGHT SYSTEMS** measure the level of granular solids in a tank or silo.

KEO 3.9. **DESCRIBE** the roll **PRESSURE** has in measuring liquid **LEVEL** using **HYDROSTATIC PRESSURE** for an open tank or a pressurized tank.

KEO 3.10. **DESCRIBE** the roll **PRESSURE** has in measuring liquid **LEVEL** using a **BUBBLER SYSTEM** for a tank that is not pressurized and for one that is pressurized.

KEO 3.11. **DESCRIBE** how **FLOATS** are used to measure liquid level.

KEO 3.12. **DESCRIBE** what a **DISPLACER LIQUID LEVEL MEASUREMENT SYSTEM** is and how it measures level.

KEO 3.13. **DESCRIBE** how a **PADDLE WHEEL SWITCH** is used as a point level measuring device.

KEO 3.14. **DESCRIBE how BEAM-BREAKING PHOTOMETRIC SENSORS provide point level measurement and how false signal can affect them.**

KEO 3.15. **DESCRIBE** how **OPTICAL LIQUID-LEVEL SENOSRS** provide point level measurement.

KEO 3.16. **DESCRIBE** how **CONDUCTIVITY PROBES** provide a point level measurement of liquid level.

KEO 3.17. **DESCRIBE** how **MAGNETOSTRICTIVE SENSORS** provide continuous measurement of liquid level.

KEO 3.18. **DESCRIBE** how **CAPACITANCE PROBES** provide point level and continuous measurement of liquid level.

KEO 3.19. **EXPLAIN** how a **THERMAL DISPERSION SWITCH** provides a point level measurement.

KEO 3.20. **DESCRIBE** how an **INDUCTIVE PROBE** provides a point level measurement of a conductive solution.

KEO 3.21. **DESCRIBE** how **ULTRASONIC SENSORS** provide a continuous level measurement.

KEO 3.22. **DESCRIBE** how **TUNING FORKS** provide a point level measurement of a liquid.

KEO 3.23. **DESCRIBE** how **RADAR** systems utilize **PULSED, FREQUENCY, MODULATED CONTINOUS WAVE,** and **GUIDED WAVE RADAR** to measure level.

KEO 3.24. **DESCRIBE** how **LASERS** measure level.

KEO 3.25. **DESCRIBE** how **NUCLEAR LEVEL INSTRUMENTS** provide point and continuous level measurement.

KEO 3.26. **DESCRIBE** how **ELECTRONIC LOAD CELLS** measure level of liquids or solids.

KEO 3.27. **DESCRIBE how HYDRAULIC LOAD CELLS are used to measure level.**

KEO 3.28. **DESCRIBE TWO DISADVANTAGES** associated with **HYDRAULIC LOAD CELLS** that are not associated with **ELECTRONIC LOAD CELLS**.

KEO 3.29. **DESCRIBE** difficult or complicated **SITUATIONS** associated with the measurement of level for **BULK SOLIDS IN SILOS AND TANKS** and what can be done to ensure safe and reliable operation of level sensors.

KEO 3.30. **DESCRIBE** difficult or complicated **SITUATIONS** associated with the measurement of level for **WATER LEVEL IN A BOILER** and what can be done to ensure safe and reliable operation of level sensors.

KEO 3.31. **DESCRIBE** difficult or complicated **SITUATIONS** associated with the measurement of level for **CORROSIVE FLUIDS** and what can be done to ensure safe and reliable operation of level sensors.

KEO 3.32. **DESCRIBE** how to compensate for level measurement of a level transmitter using a **CAPILLARY FIELD SYSTEM REQUIRING SUPPRESSION**.

KEO 3.33. **DESCRIBE** how to compensate for level measurement of level transmitter REQUIRING ELEVATION when the solution being measured is applied to the transmitter located below the tank being measured.

LEVEL MEASURMENT

KEO 3. 1 **DEFINE LEVEL** and its importance as a process variable.

Level Measurements earliest and simplest method of measurement was to insert a pole into a solution and retracting it to measure the wetted part of the pole. Another earlier method was to tie knots in a rope and attach a weight to the rope dropping it into a solution and retracting it to see how many knots were wet to measure the depth of the solution. The pole method is still used today by fuel stations when fuel is delivered into an underground tank to see how much fuel was delivered or needs to be delivered. The rope with knots method has been replaced with a float device attached to the end of a tape or wire and as the float moves up and down, this movement is indicated on the outside of the tank with a gauge device showing the level of the vessel.

The amount of water, fuel, solvent, bulk solids, or other materials is important when operating manufacturing processes on the generation of power. **Level and control of level is essential for safety of boilers and overflow and spill prevention of tanks and silos**. In the nuclear power generation, level measurement and control is critical to prevent serious accidents or incidents associated with steam generation and to prevent releases of radioactive contamination into the environment.

KEO 3. 2 **DESCRIBE** how **LEVEL** is used to measure the VOLUME of material in a tank or vessel.

Level Measurement is often used to not only measure the level, but to also **measure** the **volume** of material in a vessel or a tank. Tank or vessel configurations are important, as the shape and position of a tank or vessel affect the relationship between level and volume. For a vertical cylindrical tank with a flat bottom, the relationship is uniform and each unit level represents an equal unit of volume.

Many vertical cylindrical tanks have a dished bottom that has a special convex shape when viewed from the outside to handle internal pressures of vessel. A flat bottom could bulge when pressure is applied. This pressure can be due to the solution height or the internal pressure of the tank itself. The dished bottom will not bulge or distort with this volume or internal pressure.

The following picture depicts different shapes of storage tanks that will determine the relationship between level and volume:

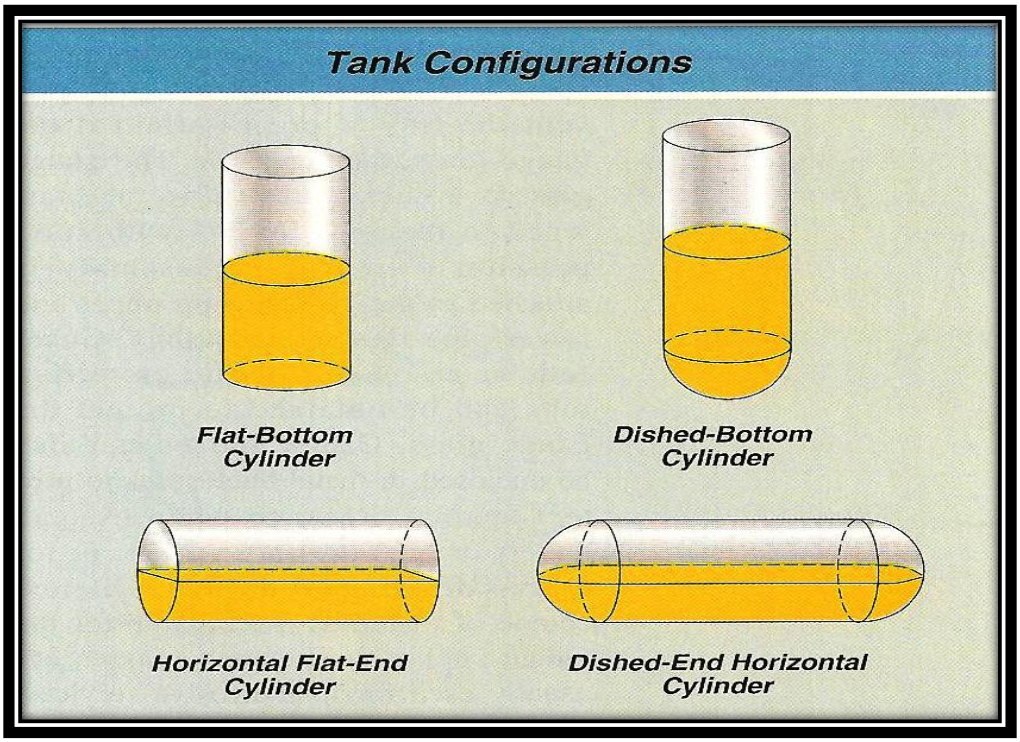


Figure 4-1 page 125

With vertical dish-bottom tanks, the relationship of level to volume is uniform for all levels except for the dished end. For horizontal tank cylindrical tanks, this is not the case as one unit of level at the middle of the tank will represents more volume than one at the bottom or top of the tank. Horizontal tanks add more complexity to the level-volume relationship because the ends are dished or hemispherical. The calculation of such tanks can be difficult and generally tank manufactures will usually provide a table showing volume for a specific level of that individual tank and its solution weight or specific gravity.

IMPORTANT LEVEL NOTE:

In order to accurately measure liquid level, the specific gravity of that solution must be known and the instrumentation has to be calibrated according to this information. Specific gravity of a solution is based on its weight. Water has a specific gravity of approximately 1.0 so anything heavier than 1.0 is an aqueous solution and anything lighter than 1.0 is an organic solution (oil is an organic solution as it floats on top of water and will not mix into water).

SUMMARY

- Level and control of level is essential for safety of boilers and overflow and spill prevention of tanks and silos.
- Tank or vessel configurations are important when determining volume, as the shape and position of a tank or vessel affect the relationship between level and volume.
- Level is a measurement of the fluid level in a vessel or tank.
- Volume measurement is a measurement of how much fluid is inside of a vessel or tank and the tank and or vessel configurations determine what its volume is – vessel or tank manufactures generally supply a table to convert the level to an equivalent volume based on the weight or specific gravity of the solution being measured.
- Volume measurement is important to determine quantity.
- Level measurement is important to provide control of solution being measured.

KEO 3. 3 **DESCRIBE** what a **POINT LEVEL MEASUREMENT** is and how it is accomplished.

POINT LEVEL MEASUREMENT is a measurement identified where the only concern is whether the amount of material is within the desired limits. This measurement is one commonly used to sound an alarm or to determine when to activate a control device to increase or decrease the level. This is a level that is critical to maintain or to report its status.

This is accomplished by placing a level sensing element at the selected level position. If high and low level operation is required, one sensor is required at each location. Examples of **POINT LEVEL MEASURMENT** could include the prevention of a tank or silo from overfilling, to avoid running a pump dry when emptying a tank, or to sound an alarm when a surge tank is above or below a normal level. Another example would be a **SAFETY BACKUP** to a process control of level if the controls were to fail.

KEO 3. 4 **DESCRIBE** what a **CONTINUOUS LEVEL MEASRURMENT** is and how it is accomplished.

CONTINUOUS LEVEL MEASRURMENT is a method to track the changes of a level over a range of values to monitor inventory or for determining when to add or remove material from containers.
Examples of **CONTINUOUS LEVEL MEASRURMENT** could include maintaining a level at a safe level when transferring material, or the water level of a boiler must be known at all times to prevent a low-water condition that could result in boiler damage or an explosion. These are examples of maintaining levels at a safe limit at all times and if these levels cannot be

maintained a system shut down must be initiated to prevent equipment damage, injury to personnel, or unsafe releases to the environment.

SUMMARY

- Point level measurement is a measurement identified where the only concern is whether the amount of material is within the desired limits.
- Continuous level measurement is a method to track the changes of a level over a range of values to monitor inventory or for determining when to add or remove material from containers.

KEO 3. 5 **DESCRIBE** how **GAUGE GLASSES** are used to measure liquid level.

GAUGE GLASSES are devices used to provide a visual indication of a liquid level that consist of a glass tube connected above and below the liquid level in a tank that allows the liquid level to be observed visually. **GUAGE GLASSES** are used as a visual indication right at the tank location. As the level of the tank increases or decreases, the liquid level is observed inside the glass tube. The liquid level is the same as the level inside the tank. A **GUAGE GLASS** is depicted below:

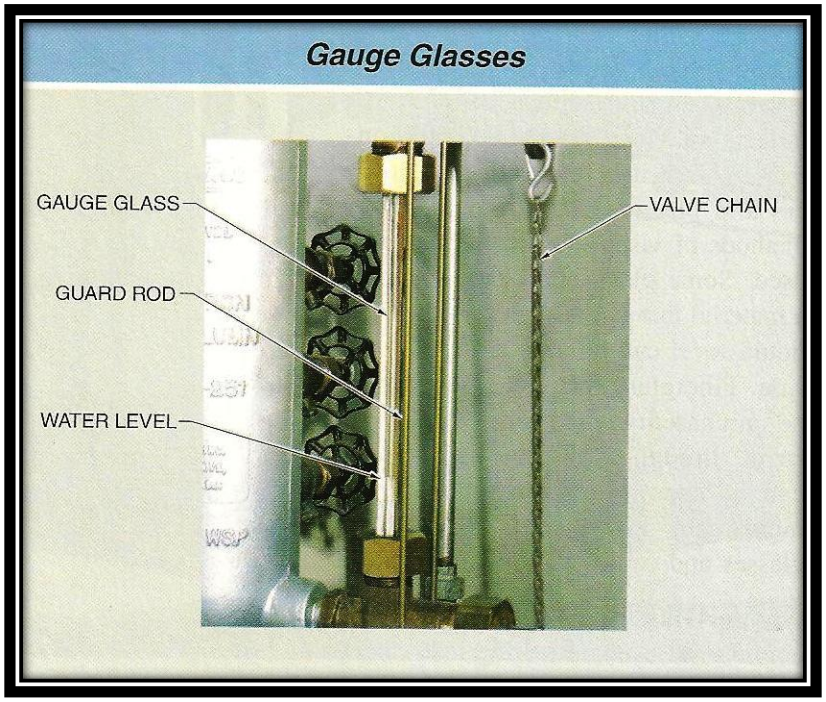


Figure 4-2 page 126

The gauge glass occupies the vertical space between the gauge cocks. The gauge cocks include ball check valves to prevent the loss of process fluid if the gauge glass should break. The gauge class is a thick-walled glass tube fastened to the gauge cocks with a compression fitting. The gauge glass assembly is attached to the vessel using upper and lower flanges or fittings.
A guard rod is attached above and below the gauge glass tube to help protect the tube. In some cases, a thicker plastic tube encloses the glass tube for added protection against breakage.

Because of a limited choice of materials for gauge cocks, gauge glasses are usually used for non corrosive solutions that can discolor the glass tube or damage the gauge cock materials.

An armored gauge glass assemble is also available for high pressure systems like boilers. These assemblies use a thick flat gauge glass inside an armored enclosure to provide high pressure protection and safety protection against breakage in high pressure vessels or boilers. If a very high tank uses armored gauges, they will use several of them at overlapping locations so all levels can be visible.

KEO 3. 6 **DESCRIBE** how **REFLEX GAUGE CLASSES** are used to measure liquid level.

REFLEX GAUGE CLASSES are similar to an Armored Gauge Glass in construction using the thick flat glass. These devices are used for applications where the liquid is hard to see in a standard gauge class and uses light refraction to show level. A reflex gauge is depicted below:

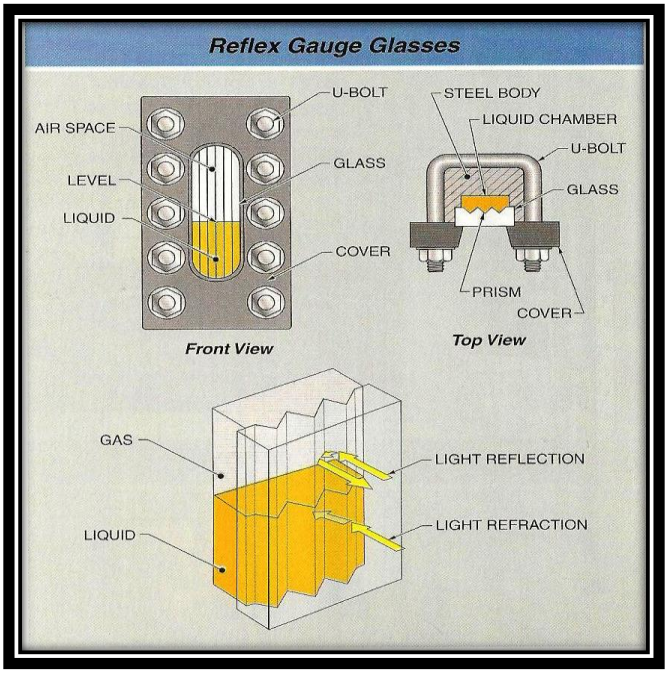
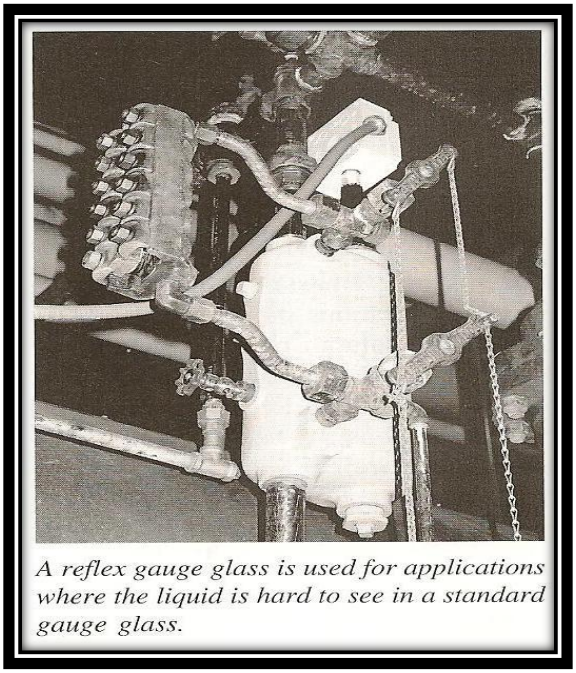


Figure 4-3 page 127



A reflex gauge glass is used for applications where the liquid is hard to see in a standard gauge glass.

Picture page 127

The **REFLEX GUAGE** is a flat gauge with a special vertical saw-tooth surface that acts as a prism to improve readability. The light entering the portion of the prism in contact with the liquid is refracted into the tank and the glass appears dark. The light entering the portion of the prism above the liquid is refracted back out of the gauge and the glass appears silvery white. This feature is useful with clear or translucent liquids that are hard to see in a conventional gauge glass.

KEO 3. 7 **DESCRIBE** how **MAGNETIC-COUPLED** level gauges measure liquid level.

MAGNETIC-COUPLED level gauges are used for more corrosive applications. These gauges use stainless steel floats containing a magnet riding in a stainless steel tube. The level indicator consists of horizontally pivoted magnetized vanes painted yellow or white on one side and black on the other in a housing bolted to the level tube.

As the liquid level raises the float, the vanes flip from showing the black side to showing the yellow or white side. Additionally this device can also include a floating bob to indicate level. The floating bob is colored white or yellow and has a black center which is the mark used to read the level. Both the flip colored flags and the floating bob **MAGNETIC COUPLED LEVEL GAUGES** are depicted below:

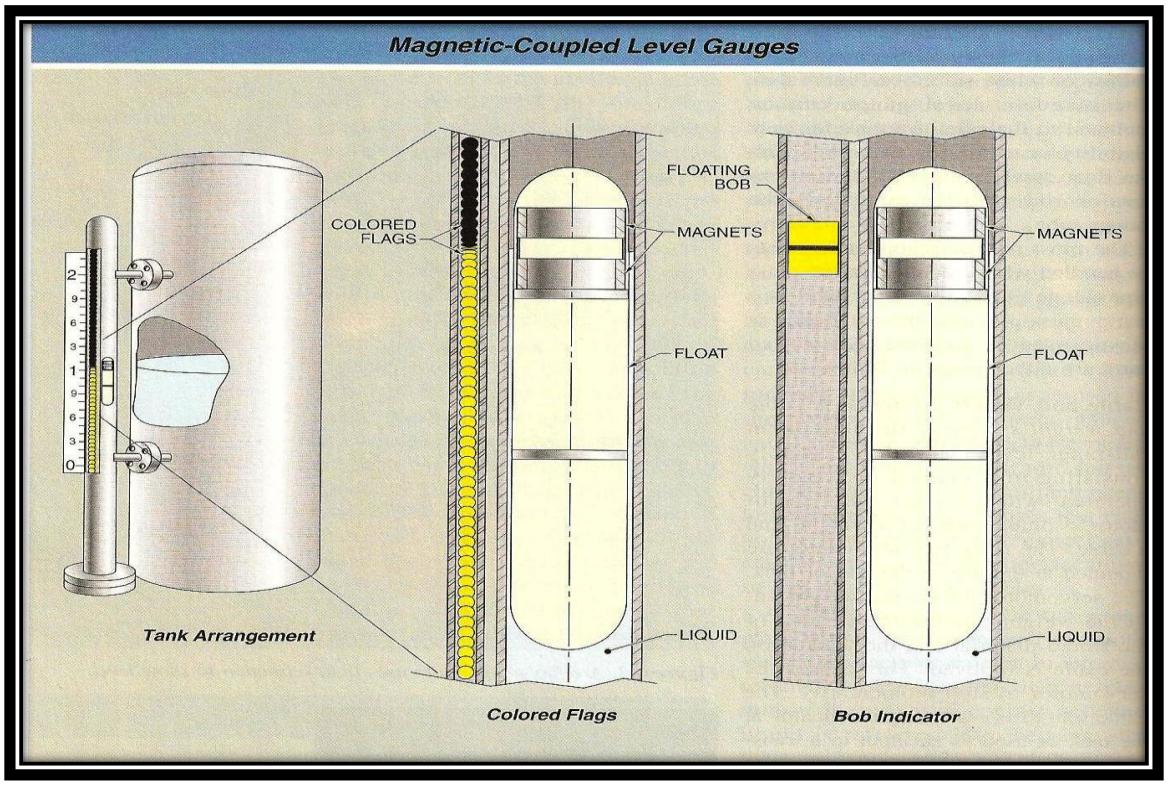


Figure 4-4 page 128

The magnetic level gauge can be mounted to the side of the tank or to the top the tank with the float down in the tank and an extension rod with the magnet in the gauge tube assembly.
The most corrosive applications can be handled with a double-walled glass pipe gauge glass. The inner wall is a heavy glass pipe and the outer pipe is plastic used as a shield. All wetted parts are either glass or Teflon.

KEO 3. 8 **DESCRIBE** how **CABLE and WEIGHT SYSTEMS** measure the level of granular solids in a tank or silo.

A **CABLE and WEIGHT SYSTEM** is an intermittent full-range level measuring assembly consisting of a manual or remotely operated switch, a relay and a servomotor, a plumb bob for a weight and a cable as depicted below:

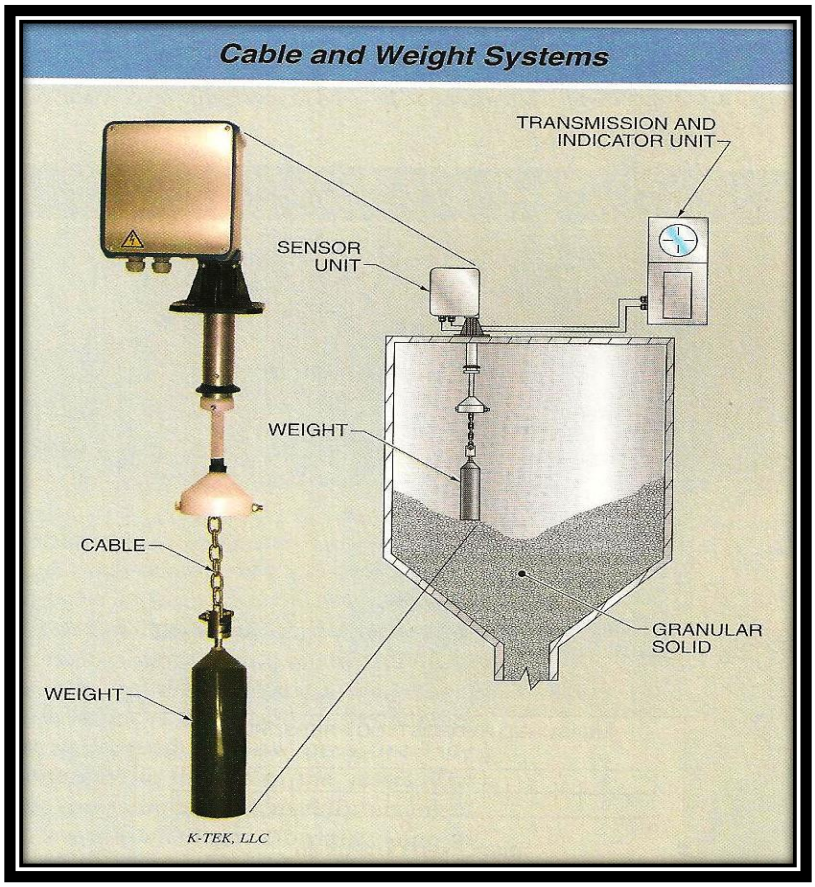


Figure 4-5 page 129

The relay and servomotor for **CABLE and WEIGHT SYSTEM** are mounted at the top of the silo or tank. The servomotor lowers the weight until the plumb bob touches the surface of the material stored in the silo or tank and the tension on the cable is relieved.

This relieved tension causes the servomotor to stop momentarily. The cable length is then read on an indicator at the tank or silo or is transmitted to a remote level indicator. The plumb bob is the returned to a rest position above the maximum tank or silo level.

A **CABLE and WEIGHT SYSTEM** is commonly used to measure granular materials in bins or silos. A needed maintenance condition can occur if dusty material comes in contact the drive mechanism and would require preventative maintenance to be performed to maintain the operability of this system.

SUMMARY

- Gauge glasses are used as a visual indication right at the tank location.
- Reflex gauge glasses are used for applications where the liquid is hard to see in a standard gauge class and uses light refraction to show level.
- Magnetic-coupled level gauges are used for more corrosive applications. These gauges use stainless steel floats containing a magnet riding in a stainless steel tube.
- A Cable and Weight System is commonly used to measure granular materials in bins or silos.

KEO 3. 9 **DESCRIBE** the roll **PRESSURE** has in measuring liquid **LEVEL** using **HYDROSTATIC PRESSURE** for an open tank or a pressurized tank.

Using pressure as means to determine level of a liquid is widely used throughout industry and is one of the most common used options for measuring level. This is because there are many process level applications where it is more convenient to measure the pressure at the bottom of a tank than to measure the actual location of the top of the liquid in a tank or vessel. Whether a tank or vessel is open or sealed to prevent the escape of volatile or toxic fluids, using pressure at the bottom of the tank is a preferred option for measuring liquid level.

The two most used methods of using pressure to measure level include: **HYDROSTATIC PRESSURE (**also referred to as **HEAD PRESSURE**) and the use of **BUBBLER SYSTEMS** to assist in detecting the **HYDROSTATIC/HEAD PRESSURE**.

When using **HYDROSTATIC PRESSURE** to measure level, **PRESSURE** is present at the base of a liquid column. This pressure provides a means of determining liquid level in a vessel or tank. With a known constant density, variations in pressure are caused only by variations in level of that liquid.

HYDROSTATIC PRESSURE of a liquid in an open vessel can be measured by connecting a pressure gauge, switch, or transmitter device to the side of a vessel at the lowest practical level so that any rise in liquid level creates an increase in **HYDROSTATIC PRESSURE**. Pressure sensing devices connected at the lowest level of the vessel can be calibrated to measure units of liquid level.

The picture below depicts how the height of a liquid creates a FLUID HEAD PRESSURE and is the basis for measuring the pressure at the bottom of the fluid being measured:

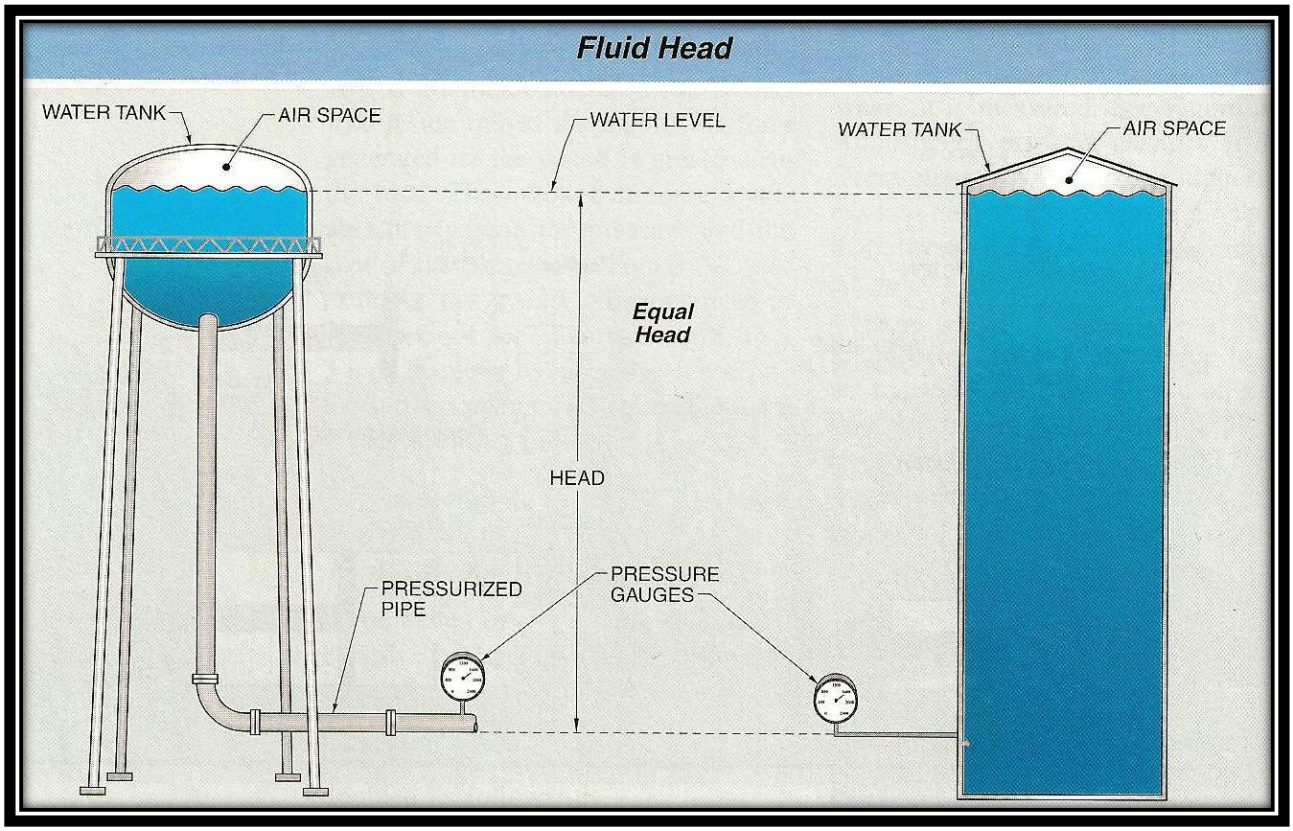


Figure 3-3 page 90

A rule to remember is that when measuring a fluid head pressure, the head of a column of liquid depends only on the height of the column, not the shape of the container.

When measuring a liquid level, the instrument is detecting the weight of that solution against the earth's gravity. A glass of water with 6 inches of water would have the same fluid head pressure that a 5 gallon bucket of water with 6 inches of water would have. The differences between the glass of water and the bucket of water would be its volume, or how much water there is actually in each container. All level tells us is how full or empty the container is.

As a general practice, the location of the pressure sensing device connection on the vessel is established as the zero level point so the calibration of the sensing device is set at zero.

An open or vented tank is depicted below showing how pressure of the height of liquid is sensed:

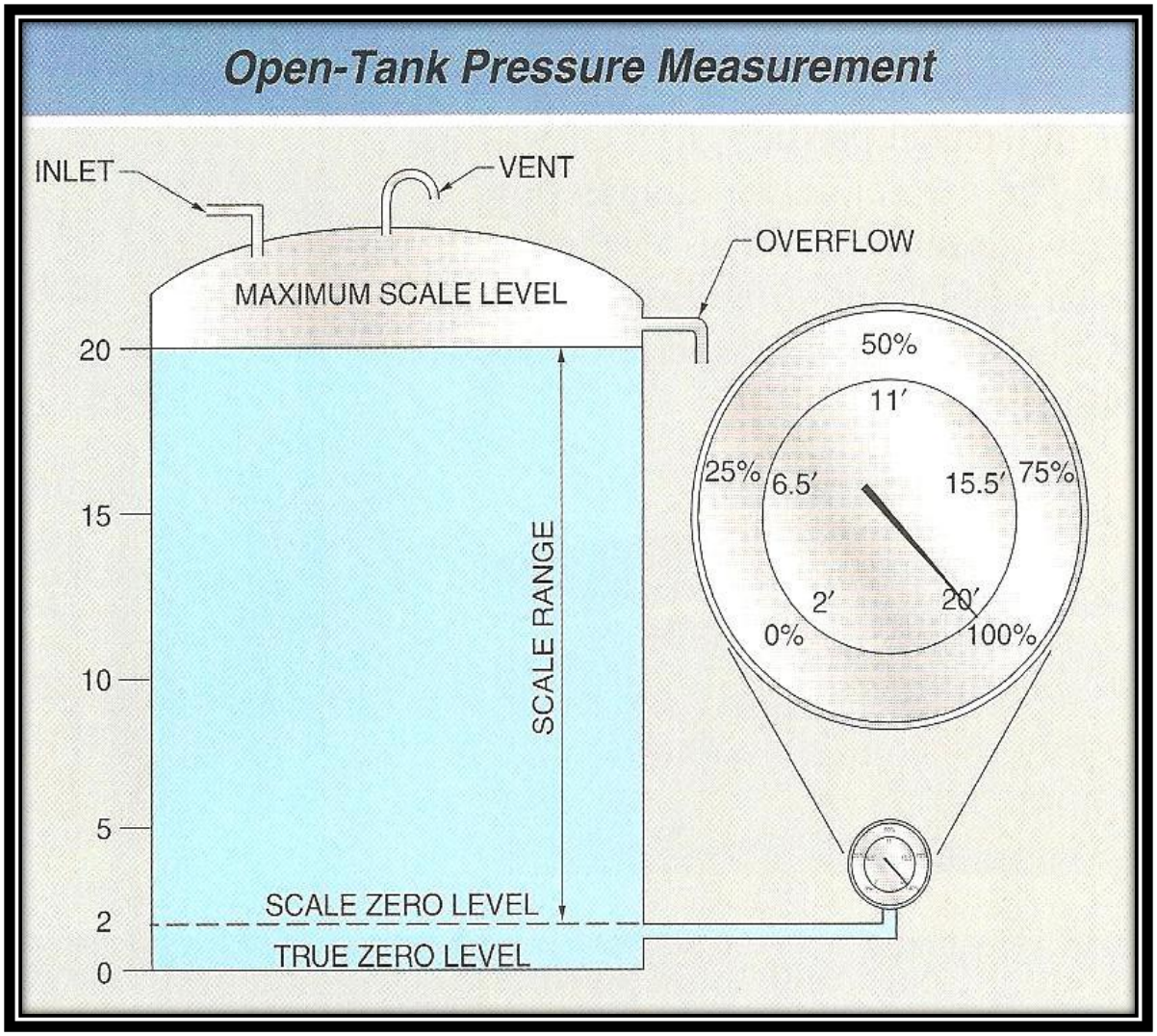


Figure 4-6 page 130

For the picture above, the scale on the left is from 0 to 20 feet in height and the solution in the vessel is water with a specific gravity of 1.0. If the pressure gauge or device measuring pressure were reading in inches of water it would read the difference of 2-20 feet of 18 feet or 0-216 inches of water column pressure.

If the tank being measured is pressurized, then a differential pressure device connection needs to be configured as the following picture depicts:

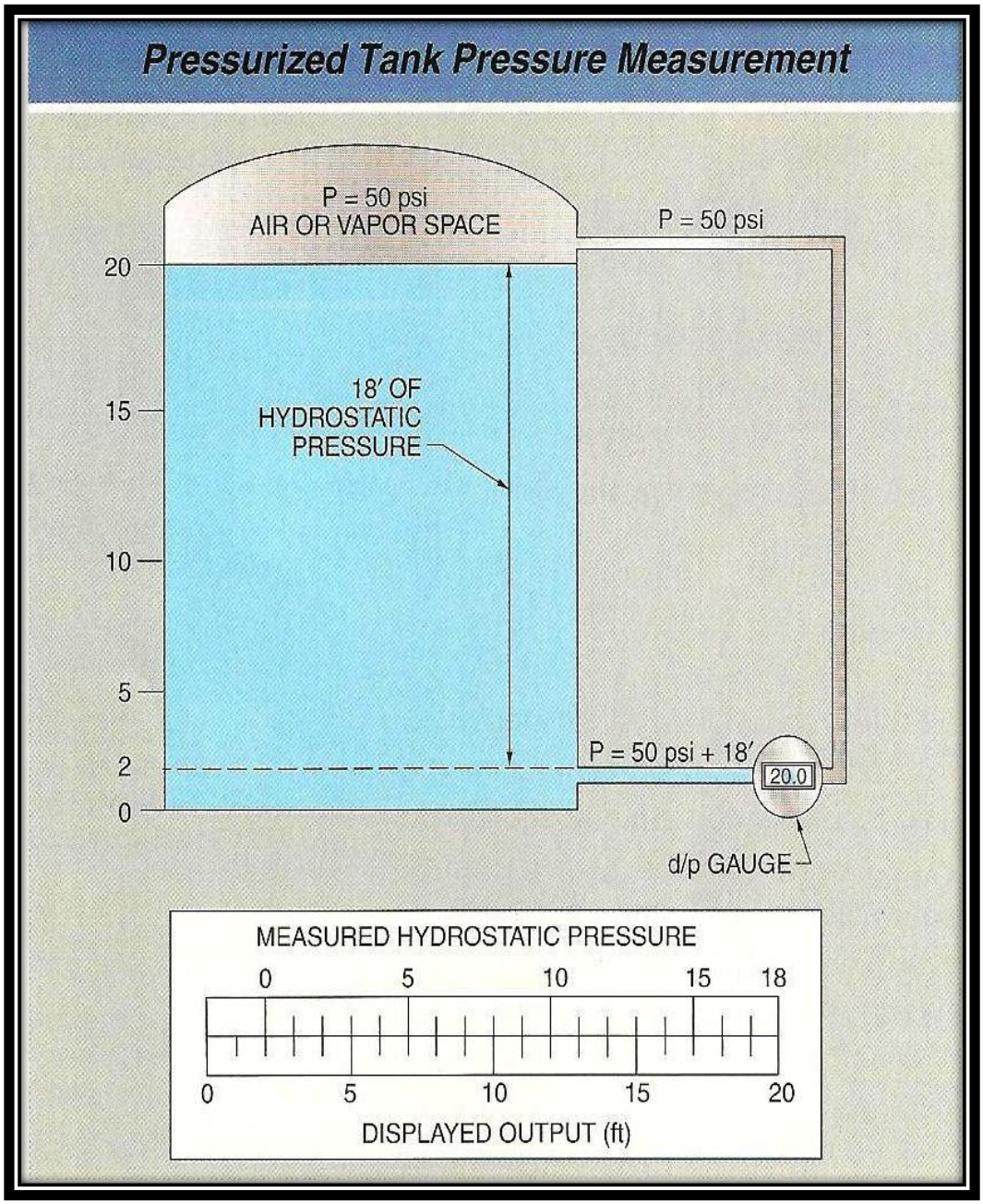


Figure 4-7 page 130

The d/p gauge depicted above could be a differential pressure transmitter providing the capability of transmitting this signal to a remote location.

Differential Pressure Cells come in a variety of sizes and shapes as depicted below:

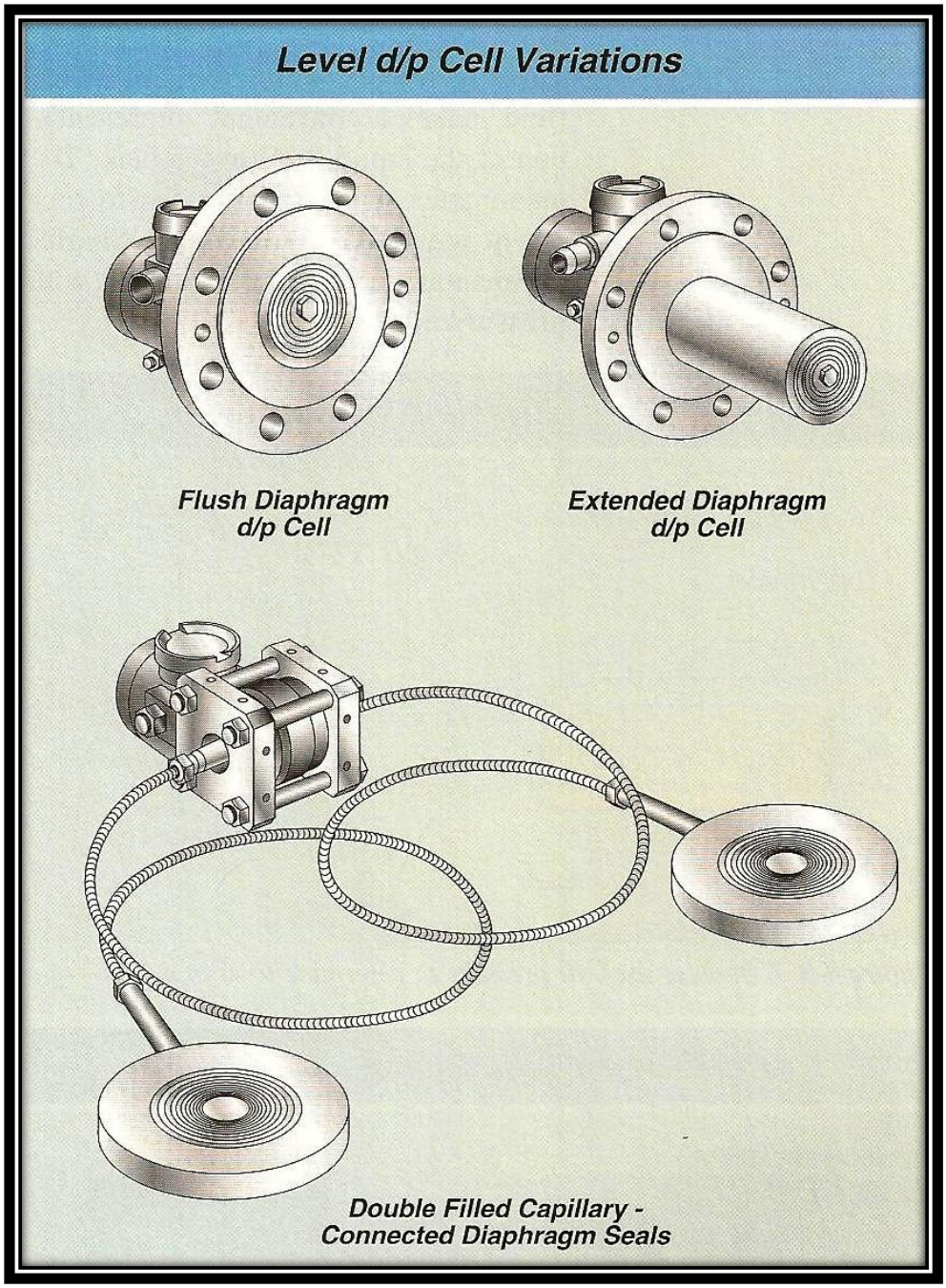


Figure 4-8 page 131

The above DP Cell variations allow for the isolation of corrosive fluids or blockage by fluids with solids when measuring liquid level.

SUMMARY

- Using pressure as means to determine level of a liquid is widely used throughout industry and is one of the most common used options for measuring level.
- When using HYDROSTATIC PRESSURE to measure level, PRESSURE is present at the base of a liquid column. This pressure provides a means of determining liquid level in a vessel or tank. With a known constant density, variations in pressure are caused only by variations in level of that liquid.
- Measuring liquid in an open vessel can be measured by connecting a pressure gauge, switch, or transmitter device to the side of a vessel at the lowest practical level so that any rise in liquid level creates an increase in HYDROSTATIC PRESSURE.
- Measuring liquid in a pressurized vessel can be measured by connecting a differential pressure device to the lowest practical level to the High Pressure port of the device and connecting the Low Pressure port of the device to the top portion of the vessel above the liquid level to allow compensation for the pressurization of that vessel.
- A rule to remember is that when measuring a fluid head pressure/hydrostatic pressure, the head of a column of liquid depends only on the height of the column, not the shape of the container.
- The use of differential pressure cells can isolate process solutions from making contact with the pressure sensing or transmitting device and allows compensation for process pressure differences.

KEO 3. 10 **DESCRIBE** the roll **PRESSURE** has in measuring liquid **LEVEL** using a **BUBBLER SYSTEM** for a tank that is not pressurized and for one that is pressurized.

A **LEVEL BUBBLER SYSTEM** for a non pressurized vessel consists of a tube extending to the bottom of a vessel, a pressure gauge, single-leg manometer, transmitter, or recorder; a flow meter to adjust the flow rate of air or nitrogen through the tube; and a pressure regulator to limit the inlet pressure as depicted below:

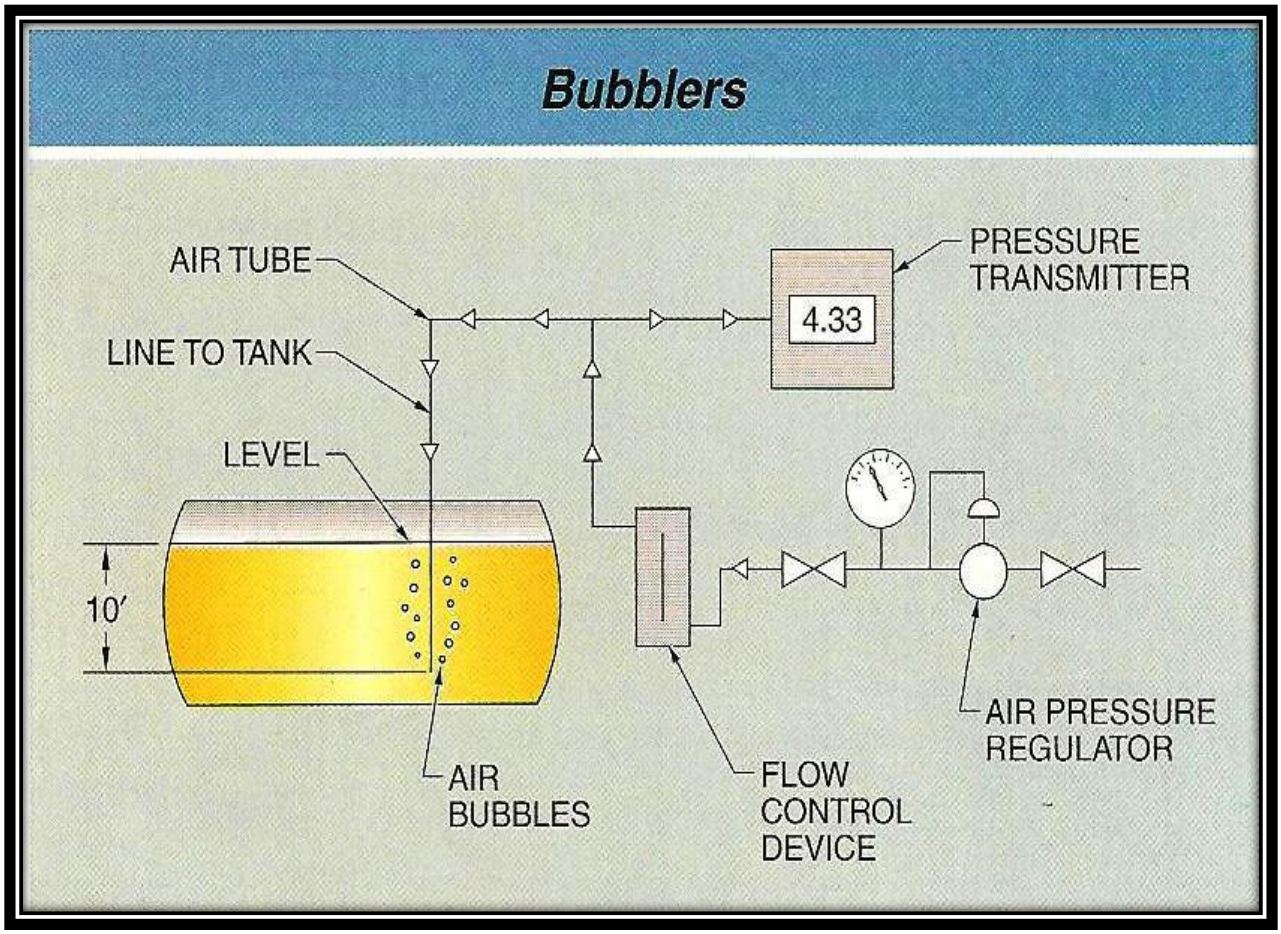


Figure 4-9 page 132

The air or nitrogen is slowly fed into the bubbler system until the pressure is equal to the hydrostatic pressure of the liquid in the tank. At this point, the flow of bubbles goes out of the end of the tube and rises to the top of the tank. For every inch of water height the bubbles raise, it equals the amount of hydrostatic pressure it takes for the bubbles to reach the top of the solution to equal the height or level of that solution.

To summarize, the hydrostatic head pressure is easily converted to level; as the level changes, the pressure in the tube and the pressure measuring instrument changes proportionally. Bubbler principle can be compared to blowing air through a straw and your checks feeling no resistance as you blow. Then you keep that same air flow and slowly immerse this straw into a glass of water. You will now feel the pressure it takes to allow the air to flow up through the water as the bubbles rise to the top of the glass.

Bubbler Level Systems are typically used with tanks that are open to atmosphere, but the system can be adapted to a closed pressurized tank by using a differential pressure gauge or transmitter as depicted below:

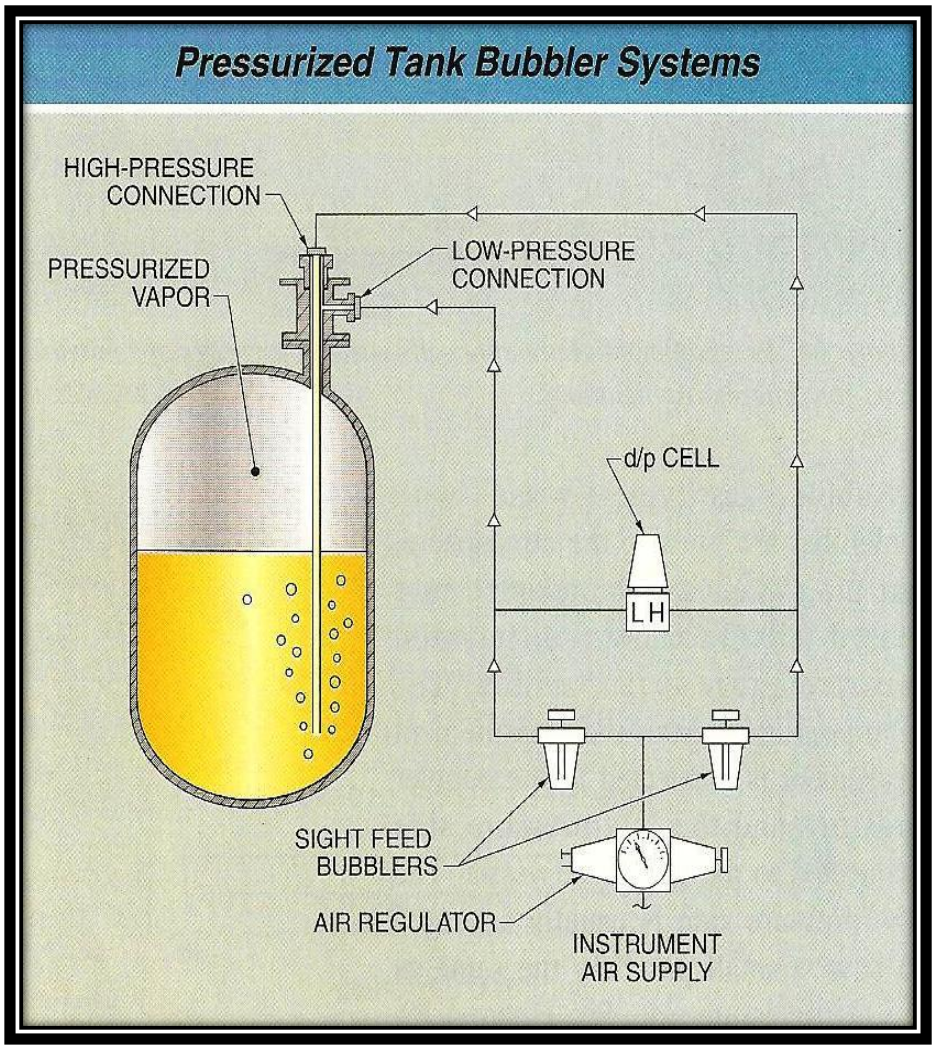


Figure 4-10 page 132

The pressurized system now reads the differential pressure from the bottom of the solution to the top of the pressurized vessel allowing for compensation of the pressure applied to this vessel.

A disadvantage to the use of bubbler tubes to measure level is that the air or nitrogen can allow for liquid drying in the end of the tube and creating a greater back pressure causing a false reading to be given. When this happens, the tubes need to be cleaned with a pressure or solution flush to keep the end of the tube fully open.

The advantage of using bubbler level systems is that the transmitter can be remotely located and not in the same vicinity as the tank being measured. The level is detected by the back pressure from the tubes to the tank via the purge air system and there is no need to compensate for the difference in where the transmitter is actually located. Solutions that are hazardous can remain in a confined location and all that is run to the vessel is tubing to detect the level. Wires and electronic sensors do not have to be in the same vicinity as the vessel to get contaminated or radiated by nuclear processes.

SUMMARY

- A **LEVEL BUBBLER SYSTEM** for a non pressurized vessel consists of a tube extending to the bottom of a vessel, a pressure gauge, single-leg manometer, transmitter, or recorder; a flow meter to adjust the flow rate of air or nitrogen through the tube; and a pressure regulator to limit the inlet pressure.
- A **LEVEL BUBBLER SYSTEM** for a pressurized vessel consists of a tube extending to the bottom of a vessel, a second tube or tubing terminated in the top of the vessel under pressure, a differential pressure gauge, dual leg manometer, transmitter or recorder, two flow meters to adjust the flow rate of air or nitrogen through both tubes, and a pressure regulator to limit the inlet pressure.
- The **DIFFERENTIAL PRESSURE DEVICE** allows for compensation of the pressurization of a vessel allowing the level to be detected without interference from the pressurization of that vessel.
- A disadvantage to the use of **BUBBLER TUBES** to measure level is that the air or nitrogen can allow for liquid drying/crystallizing in the end of the tube, creating a greater back pressure causing a false reading to be given. When this happens, the tubes need to be cleaned with a pressure or solution flush to keep the end of the tube fully open.
- The advantage of using **BUBBLER LEVEL SYSTEMS** is that the transmitter can be remotely located and not in the same vicinity as the tank being measured.

KEO 3. 11 **DESCRIBE** how **FLOATS** are used to measure liquid level.

Level devices using **FLOATS** are dependent of the buoyancy of an object to measure level. A floating object determines the surface of a liquid, whereas a solid object lowered down to top of material in a silo determines the level of a that product.

A **FLOAT** is a point level measuring instrument consisting of a hollow ball that floats on top of a liquid in a tank. Floats are attached to the instrument by a lever to an **On/Off Switch** activated by the movement of the float as depicted below:

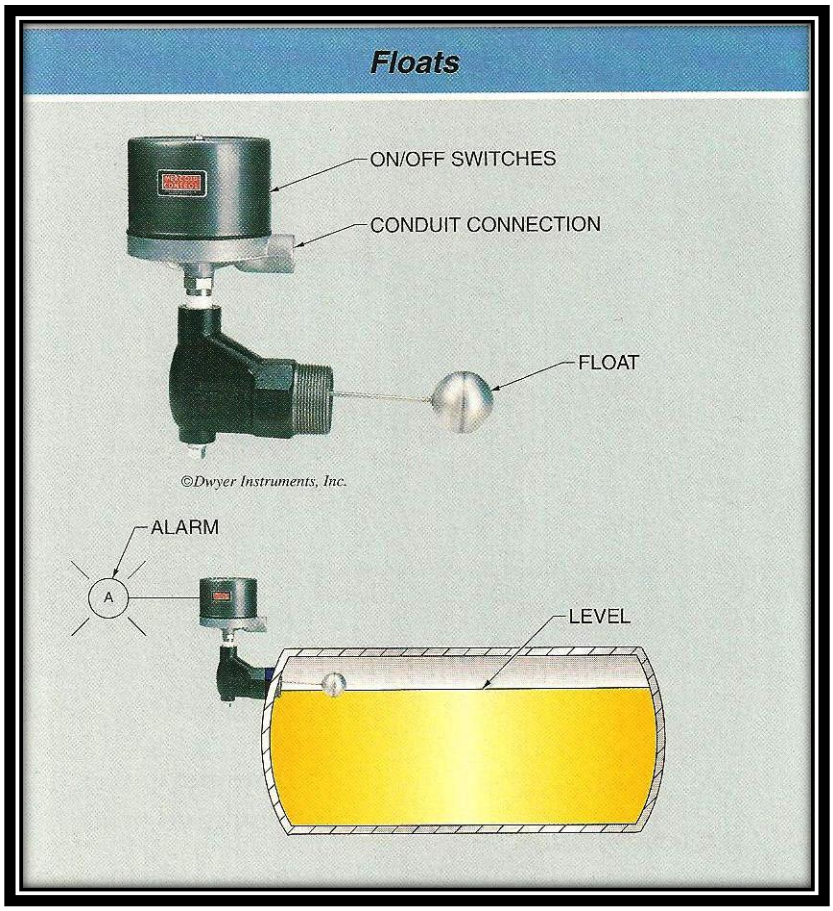


Figure 4-11 page 133

Floats are used to indicate a specific tank level, actuate alarms or shutdown switches, or even mechanically control valves. Switches can start a pump when the float is at one position and stop the pump at another position.

Floats can be located inside of a tank or enclosed in an attached cage or in a stilling well to minimize turbulence and could also include alarm contacts.

Tape Flow Level Instruments using cables, pulleys, and a float with the float located inside of a vessel, as the level raises or lowers, the float attached to a cable will cause an external indication on the outside of a tank to indicate the level of the solution in that vessel.

Tape Floats are typically used as indication devices only, but they can be used with a transmitter for continuous level measuring. Both examples are depicted below:

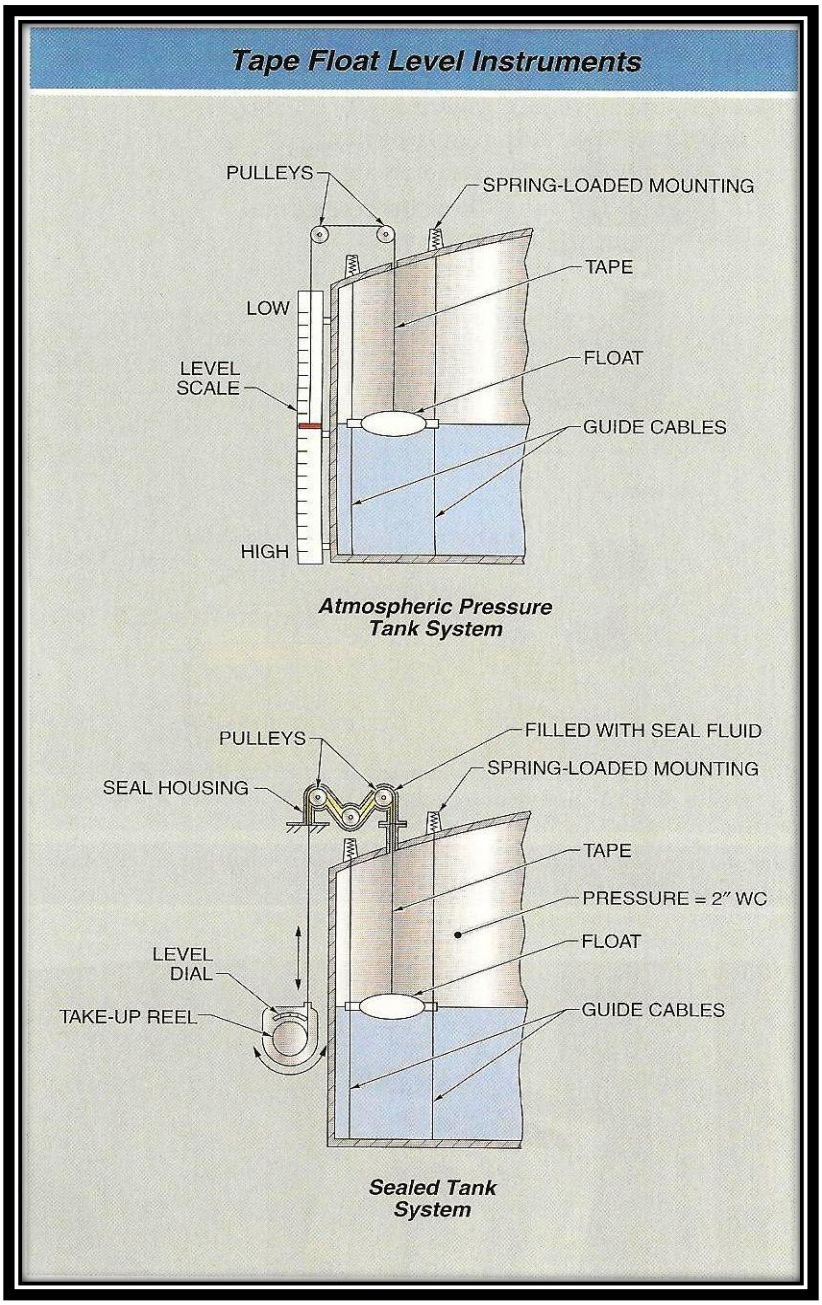
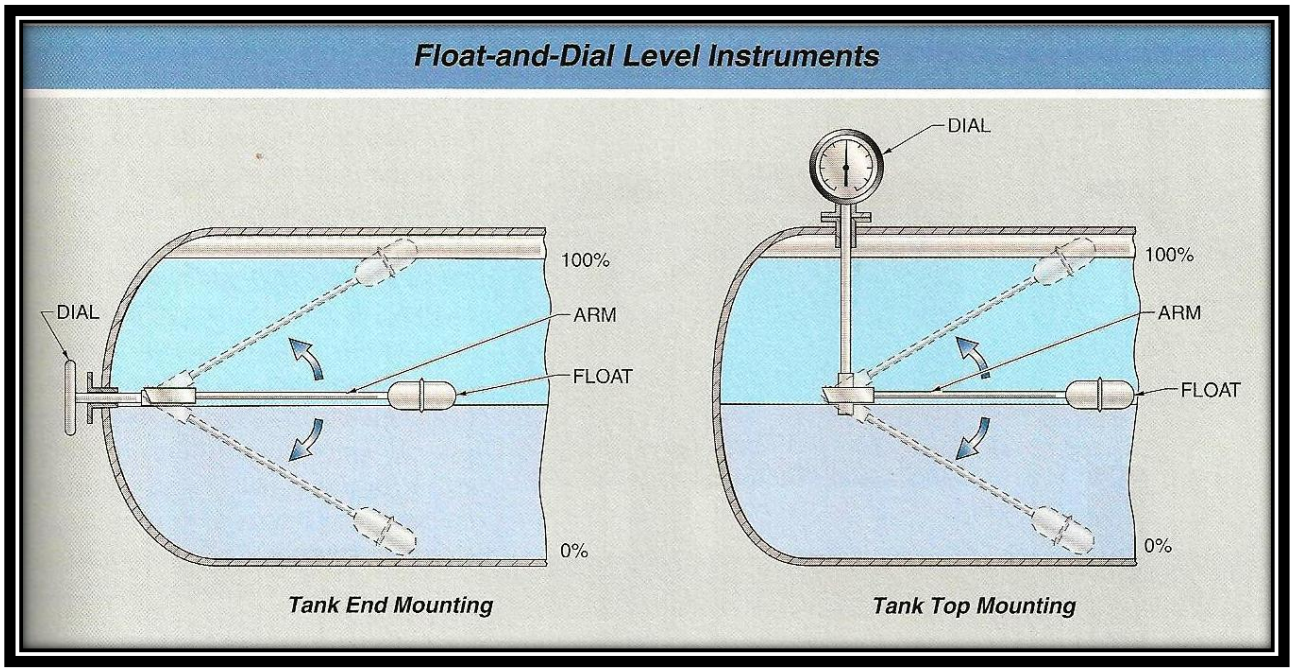


Figure 4-12 page 134

TAPE FLOATS are continuous level measuring instruments consisting of a floating object connected by a chain or cable or tape to a counterweight which is the level pointer. The float rides up and down on two guide wires that keep the float in a specific position. A scale fastened to the outside of the tank shows the reversed tank level with 100% being at the bottom and 0% being at the top. When Float is at the top, the tank if full and when it is at the bottom the tank is empty. The counterweight keeps tension on the tape and the pointer moves up or down to indicate the level.

Tape Floats are subject to mechanical problems due to corrosion and buildup of solutions on the tape causing the device to hang up and give a false indication. Sometimes the float actually develops a leak and falls to the bottom of the tank providing a false indication.

Float and Dial Level Instruments are used with horizontal tanks. A float attached to a long arm, long enough for the float to reach the top and bottom of the tank, and is coupled through a seal to a dial level indicator as depicted below:



These devices are used for measuring clean non corrosive liquids stored under pressure such as ammonia or methyl chloride, and must be ordered for each specific tank application.

The major problem with all float devices is that they are subject to mechanical problems due to moving parts that become worn and are subject to breakage or defects over time.

KEO 3. 12 **DESCRIBE** what a **DISPLACER LIQUID LEVEL MEASUREMENT SYSTEM** is and how it measures level.

A **DISPLACER LIQUID LEVEL MEASUREMENT SYSTEM** is a liquid level measuring system consisting of a buoyant cylindrical object, heavier than the liquid, immersed in the liquid and connected to a spring or torsion device that measures the buoyancy of the cylinder as depicted below:

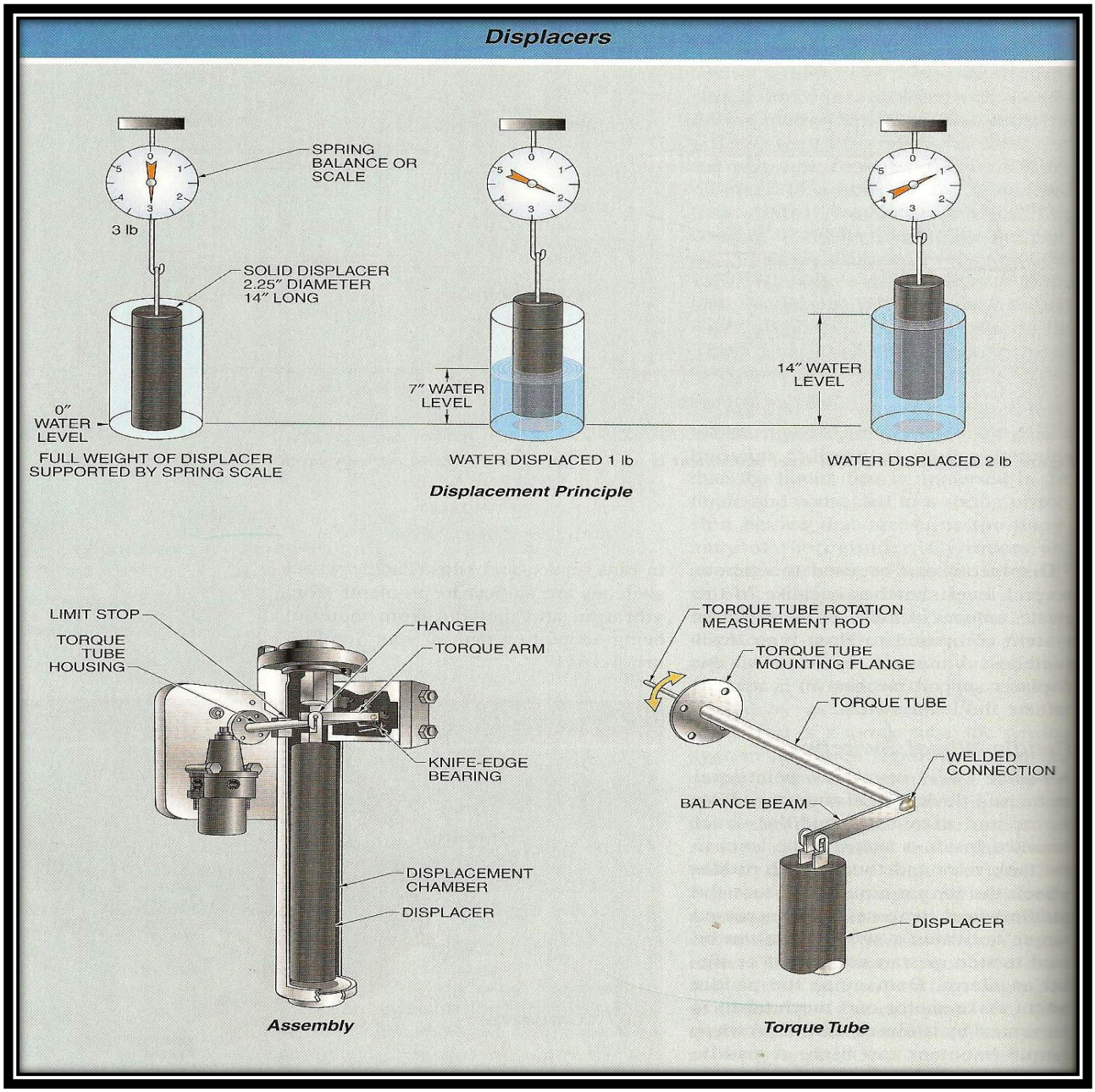


Figure 4-15 page 136

The advantage of using the Displacer Level Instrument is that the movement from the torque tube assembly can easily be transmitted via a pneumatic 3-15 psig or a 4-20 mA signal for remote level indication and control of the liquid level.

SUMMARY

- A **FLOAT** is a **Point Level Measuring Instrument** consisting of a hollow devicel that floats on top of a liquid in a tank.
- **Floats are used** to indicate a specific tank level, actuate alarms or shutdown switches, or even mechanically control valves.
- **TAPE FLOATS** are **Continuous Level Measuring Instruments** consisting of a floating object connected by a chain or cable or tape to a counterweight which is the level pointer.
- **Float and Dial Level Instruments** are **Continuous Level Measuring Instruments** used with horizontal tanks and uses a float attached to a long arm, long enough for the float to reach the top and bottom of the tank, and is coupled through a seal to a dial level indicator.
- **All level float devices** are subject to mechanical problems due to moving parts that become worn and are subject to breakage or defects over time.
- A **DISPLACER LIQUID LEVEL MEASUREMENT SYSTEM** is a liquid level measuring system consisting of a buoyant cylindrical object, heavier than the liquid, immersed in the liquid and connected to a spring or torsion device that measures the buoyancy of the cylinder as level increases or decreases.
- The **Displacer Level** system is a **Continuous Level Measuring Instrument**.
- The **Displacer Level Instrument Using The Torque Tube Assembly** can easily be transmitted via a pneumatic 3-15 psig or a 4-20 mA signal for remote level indication and control of the liquid level.

KEO 3. 13 **DESCRIBE** how a **PADDLE WHEEL SWITCH** is used as a point level measuring device

A **PADDLE WHEEL SWITCH** is a point level measuring device which utilizes a motor and a rotating paddle wheel mounted inside of a tank as depicted below:

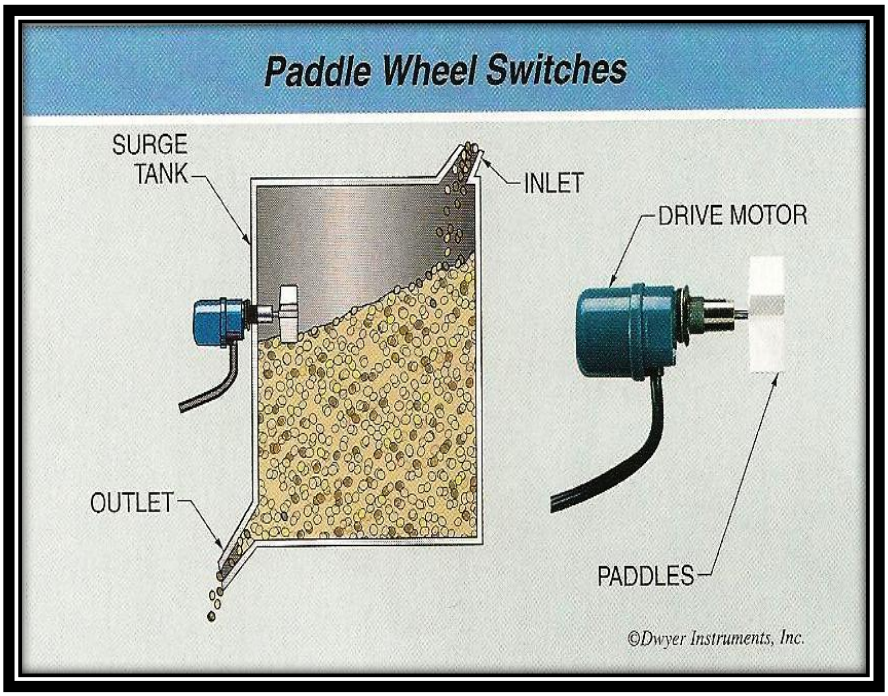


Figure 4-16 page 137

When the level in the tank rises and comes in contact with the paddle wheel, the torque required to turn the wheel increases. The increase in torque activates a switch circuit that can be used to stop or start equipment to increase or decrease the material flow in or out. This torque can also signal an alarm to provide operations with a high or low level alarm status of the product in the tank.

Damage to the motor and paddle wheel assembly is prevented by a slip clutch which allows the motor to continue rotating, but allows the paddle to slip with the increase in torque.

Paddle wheels are commonly used to measure the level of granular solids in pneumatic conveying equipment and in bins and tanks for silo collection and storage. They are subject to problems from vibration and damage from material being added to the tank or silo. Location of the paddle switch should be in direct flow of material being added to the tank or silo as depicted in above picture.

KEO 3. 14 **DESCRIBE** how **BEAM-BREAKING PHOTOMETRIC SENSORS** provide point level measurement and how false signal can affect them.

A **BEAM-BREAKING PHOTOMETRIC SENSORS** provide point level measurement with a light source and a detector. The light source shines a beam of light to the detector (called a reflector) that indicates a point level when the product level blocks the beam indicating a full level has been achieved as depicted below:

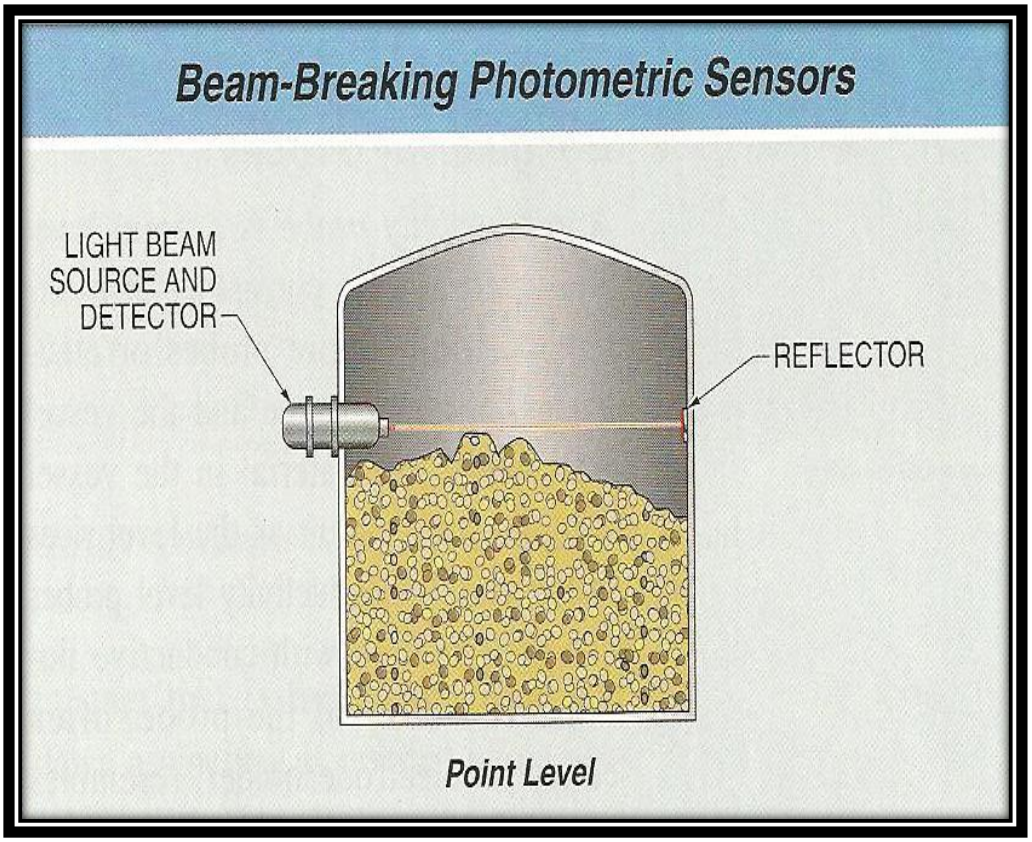
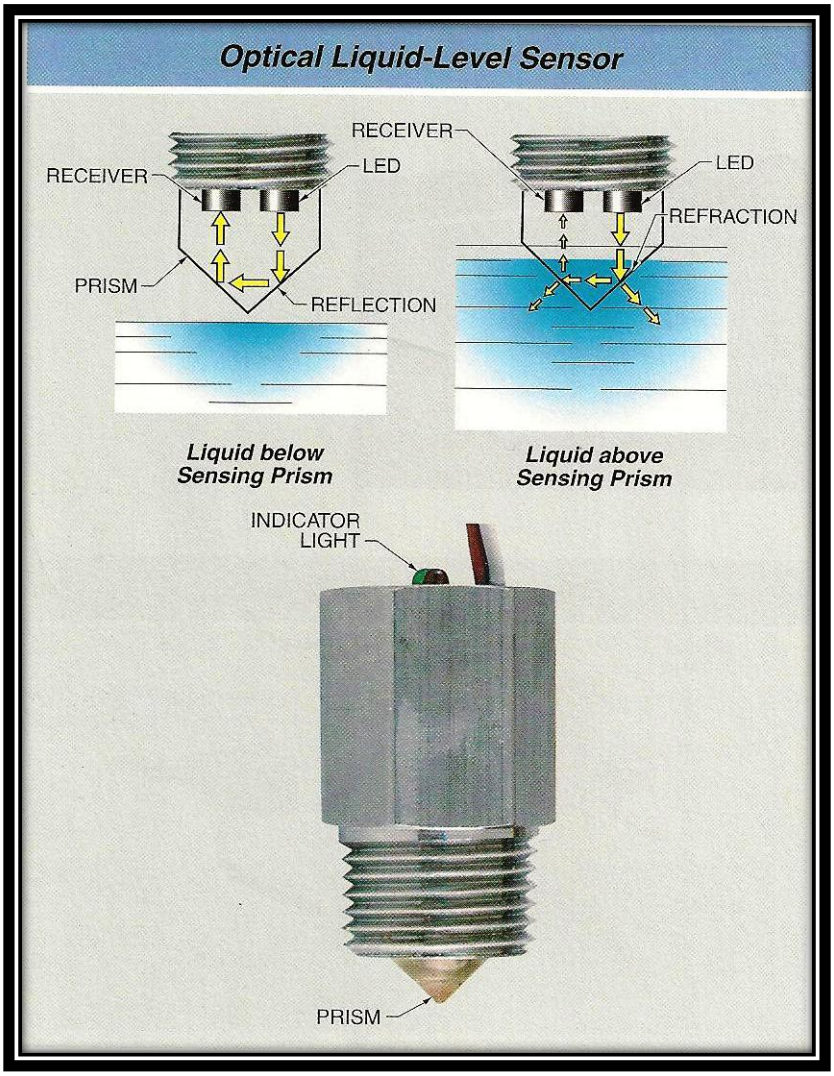


Figure 4-17 page 137

These sensors are subject to giving a false indication from an outside light source or from dust of splashes of liquid on the reflector or light beam source. The light source and reflector lenses must be kept clean to maintain the strength of the light beam and the ability of the detector to accurately sense the beam until the product blocks it.

KEO 3. 15 **DESCRIBE** how **OPTICAL LIQUID-LEVEL SENSORS** provide point level measurement.

OPTICAL LIQUID-LEVEL SENSORS provide point level measurement by utilizing the same principle as the **REFLEX GAUGE GLASS**. A light source and a light detector, shielded from each other, are mounted in an enclosed housing as depicted below:



The light beam is directed against the inside of a glass or plastic cone-shaped prism. If the cone is above the liquid, the light is reflected from the cone back to the detector. If the cone is submerged in the liquid, the light is refracted into the liquid and is not sensed by the detector. The light detector circuit can activate a control relay contact for alarms or control.

SUMMARY

- A paddle wheel switch is a point level measuring device which utilizes a motor and a rotating paddle wheel mounted inside of a tank.
- Paddle wheels are commonly used to measure the level of granular solids in pneumatic conveying equipment and in bins and tanks for silo collection and storage.
- Beam-breaking photometric sensors provide point level measurement with a light source and a detector.
- Beam-breaking sensors are subject to giving a false indication from an outside light source or from dust of splashes of liquid on the reflector or light beam source.
- The light source and reflector lenses must be kept clean to maintain the strength of the light beam and the ability of the detector to accurately sense the beam until the product blocks it.
- Optical Liquid-Level sensors provide point level measurement by utilizing the same principle as the Reflex Gauge glass.

KEO 3. 16 **DESCRIBE** how **CONDUCTIVITY PROBES** provide a point level measurement of liquid level.

CONDUCTIVITY PROBES provide a point level measurement of liquid level through the electrical conductivity of a liquid. The liquid has to be conductive to be able to provide this level measurement. **CONDUCTIVITY PROBES** consist of an electrical circuit of two or more probes (electrodes) inserted in a metal conductive tank where the metal in the vessel completes the circuit as the liquid level rises to immerse the electrode(s) as depicted below:

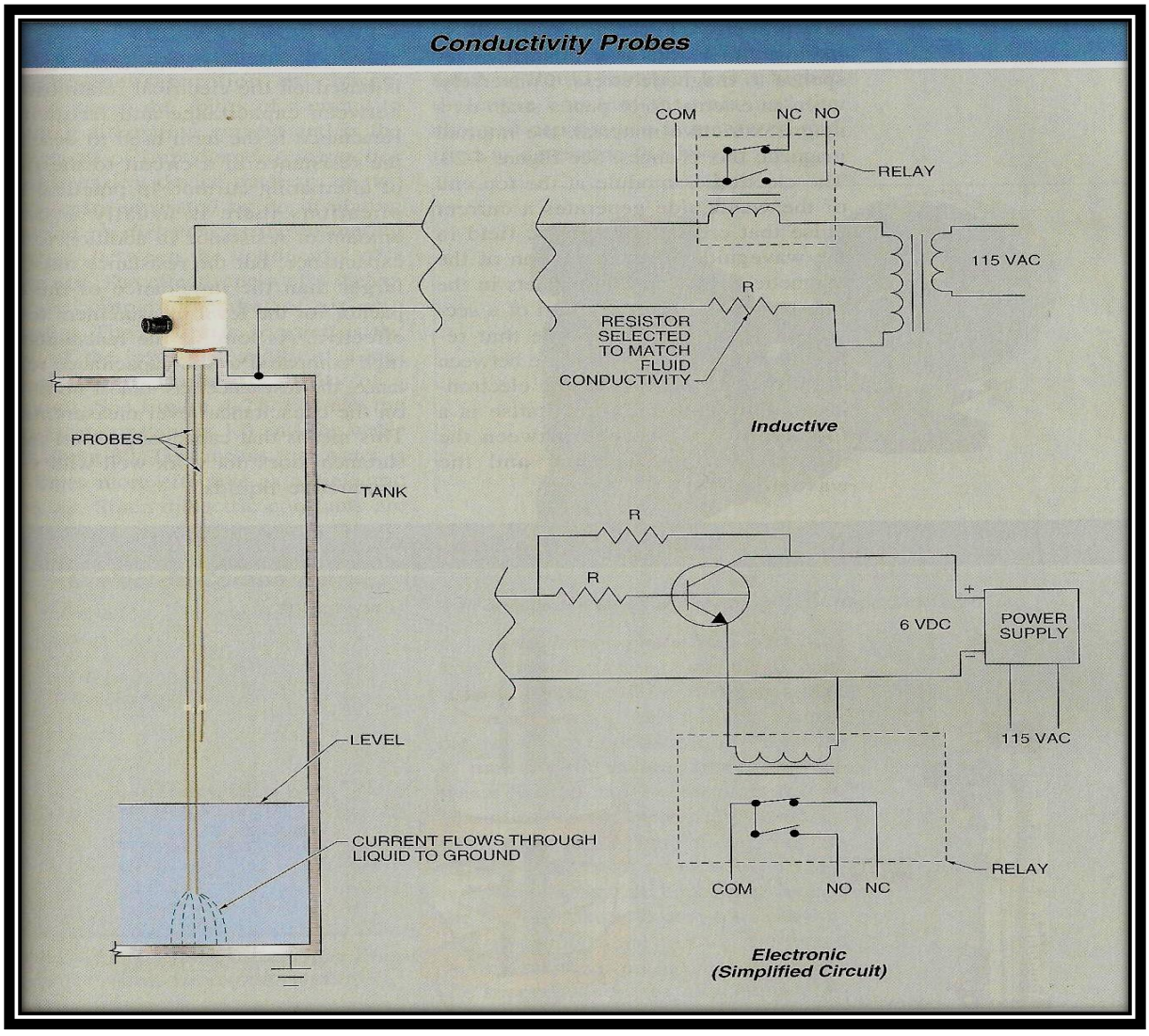


Figure 4-19 page 139

The circuit above shows using both an AC and a DC power source to accomplish two ways of sensing the conductivity of the liquid. The body of the electrode/probe is often called an electrode holder and resembles an automotive spark plug.

This holder is threaded into a receptacle or bolted to a flange at the top of the tank to act as an electrical ground for the circuit. There is an electrical hazard associated with the utilization of conductivity probes and a high level AC power source.

In the conductivity probe picture above, the first example shows how the inductive use of AC power can open and close a set of contacts. As the 115 Volt AC power is applied to the circuit, one leg of the power goes to the electrode and the other leg goes directly to the metal tank that is grounded back to the power source. If a person were to touch the side of the tank with a bare hand when the liquid level allows a current to flow to ground, there is an electrical shock hazard of 115 Volts AC. For this reason, when utilizing AC Power electrical hazards must be mitigated so this potential is minimized.

The alternative method shown in the above picture show how a power supply of 6 Volts DC from a power supply can reduce this electrical shock hazard and still accomplish the task of measuring liquid level conductivity. With this method of a transistorized sensing circuit, a person touching the tank wall with a bare hand would not receive a shock as the 6 Volt DC power source eliminates any danger of an electrical shock hazard.

The picture also depicts the possibility of more than one electrode to provide different point level detection of the liquid level. With this configuration, a tank level alarm could signal a level alarm at different levels.

A common application for conductivity probes is for sump pump applications. For example if a tank where to overfill or a flooding condition occurred in an undesired location, as the sump detects a level, it will activate a pump to pump the sump dry to prevent a possible flood or the spread of a solution to the environment.

KEO 3. 17 **DESCRIBE** how **MAGNETOSTRICTIVE SENSORS** provide continuous measurement of liquid level.

MAGNETOSTRICTIVE SENSORS provide continuous measurement of liquid level with an electronics module, a waveguide, and a float containing a magnet that is free to move up and down a pipe that is attached to a vessel.

Within the pipe is a waveguide constructed of magnetostrictive material. “Magnetostrictive” refers to a property of certain ferrous alloys having dimensions that change in response to magnetic stress. In opposition, when an external force puts a strain on a magnetostrictive material, the internal magnetic flux changes.

A **MAGNETOSTRICTIVE SENSOR** uses a magnetic pulse to determine the location of a moving float as depicted below:

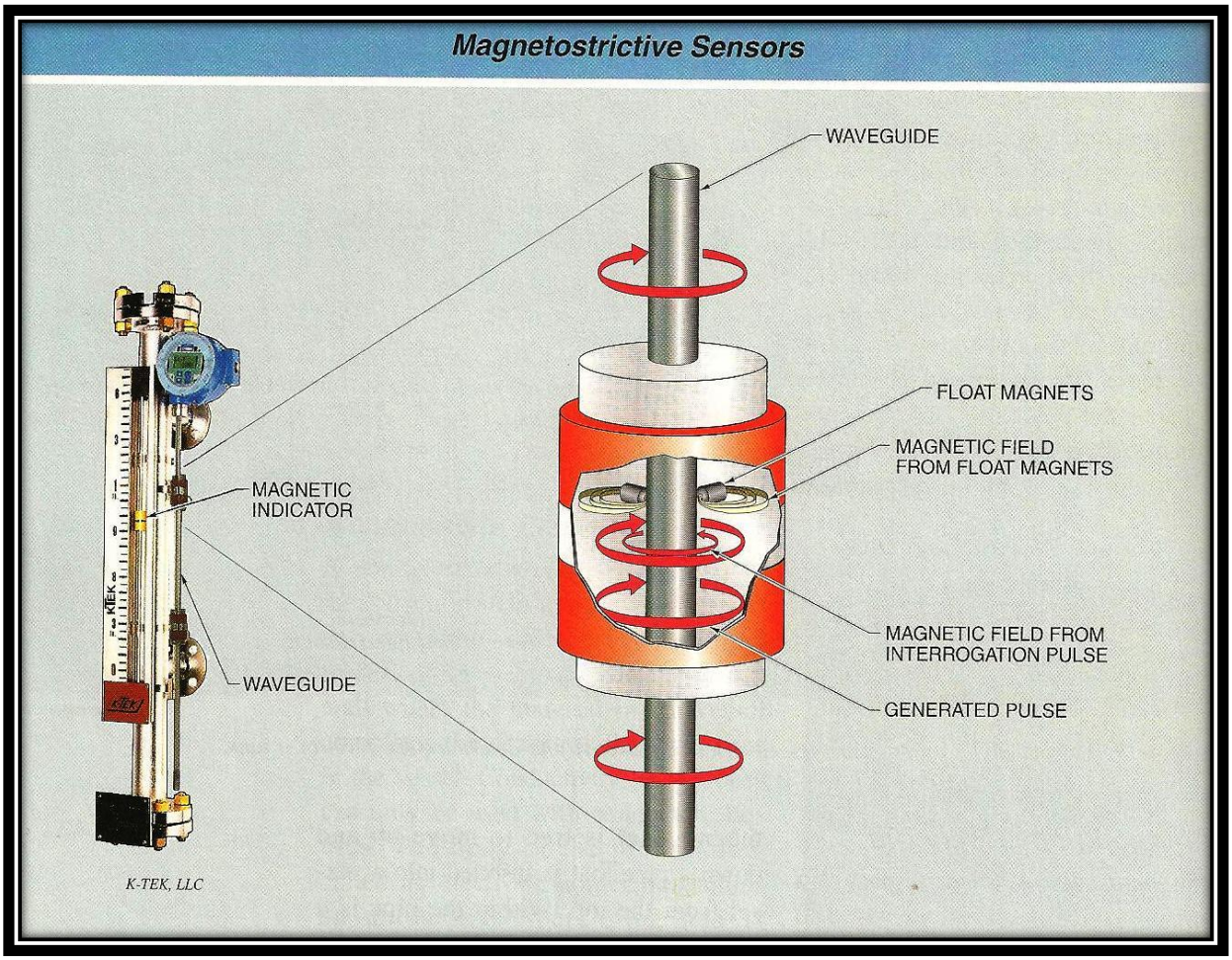


Figure 4-20 page 140

The electronics module at the top end of the waveguide generates a current pulse that creates a magnetic field in the waveguide. The interaction of the magnetic field with the magnets in the float results in the generation of a second pulse in the waveguide that reflects back to the top.

The time between the generated pulse from the electronics module and the return pulse is a function of the distance between the magnets within the float and the waveguide.

SUMMARY

- Conductivity Probes provide a point level measurement of liquid level through the electrical conductivity of a liquid.
- The liquid has to be conductive to be able to provide this level measurement.
- Conductivity Probes consist of an electrical circuit of two or more probes (electrodes) inserted in a metal conductive tank where the metal in the vessel completes the circuit as the liquid level rises to immerse the electrode(s).
- There is an electrical hazard associated with the utilization of Conductivity Probes and a high level AC power source; this is prevented with the use of a 6 VDC power supply an electronic transistor switching circuit.
- A common application for Conductivity Probes is for sump pump applications.
- A Magnetostrictive Sensor uses a magnetic pulse to determine the location of a moving float.
    - An electronics module at the top end of the waveguide generates a current pulse that creates a magnetic field in the waveguide.
    - The interaction of the magnetic field with the magnets in the float results in the generation of a second pulse in the waveguide that reflects back to the top.

KEO 3. 18 **DESCRIBE** how **CAPACITANCE PROBES** provide point level and continuous measurement of liquid level.

CAPACITANCE PROBES can provide both point level and continuous level measurement. **Capacitance Probes** are based on the electrical relationships between capacitance and frequency.

Reactance is the term used to describe the resistance of a circuit to the flow of alternating current. In practical applications there is usually a small amount of resistance in addition to the capacitance, but the resistance must be larger than the impedance of the capacitor for the level measurement to be effective.

As long as the resistance is high compared to the capacitance reactance, the resistance has almost no effect on the capacitance level measurement. This means that capacitance level measurement does not work well with very conductive liquids and a better option for conductive liquids would be conductivity probes as previously discussed in objective KEO 3.16.

- ***Capacitance*** is the ability of an electrical device to store charge as the result of the separation of charge.
- ***Admittance*** is the ability of a circuit to conduct alternating current and is the reciprocal of impedance.
- A ***Capacitor*** is an electrical device made up of two conductors separated by an insulating material.
- A ***Dielectric*** is the insulating material between the conductors of a capacitor. The effectiveness of a dielectric is compared to that of air or a vacuum.
- The ***Dielectric Constant*** is the ratio of the insulating ability of a vacuum.

The following depicts how the dielectric constant determines the effectiveness of a capacitor:

Dielectric Comparison

Material	Dielectric Constant*
Vacuum	1
Acetone	21
Ammonia (-27°F)	22
Calcium Carbonate	9.1
Ethanol	24
Freon 12	2.4
Kerosene	1.8
Methanol	33
Naphthalene	2.5
Sand	3.5
Sugar	3.0
Toluene	2.4
Water	80
Water (212°F)	48

*at 68°F unless noted

Figure 4-21 page 141

The above compares Vacuum (Air) to Water, and shows that water is 80 times more effective as a dielectric than Vacuum (Air).

The amount of capacitance depends on the dielectric constant, the surface area of the conductors, and the distance between the conductors. For example, when a capacitor probe is fixed in place, the surface area and the distance cannot change. When a level rises, the capacitance increases because the material in the vessel replaces the air or vapor between the conductors. A granular solid or liquid material has a higher dielectric than air, and the changed capacitance is measured and used to determine the level.

A **CAPACITANCE PROBE** is a part of the level measuring instrumentation and consists of a metal rod inserted into a tank or vessel, with a high-frequency alternating voltage applied to it as a means to measure the current that flows between the rod and the second conductor.

The metal rod (***capacitance probe***) is electrically insulated from the tank or vessel. A bare rod can only be used with nonconductive liquids and a plastic coated rod can be used with conductive liquids.

The current that flows from the rod to the second conductor, is proportional to the admittance or capacitance from the metallic rod to the second conductor. For many applications, the most convenient conductor is the metallic tank or vessel wall. Therefore, a capacitance probe measures current flow from the metal rod through the liquid material in the tank or vessel to ground as depicted below:

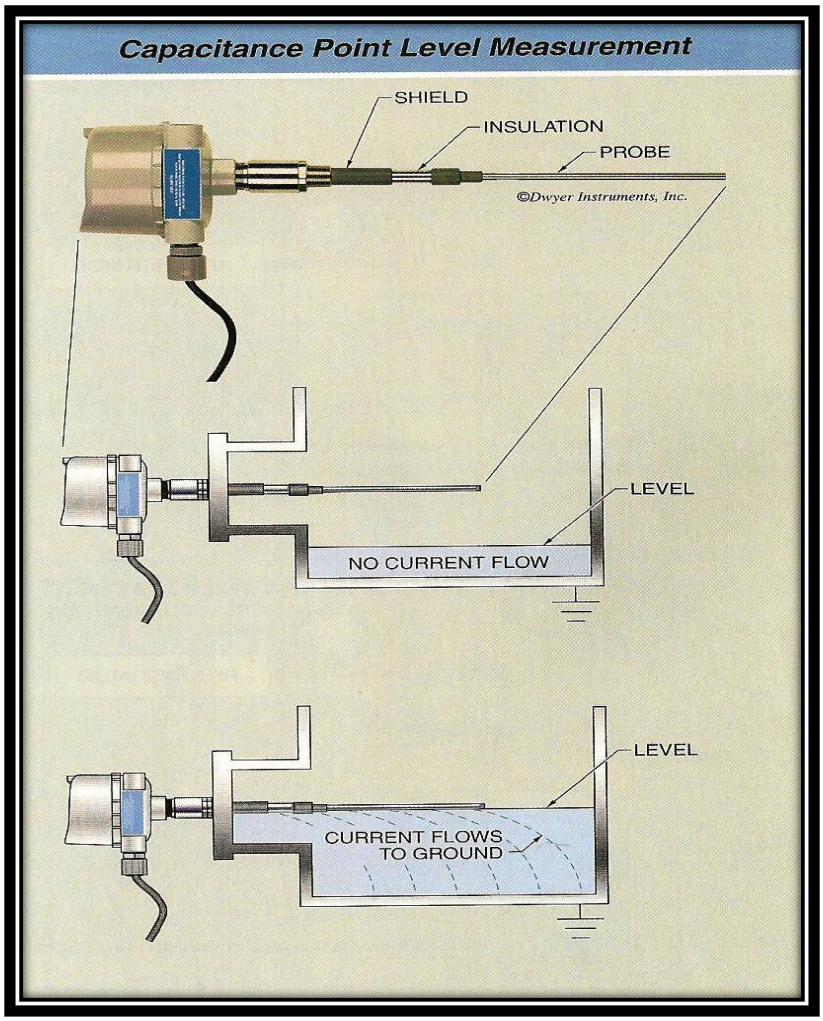


Figure 4-23 page 143

Capacitance Level Measurement works best for liquids that do not coat the probe and is generally not as effective for slurries or granular materials. Some RF Capacitance Sensor designs are effective for working with other than liquid materials.

In order for current to flow in a capacitive circuit, alternating current (AC) must be used. The typical power supply us generally a standard 120 VAC (other voltages are also available). The AC is converted by an oscillator to a 100 kHz radio frequency (RF) input to a bridge circuit as depicted below:

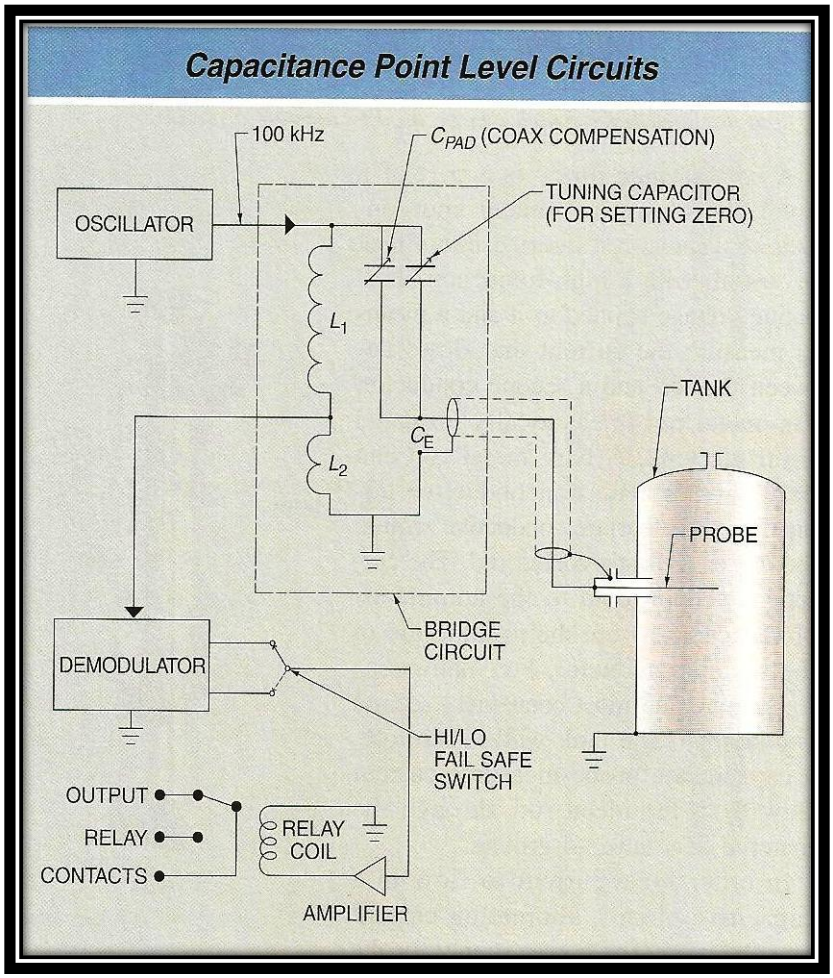


Figure 4-22 page 142

Because capacitance probes use RF, they are also called RF Capacitance Probes.
Single-Point Level-Control uses capacitance probes as a switch to signal an alarm or to actuate a circuit when the level in a tank or vessel reached a specified limit.

Continuous Capacitance Probes senses the level in the tank as it rises up the probe, the amount current that can flow increases providing a continuous level measurement. The increased current is directly proportional to the level as depicted below:

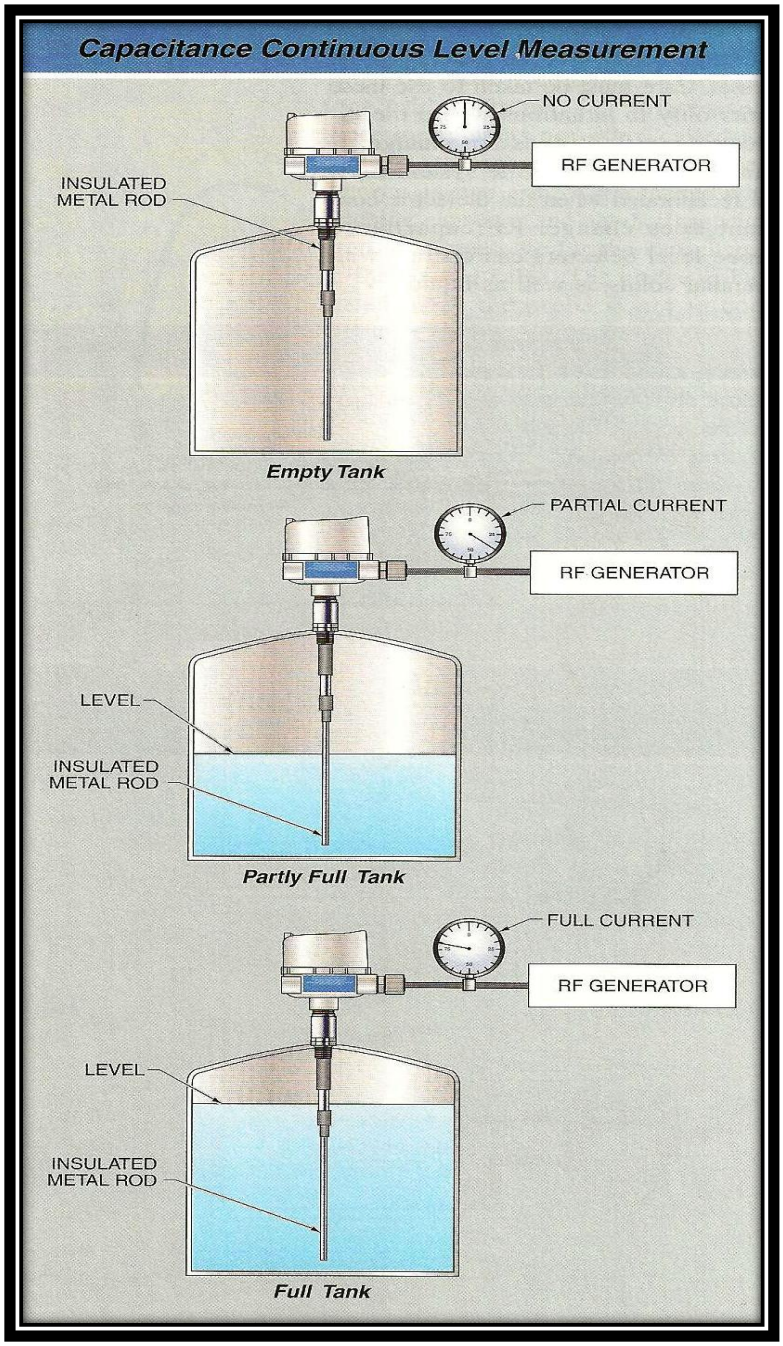
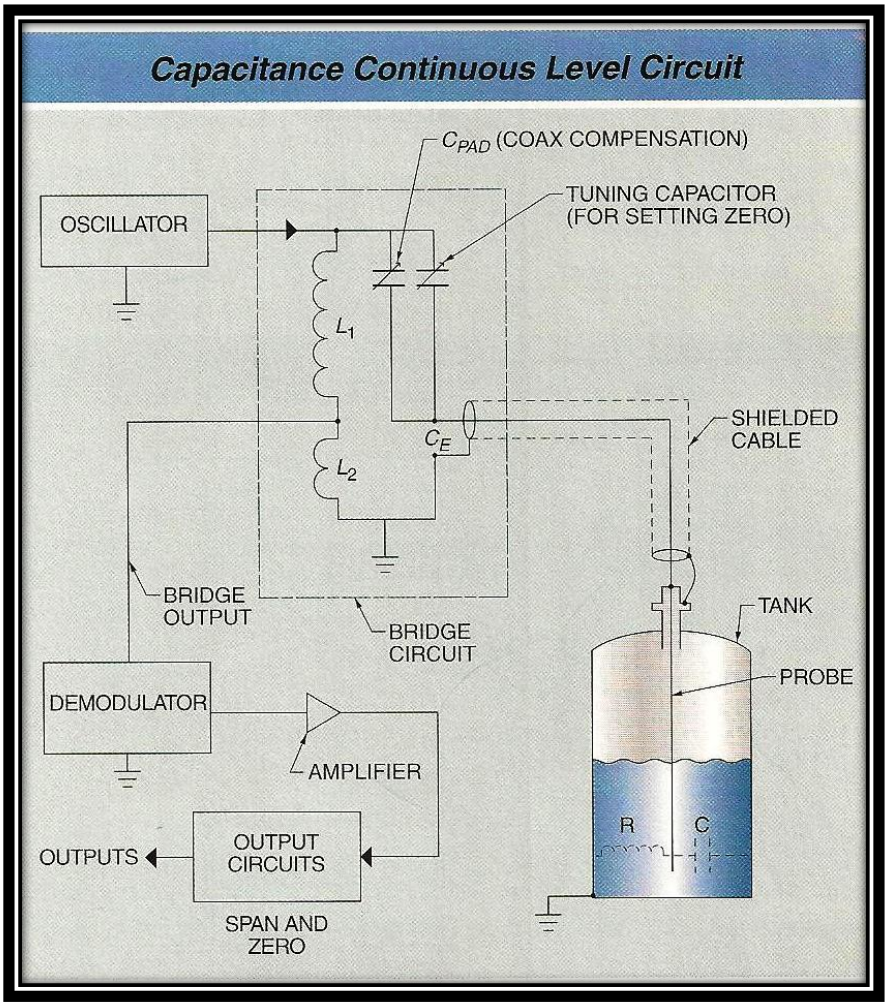


Figure 4-24 page 144

Depicted below is a Capacitance Continuous Level Circuit:



Problems with Capacitance Level Measurement include the buildup of process material on the probe can sometime present a problem as the presence of a coating can act as a dielectric even when the level is below the probe. Manufactures have developed probe designs and added adjustment specifications to diminish the effect of coating as well as changes in the physical properties of the process fluid like density or composition.

SUMMARY

- CAPACITANCE PROBES can provide both point level and continuous level measurement.
- Capacitance Probes are based on the electrical relationships between capacitance and frequency.
- ***Capacitance*** is the ability of an electrical device to store charge as the result of the separation of charge.
- ***Admittance*** is the ability of a circuit to conduct alternating current and is the reciprocal of impedance.
- A ***Capacitor*** is an electrical device made up of two conductors separated by an insulating material.
- A ***Dielectric*** is the insulating material between the conductors of a capacitor. The effectiveness of a dielectric is compared to that of air or a vacuum.
- The ***Dielectric Constant*** is the ratio of the insulating ability of a vacuum.
- Water is 80 times more effective as a dielectric than Vacuum (Air).
- For many applications, the most convenient conductor is the metallic tank or vessel wall; Therefore, a capacitance probe measures current flow from the metal rod through the liquid material in the tank or vessel back to ground.
- Capacitance Level Measurement works best for liquids that do not coat the probe and is generally not as effective for slurries or granular materials.
- Because Capacitance Probes use RF, they are also called RF Capacitance Probes.
- Single-Point Level-Control uses capacitance probes as a switch to signal an alarm or to actuate a circuit when the level in a tank or vessel reached a specified limit.
- Continuous Capacitance Probes senses the level in the tank as it rises up the probe, the amount current that can flow increases providing a continuous level measurement (the increased current is directly proportional to the level).
- Problems with Capacitance Level Measurement include the buildup of process material on the probe can sometime present a problem as the presence of a coating can act as a dielectric even when the level is below the probe.

KEO 3. 19 **EXPLAIN** how a **THERMAL DISPERSION SWITCH** provides a point level measurement.

A **THERMAL DISPERSION SWITCH/SENSOR** provides a point level measurement using two probes that extend from the detector into the vessel, with one of the probe tips being heated.

The difference detector monitors the difference between the heated probe tip and the unheated probe as depicted below:

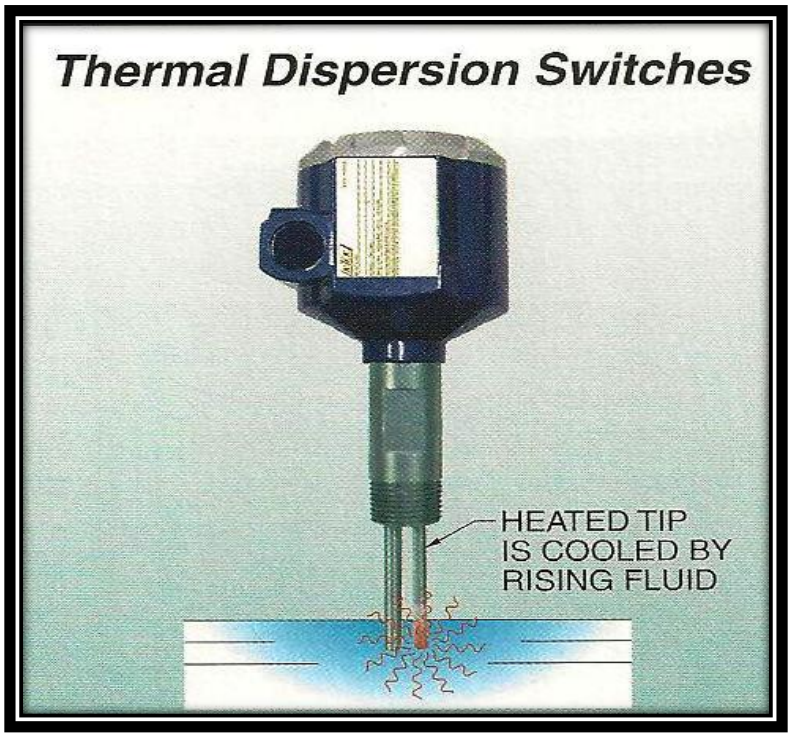


Figure 4-26 page 145

A Thermal Dispersion Sensor measures the temperature difference between two sensor tips as the heat it carried away by a fluid.

When the liquid covers the probe tips, the temperature of the heated probe drops because the heat is removed by the liquid. The decreased differential temperature is detected and activates a switch to indicate a point level detection circuit.

An alternative circuit uses a constant current flow through a thermistor in one of the probes. As the probe is immersed in the liquid, heat is conducted away from the headed probe. The resistance of the thermistor changes with a change in temperature and this is used in a bridge circuit to close a contact to energize a level indication circuit.

KEO 3. 20 **DESCRIBE** how an **INDUCTIVE PROBE** provides a point level measurement of a conductive solution.

An Inductive Probe provides a point level measurement of a conductive solution using a sealed probe containing an electrical coil attached to a bridge circuit. An electrical source generates an alternating magnetic field, and circuitry to detect changes in inductance.

The sealed probe is inserted into a vessel containing a conductive solution. As the level increases and the solution makes contact with the probe, the magnetic field of the probe interacts with the conductive material and is detected by measuring the inductive reactance X_L as depicted below:

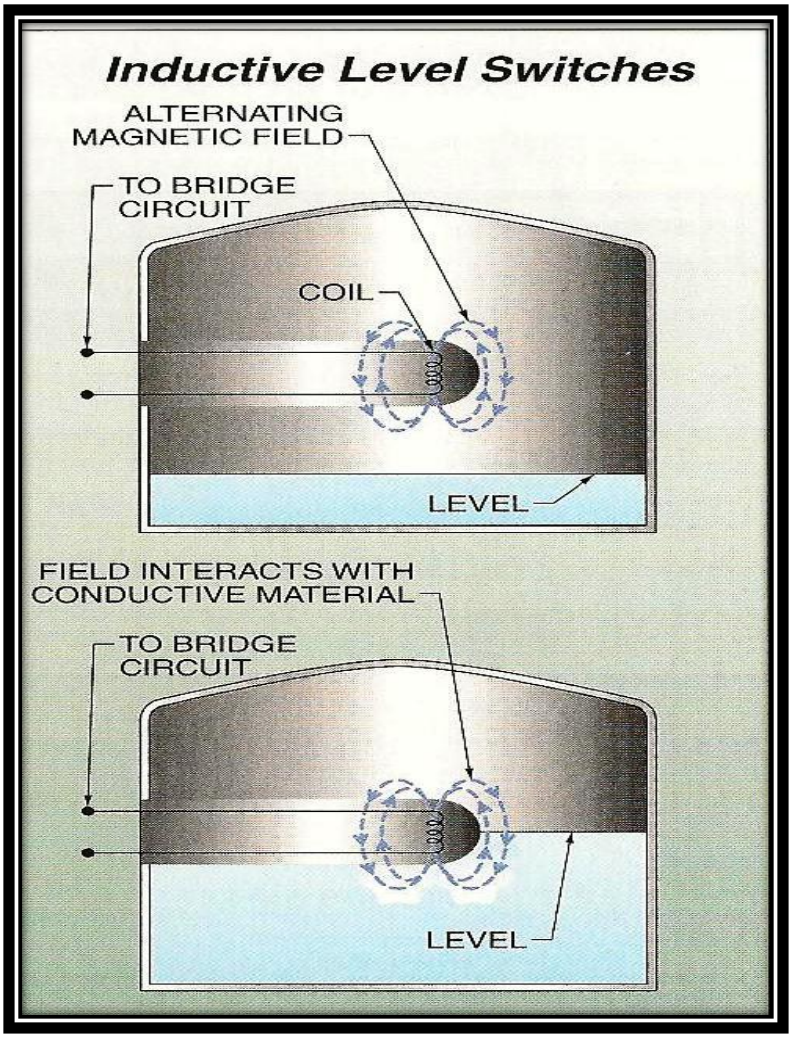


Figure 4-27 page 146

An Inductive Level Probe/Switch uses a bridge circuit to reassure changing inductance to determine when the level of material in a tank reaches the level of the switch.

KEO 3. 21 **DESCRIBE** how **ULTRASONIC SENSORS** provide a continuous level measurement.

An **ULTRASONIC SENSOR** is a continuous level measurement device consisting of two electrically energized crystals mounted above the maximum level of the material in the vessel, with one crystal used as a transmitter and the other used as a receiver.

The transmitter crystal generates a high-frequency sound directed at the surface of the material in the vessel or tank. ***Transit Time*** is the time it takes for a transmitted ultrasonic signal to travel from the ultrasonic level transmitter to the surface of the material to be measured back to the receiver. The electronic circuitry in the receiver measures the ***Transit Time*** and calculates the distance as depicted below:

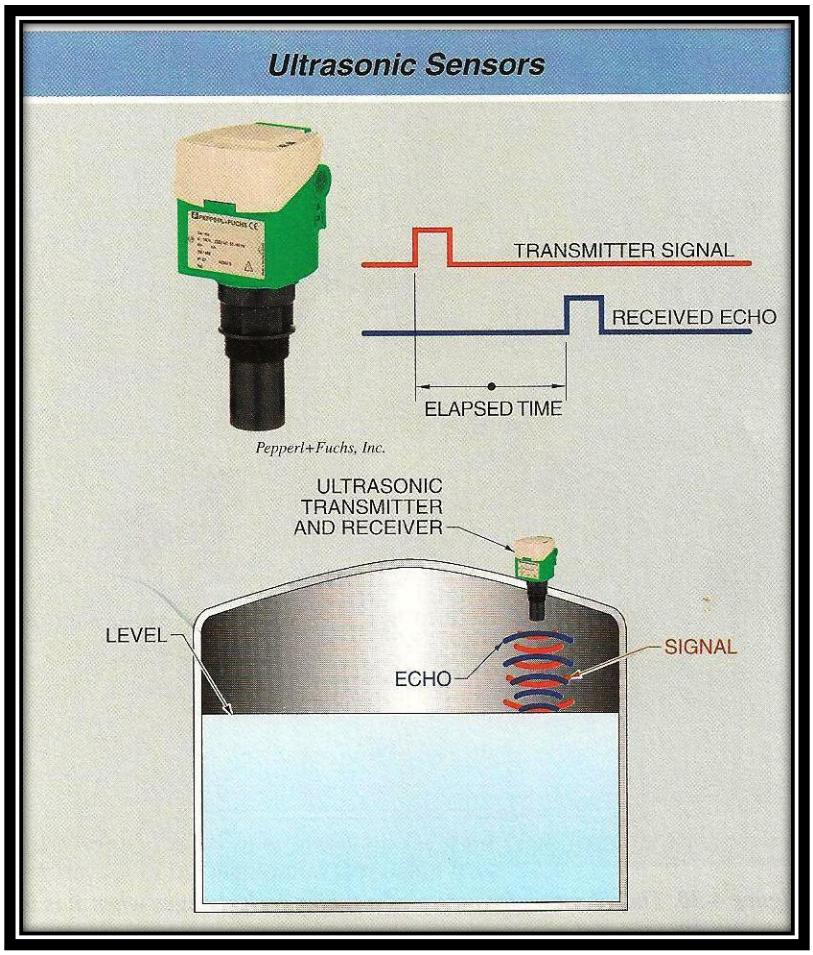


Figure 4-28 page 147

This type of sensor is primarily used for granular solids, but is also used with non corrosive liquids and slurries. Industrial noise and dust can create false signals with Ultrasonic devices. Ultrasonic Sensors can also be used to provide a point level measurement.

The design for point level measurement uses two similar crystals, one being the transmitter and the other being the receiver. Both crystals are enclosed in a probe, but are separated by a small intergral air gap. This ultrasonic sensor is called a Gap Switch as depicted below:

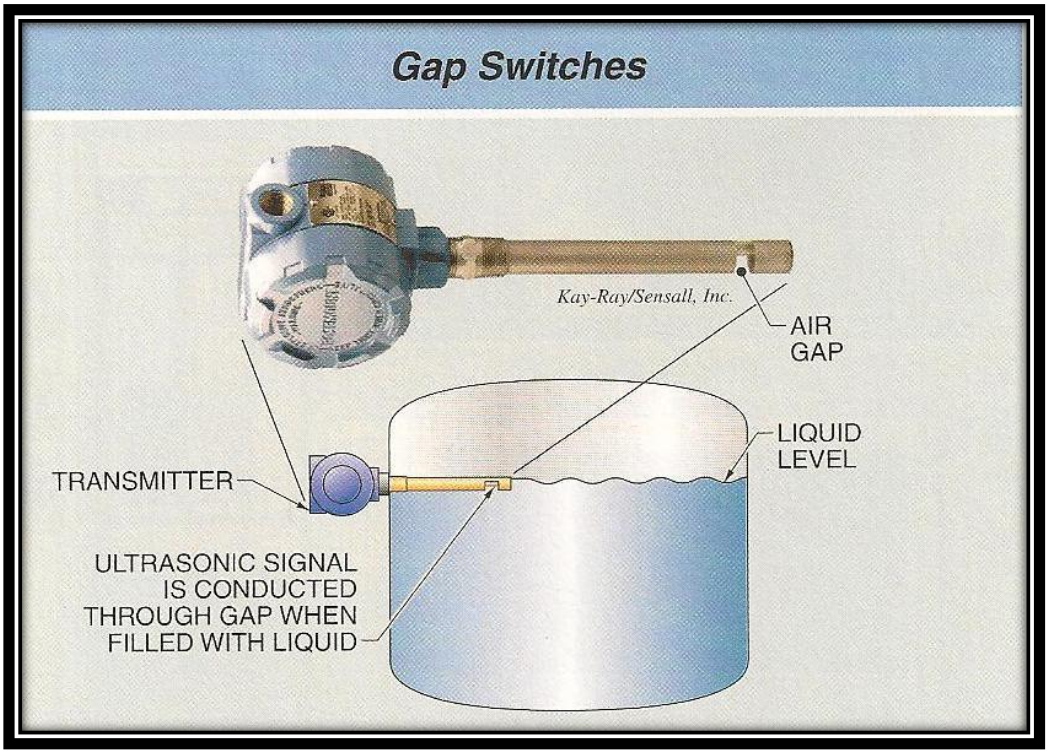


Figure 4-29 page 147

A Gap Switch measures the strength of an ultrasonic signal across a small gap to determine when material in the tank has reached the switch.

When the gap is exposed to air or vapor, the ultrasonic signal is not able to pass through in sufficient strength to be received; however, when the liquid rises and fills the gap, the ultrasonic signal from the transmitter is received. This is due to the fact that liquids carry sound waves more efficiently than air or vapor.

If liquid is in a slurry state or is sticky, a wider gap permits it to drain more readily from the gap.

The disadvantage to Gap Switches is that the material used for these devices are not suitable for corrosive liquids.

KEO 3. 22 **DESCRIBE** how **TUNING FORKS** provide a point level measurement of a liquid.

A **TUNING FORK** provides a point level measurement of a liquid using a vibrating fork that resonates at a particular frequency and the circuitry to measure that frequency as depicted below:

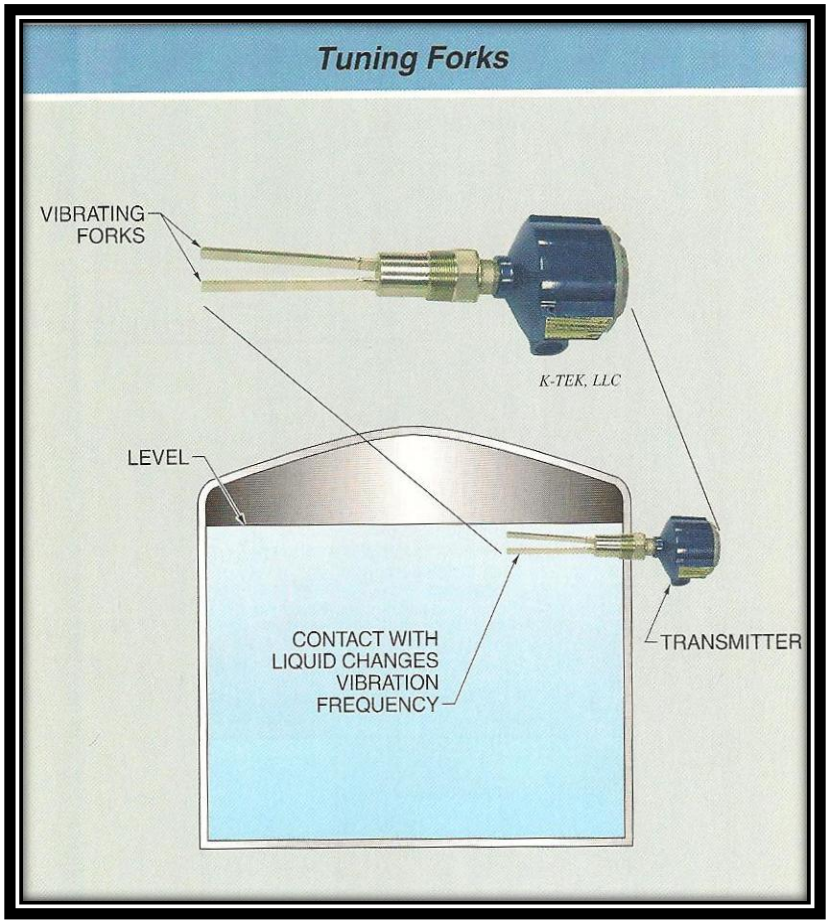


Figure 4-30 page 148

Tuning Forks are commonly used for single-point level detection of liquids or granular solids. When the Tuning Fork is uncovered, it vibrates at a fixed frequency. The vibration frequency of the Tuning Fork changes when it is in contact with the material in the tank to provide a point level measurement. This change in frequency provides an electronic signal to actuate an output relay for feeder control or an alarm circuit.

The limitation Tuning Forks have is that they cannot be used for corrosive fluids due to the material they are constructed of.

SUMMARY

- A Thermal Dispersion Switch/Sensor provides a point level measurement using two

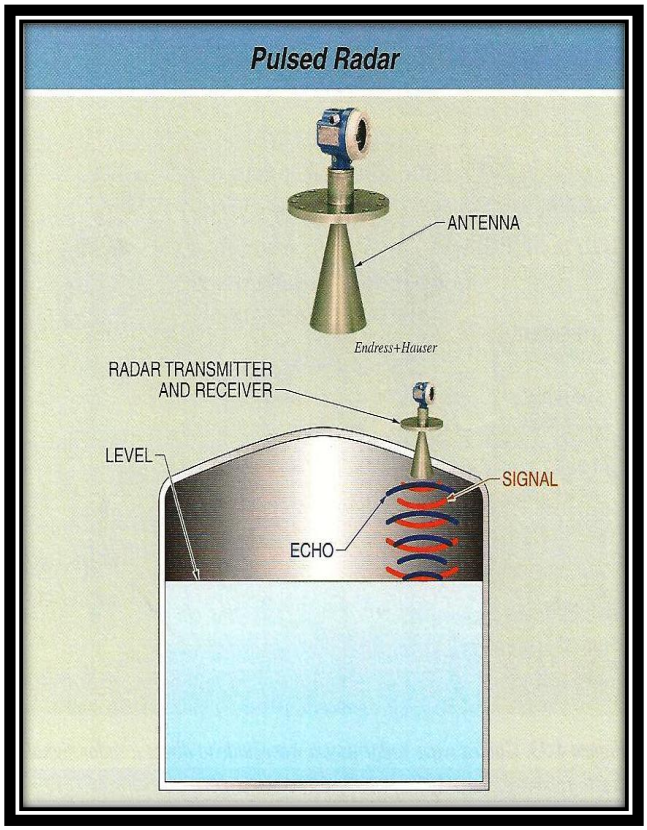
probes that extend from the detector into the vessel, with one of the probe tips being heated.

- The difference detector monitors the difference between the heated probe tip and the unheated probe and this generates a signal that activates a level indication circuit.
- An Inductive Level Probe/Switch uses a bridge circuit to reassure changing inductance to determine when the level of material in a tank reaches the level of the switch.
- An Ultrasonic Sensor is a continuous level measurement device consisting of two electrically energized crystals mounted above the maximum level of the material in the vessel, with one crystal used as a transmitter and the other used as a receiver.
- The transmitter crystal generates a high-frequency sound directed at the surface of the material in the vessel or tank.
    - ***Transit Time*** is the time it takes for a transmitted ultrasonic signal to travel from the ultrasonic level transmitter to the surface of the material to be measured back to the receiver.
    - The electronic circuitry in the receiver measures the ***Transit Time*** and calculates the distance
- A Gap Switch is used as a point level measurement and measures the strength of an ultrasonic signal across a small gap to determine when material in the tank has reached the switch.
    - The disadvantage to Gap Switches is that the material used for these devices are not suitable for corrosive liquids.
- Tuning Forks are commonly used for single-point level detection of liquids or granular solids.
- When the Tuning Fork is uncovered, it vibrates at a fixed frequency. The vibration frequency of the Tuning Fork changes when it is in contact with the material in the tank to provide a point level measurement. This change in frequency provides an electronic signal to actuate an output relay for feeder control or an alarm circuit.
- The limitation Tuning Forks have is that they cannot be used for corrosive fluids due to the material they are constructed of.

KEO 3. 23 **DESCRIBE** how **RADAR** systems utilize **PULSED, FREQUENCY, MODULATED CONTINOUS WAVE,** and **GUIDED WAVE RADAR** to measure level.

RADAR SYSTEMS use approximately 10 GHz radio waves signals aimed at the surface of material in the storage vessel or tank being measured. The radio waves are reflected off the material in the vessel or tank and returned to the emitting source. Common types of Radar Wave Systems include: Pulsed, Frequency Modulated Continuous, and Guided wave.

PULSED RADAR is a level measuring sensor consisting of a radar generator that directs an intermittent pulse with a constant frequency toward the surface of the material in a vessel or tank. There are two common antennae types used to emit a radar pulse: A Cone Antennae, which is larger and sturdier and less subject to material buildup or condensation, and a Rod Antennae which is smaller and less expensive, making them more suitable for use in smaller vessels. A Cone Antennae Pulsed Rader system is depicted below:



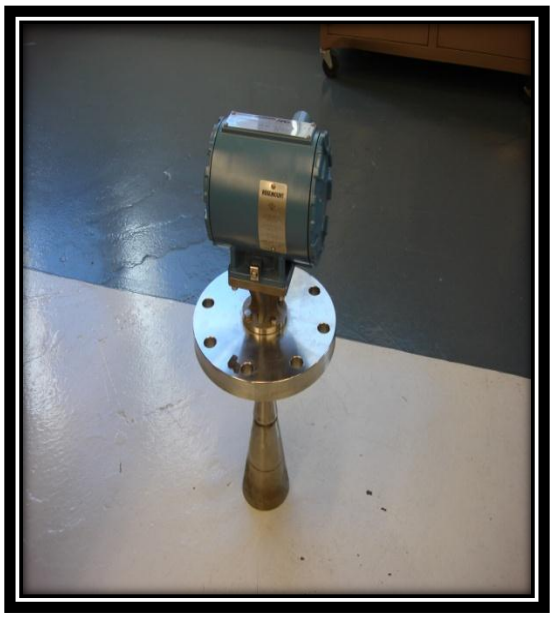


Figure 4-31 page 149

Pulsed Radar measures the transit time from the transmitter to the surface of the material to determine level.

The following picture depicts a tank level system at Idaho State University's College of Technical Education using the Pulsed Radar system:



Frequency Modulated Continuous Wave (FMCW) system is a level measuring sensor consisting of an oscillator that emits a continuous microwave signal that repeatedly varies its frequency between a minimum and maximum value, a receiver that detects the signal, and electronics that measure the frequency difference between the signal and the echo.

Frequency Modulated Continuous Wave is depicted below:

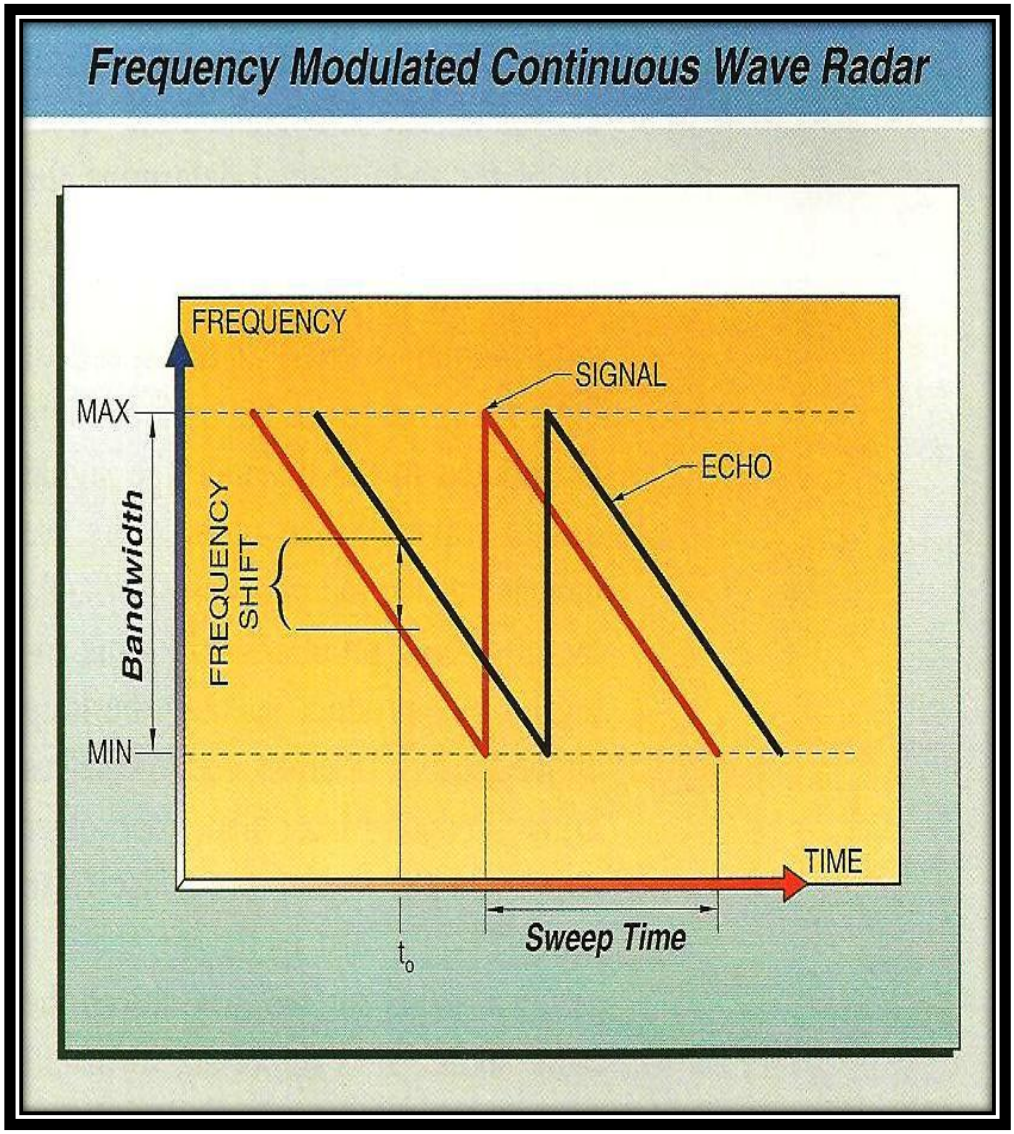


Figure 4-32 page 149

Frequency Modulated Radar measures the frequency difference of the radar signal and the echo to determine level.

GUIDED WAVE RADAR is a level measuring detector consisting of a cable or rod as the wave carrier extending from the emitter down to the bottom of the vessel or tank and electronics to measure the transit time. **Another name** for the Guided Wave Radar is a **Time Domain Reflectometer (TDR).**

With the **Guided Wave Radar**, the material in the vessel or tank reflects, or echoes, some of the microwave energy at the point where the carrier and material make contact as depicted below:

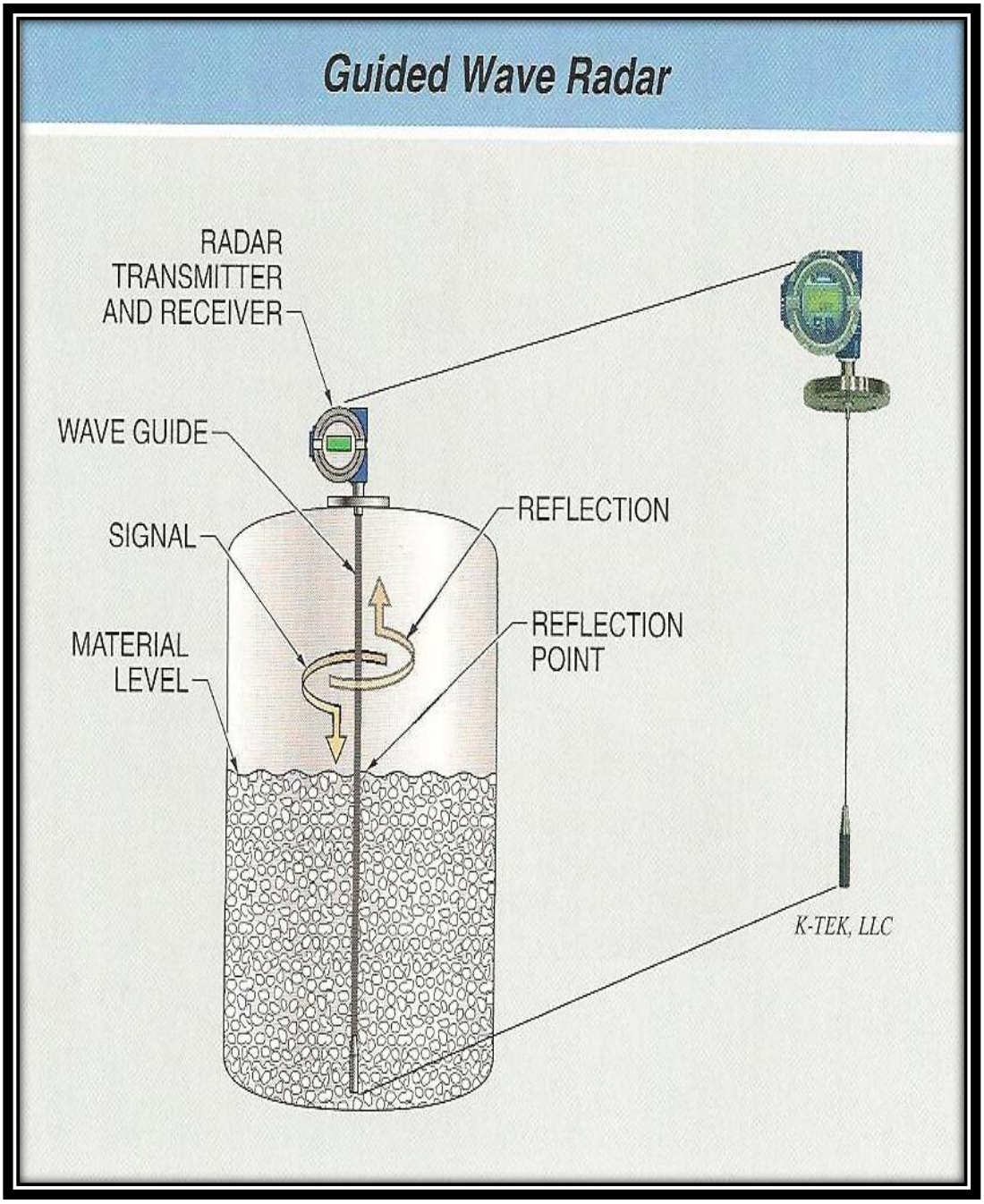


Figure 4-33 page 150

The **Guiding Rod Radar** reduces the effect dust above granular solids as well as the turbulence in some liquids.

KEO 3. 24 **DESCRIBE** how **LASERS** measure level.

A **LASER** is used to measure level with a laser beam generator, a timer, and a detector mounted at the top of a vessel. Laser beams are intense, narrow light beams that can travel long distances. The laser beam is reflected back to the emitter where a very accurate timing device measures the out-and-back interval. The travel time varies with the level of material being measured as depicted below:

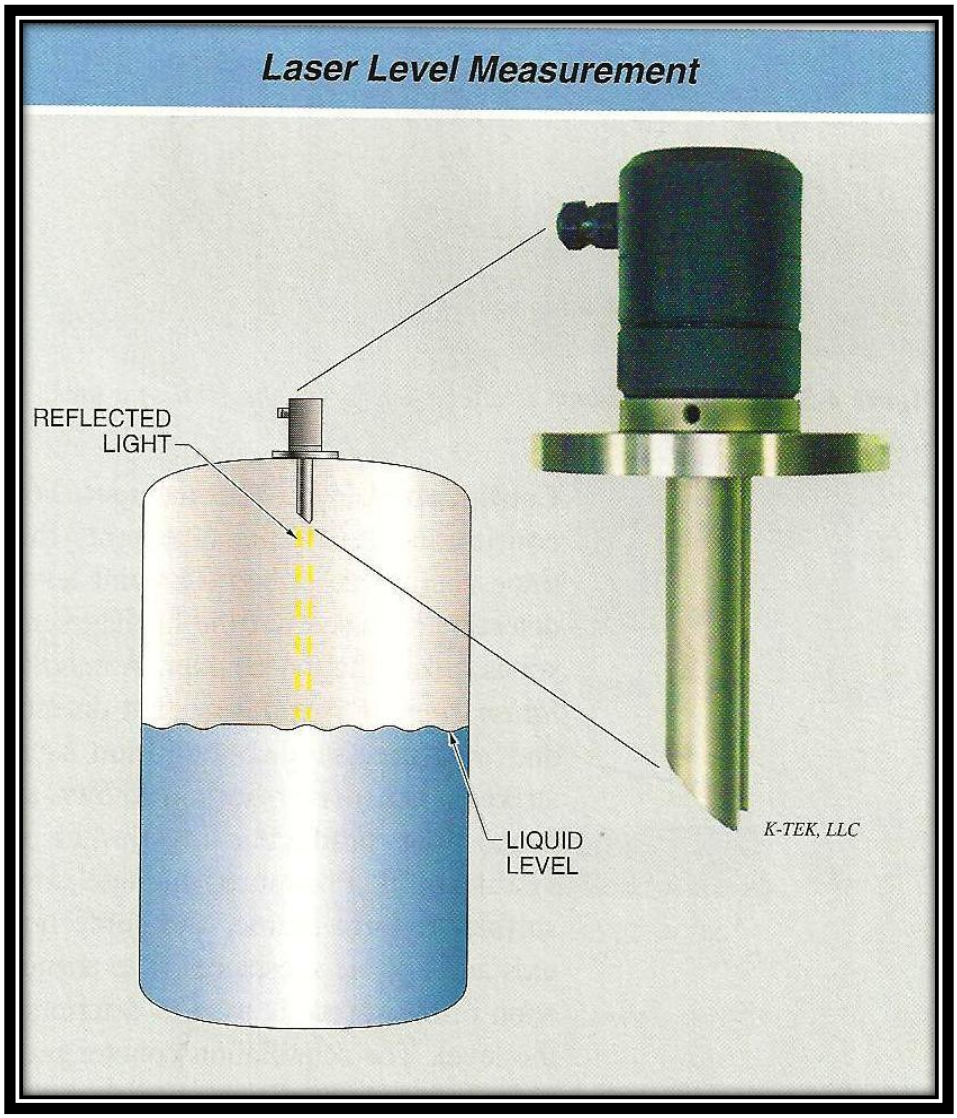


Figure 4-34 page 151

A Laser measures the transit time of reflected light to determine level. Because laser beams are light beams, dust and vapor can interfere with their transmission and reception.

SAFETY WARNING:

Laser Beams intense, narrowly focused light can have a destructive effect on eyes. Caution should be used at all time to never look at any laser beam. Laser Safety Training is required for personnel working with or near laser devices.

- **Lasers are generally divided into four basic classifications, based on potential risk for the operator and other workers in the area (the higher the classification, the greater the risk). Classification are Class I, Class II, Class III (A, B, and C), and Class IV.**
- **Personnel working on or near laser equipment need to be sure to determine the classification of the laser for determination of the proper safety precautions.**

SUMMARY

- Radar Systems use approximately 10 GHz radio waves signals aimed at the surface of material in the storage vessel or tank being measured. The radio waves are reflected off the material in the vessel or tank and returned to the emitting source.
- Pulsed Radar measures the transit time from the transmitter to the surface of the material to determine level.
- A Cone Shaped Pulsed Radar Antennae is not subject to material build up or condensation.
- A Rod Shaped Pulsed Radar Antennae is less expensive more suitable for measuring level in smaller vessels or tanks.
- Guided Wave Radar is a level measuring detector consisting of a cable or rod as the wave carrier extending from the emitter down to the bottom of the vessel or tank and electronics to measure the transit time.
- Another name for the Guided Wave Radar is a Time Domain Reflectometer (TDR).
- The Guided Wave Radar reflects, or echoes, some of the microwave energy at the point where the carrier and material make contact.
- The Guiding Rod Radar reduces the effect dust above granular solids as well as the turbulence in some liquids.
- A Laser is used to measure level with a laser beam generator, a timer, and a detector mounted at the top of a vessel.
- The laser beam is reflected back to the emitter where a very accurate timing device measures the out-and-back interval.
- A Laser measures the transit time of reflected light to determine level. Because laser beams are light beams, dust and vapor can interfere with their transmission and reception.
- **Laser Beams intense, narrowly focused light can have a destructive effect on eyes. Caution should be used at all time to never look at any laser beam! Laser Safety Training is required for personnel working with laser devices.**

KEO 3. 25 **DESCRIBE** how **NUCLEAR LEVEL INSTRUMENTS** provide point and continuous level measurement.

A **NUCLEAR LEVEL INSTRUMENT** is a level measuring system consisting of a radioactive source that directs radiation through a vessel to a detector, such as a GEIGER COUNTER on the other side of a vessel.

Nuclear level sensors are used for process materials that are extremely hot, corrosive, toxic, or under very high pressure and so are not suitable for intrusive level detectors.

Radioactive elements such as cesium 137 or cobalt 60 provide the radioactive source in the form of gamma rays. The amount of radioactive energy required is calculated based upon a vessel or tanks wall thickness and distance between source and detector.

Nuclear level sensors are relatively expensive to purchase, install, and operate. However, they are sometimes the only way to measure level under extreme conditions.

SAFETY NOTE:

Federal, State, and Local Authorities closely regulate the use of nuclear energy sources.

Radiation Safety Worker Training is required for any person working on or near radiation sources.

Work on or near Radiation Sources Must be supervised by qualified Radiation Control Technicians.

POINT LEVEL measurement is achieved with a radioactive source mounted externally on one side of a vessel at the selected level. The source must be enclosed in a protective housing with a window allowing the radiation to be directed toward the detector on the opposite side of the vessel. The nuclear energy source produces a beam of radiation whose frequency is proportional to the strength of the radiation.

When the material level blocks the Radiation Beam Path, the detected radioactive energy is reduced enough to cause an electrical relay to change its state and provide a level indication or alarm. This relay can start or stop a feeder, light a lamp, or sound an alarm as depicted below for Point Level Measurement:

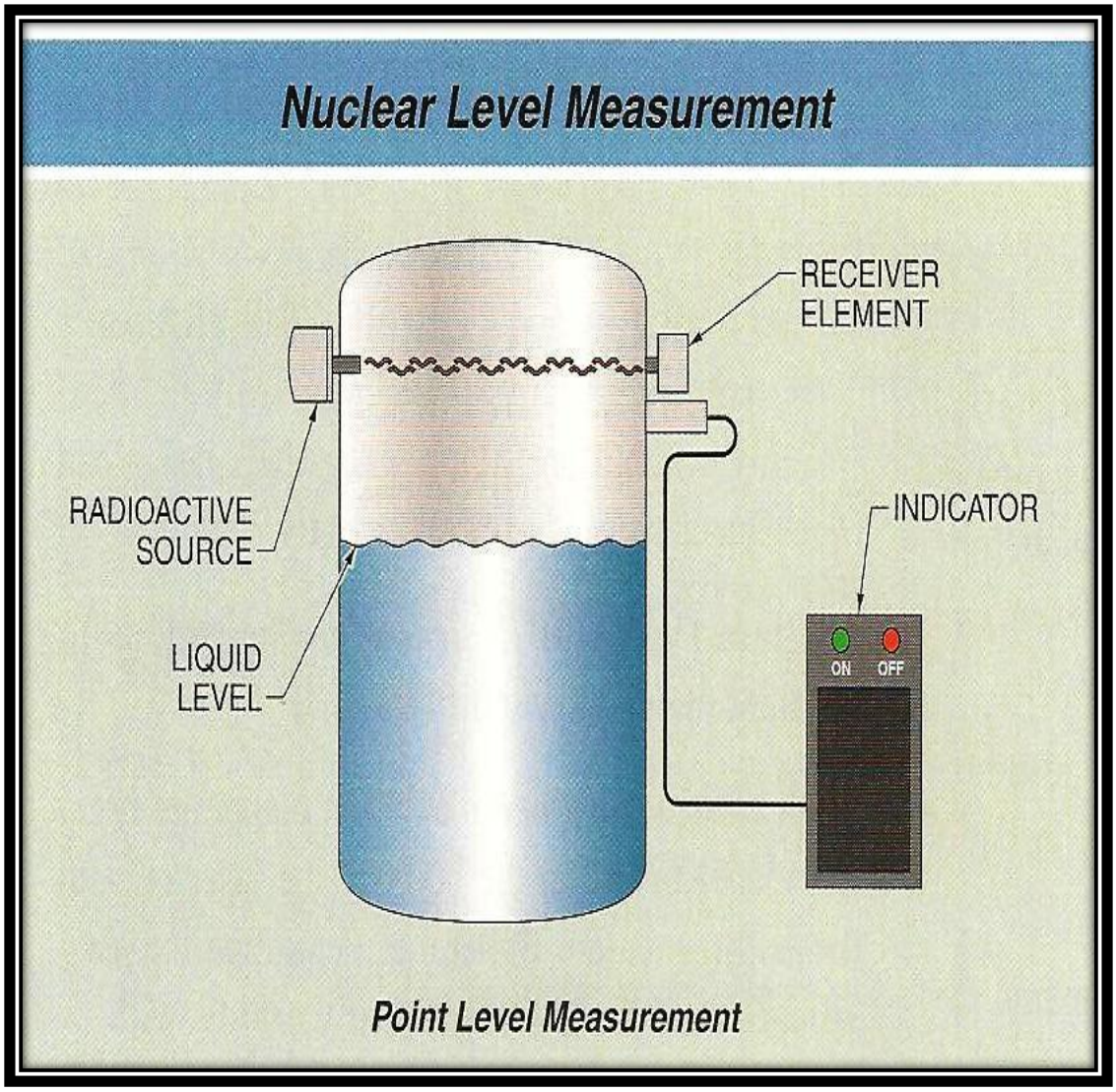


Figure 4-35 page 152

Notice in the picture above, the indicator receiving its signal from the receiver is only an indication of a level being an on or off state as the device depicts (alarm on or off or control device on or off).

Nuclear Continuous Level Measurement differs from Point Level Measurement in that it will have several receiving elements instead of just one as depicted below:

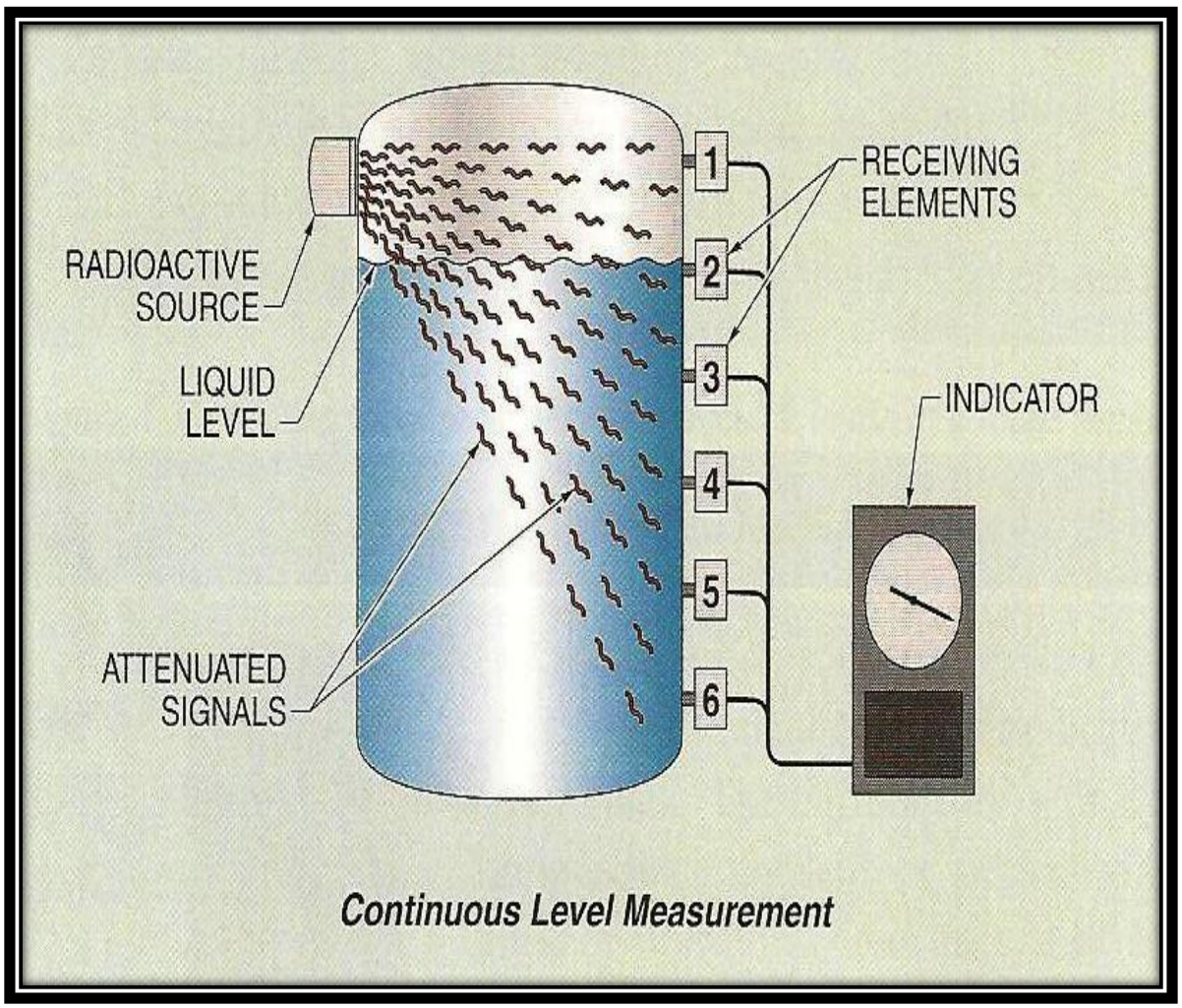


Figure 4-35 page 152

Notice in the picture above, the indicator receiving its signal from the receiver is an indication of a continuous level as all times and the device receiving its signal from the receiver is providing a level reading of 0 to 100 % level and it is not an on or off device.

For continuous level measurement, the difference from a point level device is that the receiver is a scintillation counter that detects and measures nuclear radiation as it strikes a sensitive material, know as a phosphor, producing tiny flashes of visible light. Phosphors include Zinc, Sulfide, Sodium Iodide and some liquids and organic substances. The attenuation of the source is used to determine the level.

SUMMARY

- A Nuclear Level Instrument is a level measuring system consisting of a radioactive source that directs radiation through a vessel to a detector.
- Point Level measurement is achieved with a radioactive source mounted externally on one side of a vessel at the selected level.
- Nuclear Continuous Level Measurement differs from Point Level Measurement in that it will have several receiving elements instead of just one receiving element.
- All radioactive sources must be enclosed in a protective housing with a window allowing the radiation to be directed toward the detector(s) on the opposite side of the vessel.
- **Federal, State, and Local Authorities closely regulate the use of nuclear energy sources.**
- **Radiation Safety Worker Training is required for any person working on or near radiation sources.**
- **Work on or near Radiation Sources Must be supervised by qualified Radiation Control Technicians.**

KEO 3. 26 **DESCRIBE** how **ELECTRONIC LOAD CELLS** measure level of liquids or solids.

Weighing a vessel or tank containing either liquids or solids is a very accurate method of determining level. This type of measurement requires the use of **ELECTRONINC LOAD CELLS**. Load Cells are either piston-cylinder devices that produce a hydraulic output pressure or strain gauge assemblies that provide an electrical output proportional to the applied load.

The use of **LOAD CELLS** for level measurement requires an accurate value for the density of the material being measured. If the actual density of the material is less than expected, the tank can overflow while the level reading still shows sufficient room to continue filling. Density is a very important aspect of measuring level and it must be known in order to accurately measure level of a material. Density is the weight of a material such as water. Water has a density (specific gravity) of 1.0 and is used as a basis for comparison. Material having a water base will be 1.0 or greater and materials having an organic base (like oil), will have a density of less than 1.0.

Load cells are available in Compression and Tension configurations as depicted below:

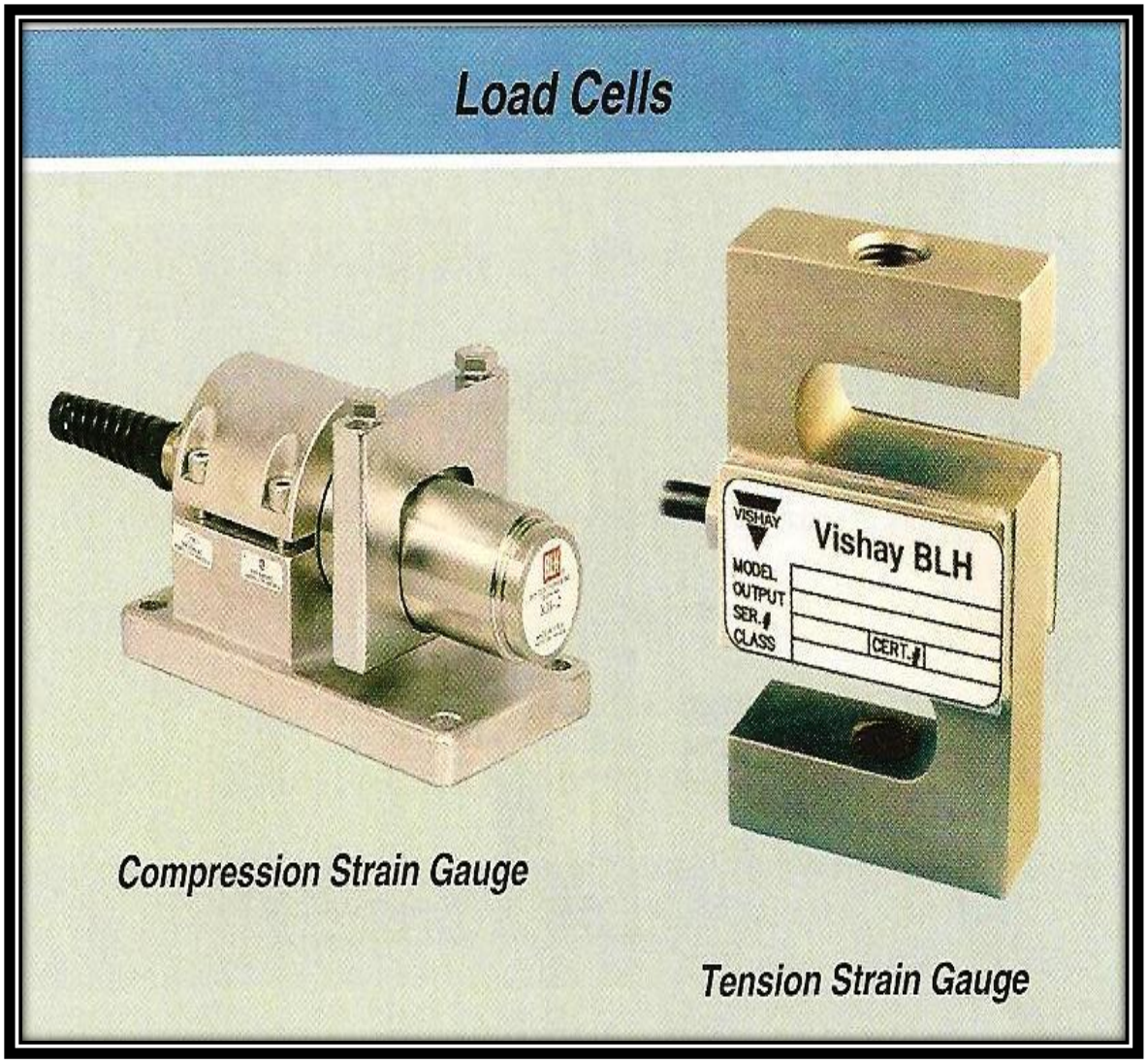


Figure 4-36 page 153

Strain gauges load cells are generally in the form of a beam, column, or other stress member with strain gauges bonded to them.

When a weight or load is impressed against a member, the strain gauge is deformed and its electrical resistance changes in a bridge circuit, which provides an output that is proportional to the force acting upon the load cell.

There are three forms of **Electronic Load Cells**: 1) Shear, 2) Compression, and 3) Tension.

1) ***Shear-Type*** are used to measure the weight in vertical vessels as depicted below:

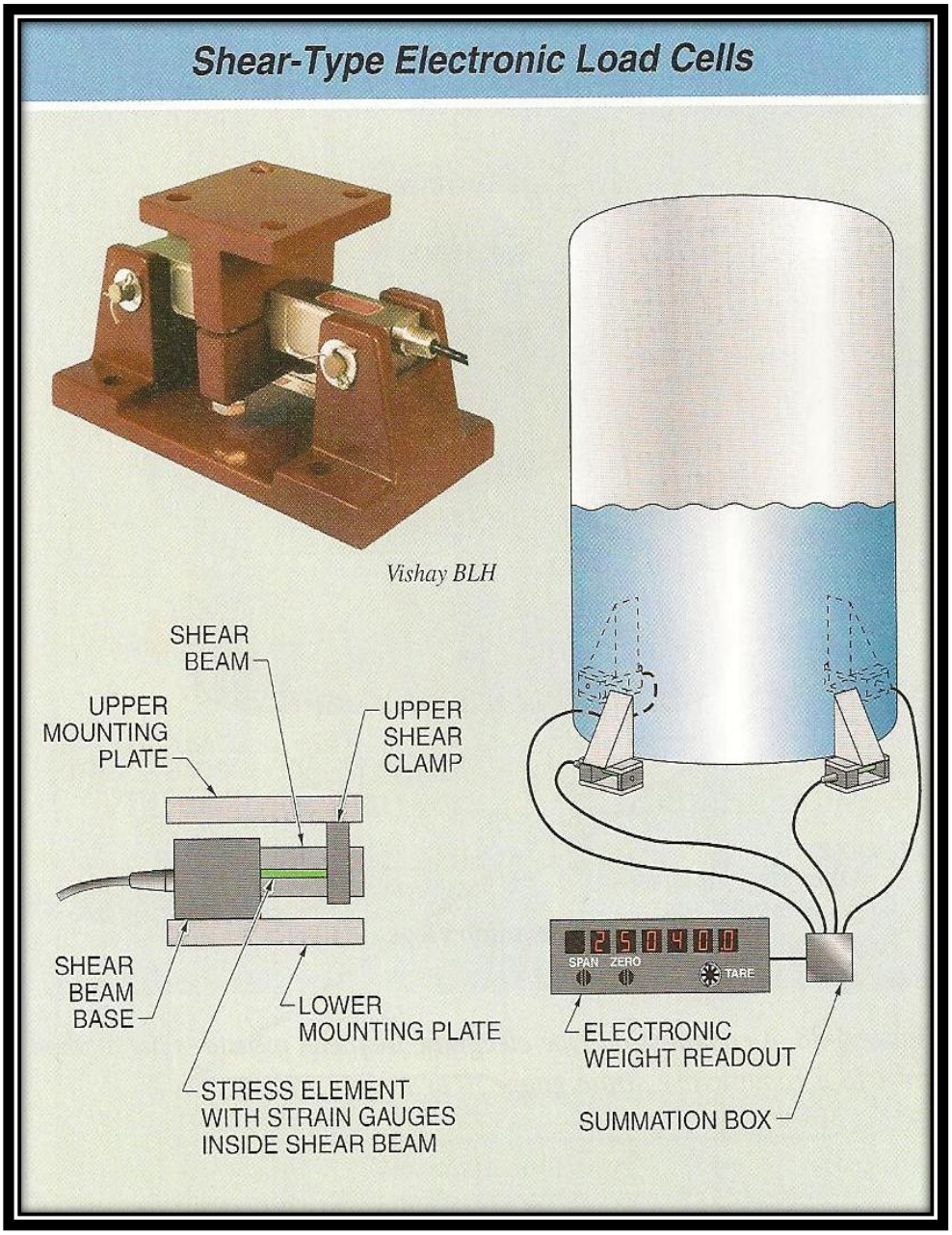


Figure 4-37 page 153

Shear-Type Load Cells are placed under the feet of a tank to measure the weight of the material in the tank.

2) ***Compression-Type*** load cells are used for long horizontal vessels where one end of the vessel needs to be free floating to allow for dimensional changes with temperature changes as depicted below:

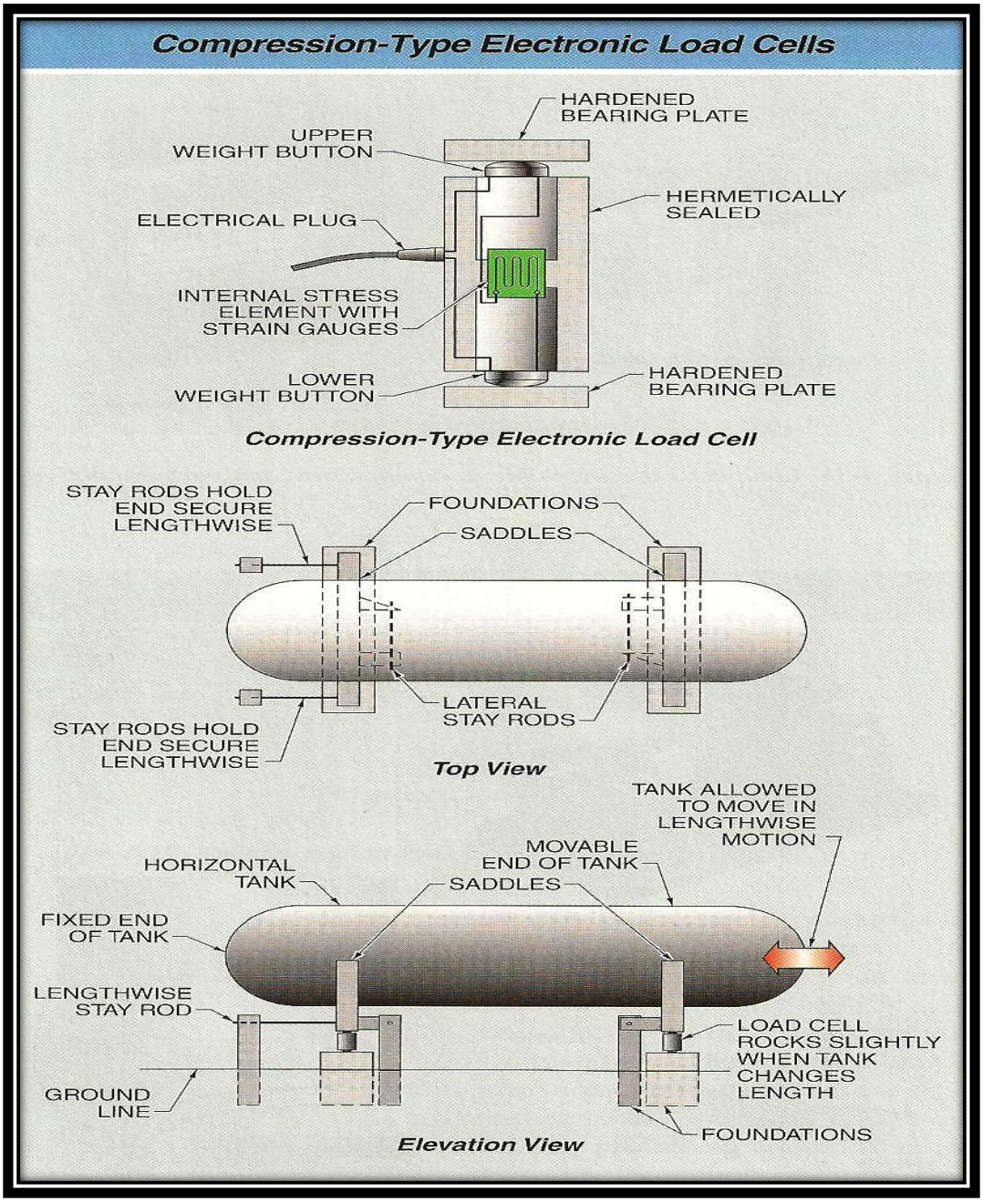


Figure 4-38 page 154

Compression-Type Load Cells allow the tank to move slightly and still provide a means to measure the weight of the material in the tank.

3) ***Tension-Type load cells*** are used where the tank hangs from a ceiling or beam and the tank is actually attached to the load cells attached to the ceiling or beam.

A significant difficulty with the use of load cells is the piping restraints to the vessel, which tend to support some of the weight of the vessel. All load cell systems require some vertical movement with increased load. Piping restraints are reduced by having long, unrestrained horizontal runs or by using flexible joints.

Individual load cells are sized for different maximum loads. If four load cells are used, the maximum total weight of the vessel is divided by four. Load Cells are selected with a safety margin that is 50% to 100% greater than maximum calculated load expected to be applied by each cell.

Load Cell Calibrations

Electronic Load Cells can be calibrated with a **Load Cell Simulator** as depicted below:

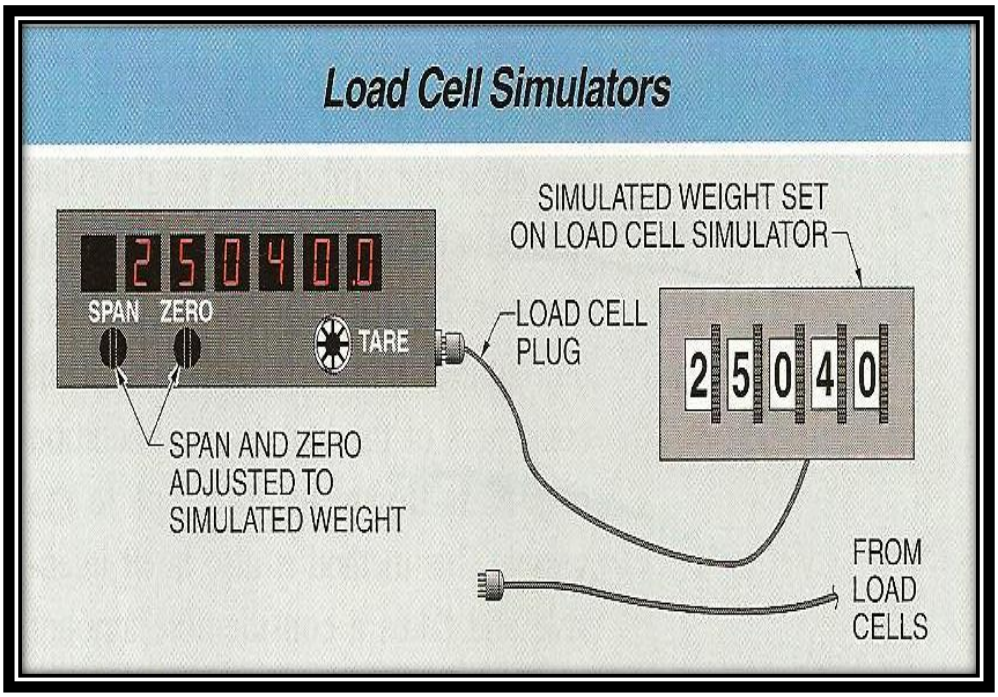


Figure 4-39 page 155

The simplest method of calibration is with a **Load Cell Simulator**. However, this method does not work well when there is piping restraints that could shift the calibration from the ideal condition.

A Load Cell can be calibrated using a **Physical Weight Calibration Method** by adding known weights to a tank as depicted below:

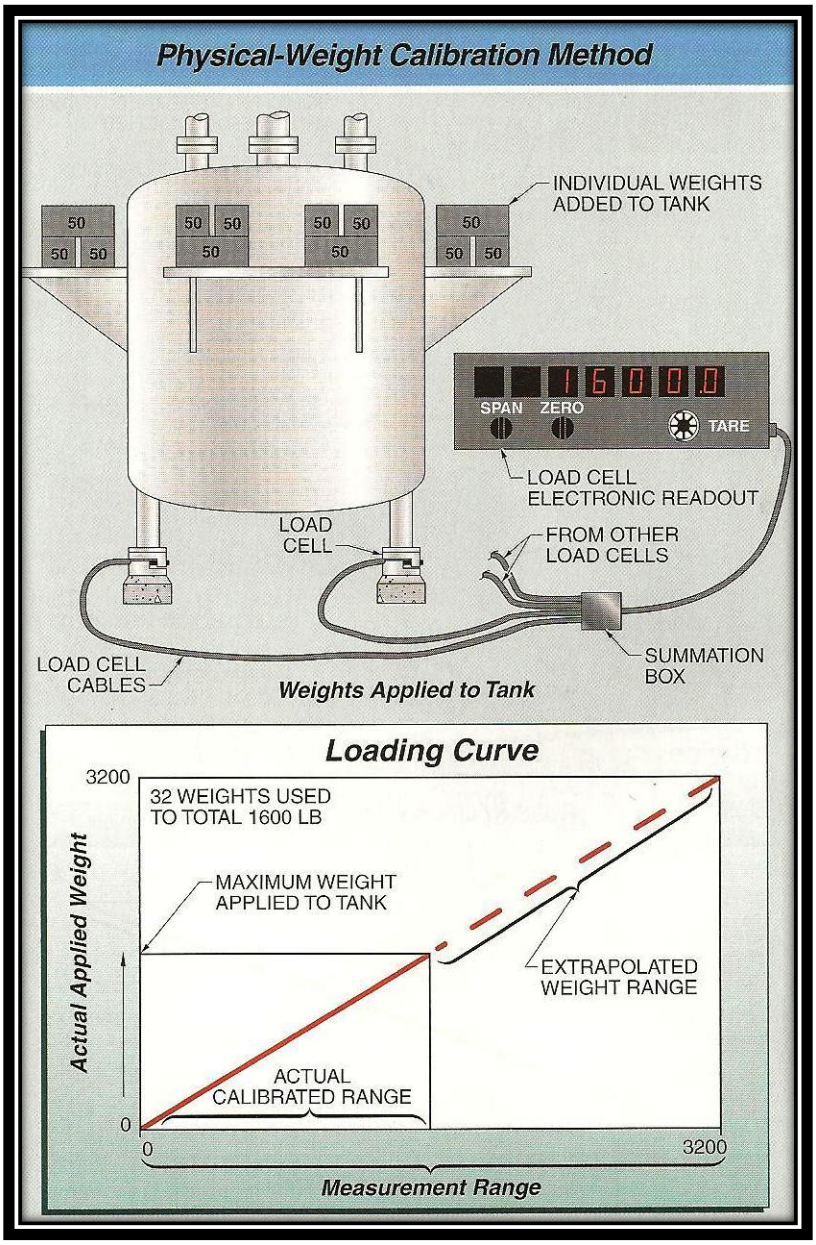


Figure 4-40 page 155

Adding know weights is an accurate calibration method; however, it is difficult to add the full weight to a vessel and requires hard work and a lot of time to accomplish and piping restraints can add significant errors that cannot be removed by recalibration.

A tank weighting system can be calibrated by a **Weighted-Liquid Calibration Method** as depicted below:

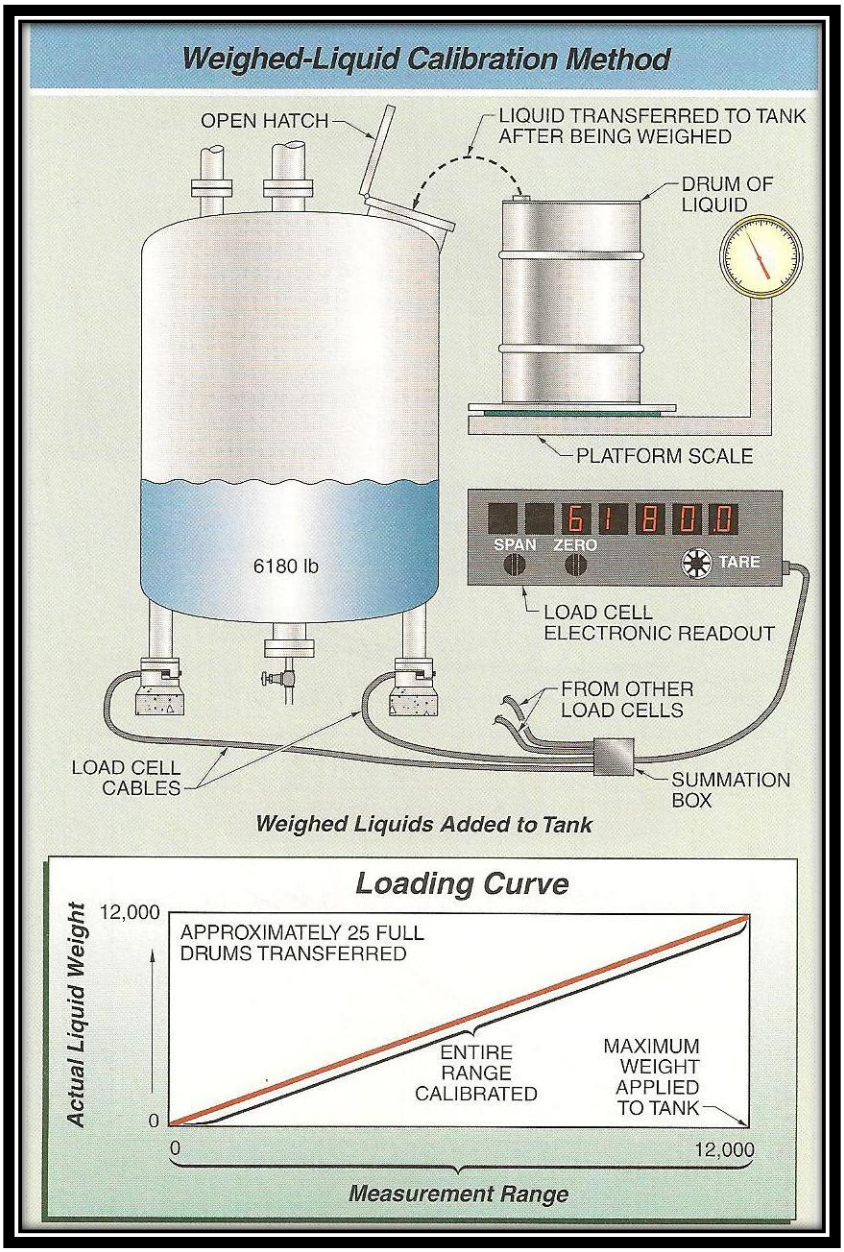


Figure 4-41 page 156

This method uses known amounts of liquid added to the vessel as the calibration weight. The **Weighed-Liquid Calibration Method** allows adding sufficient weight to cover the whole measurement range, but requires accurately weighing small containers of liquid or using a mass flow meter to add the solution to the vessel. This method is also labor intensive and time consuming.

The final method uses a **Portable Load Cell Calibration Method**. This system uses a readout instrument, and hydraulic jacks. The jacks are positioned in line with the calibration load cells to either simulate loads on the vessel or shift weight from the application weigh vessel to the **Portable Load Cell Calibration System**. This method is depicted below:

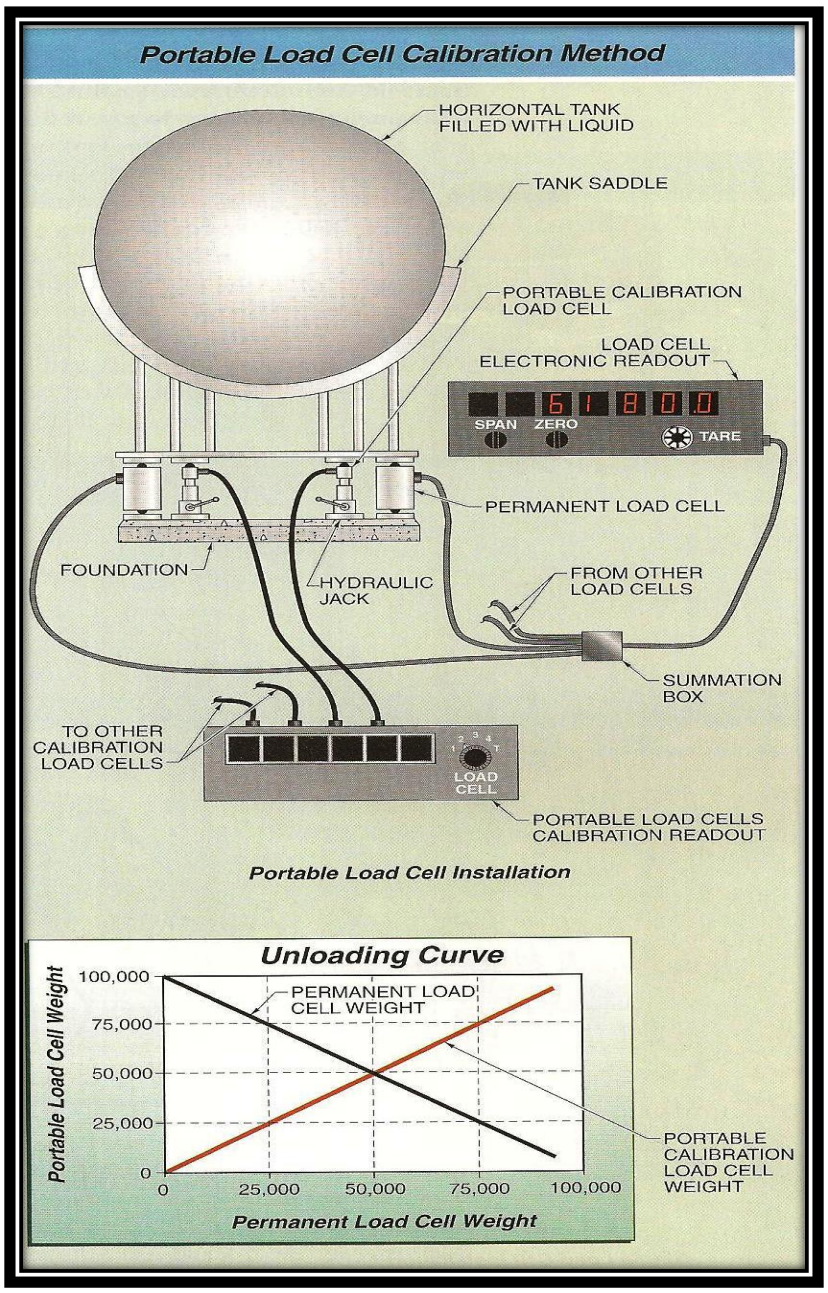


Figure 4-42 page 157

The **Portable Load Cell Calibration Method** is quick and accurate, but is expensive. In addition, the supporting structure must be designed for use with this calibration system.

KEO 3. 27 **DESCRIBE** how **HYDRAULIC LOAD CELLS** are used to measure level.

HYDRAULIC LOAD CELLS are part of a closed hydraulic pressure system in which the load cell transfers the pressure acting on the cell from the weight of the vessel and its contents to a piston. The piston compresses the system hydraulic fluid in a diaphragm chamber. This change in pressure varies with the load acting on the load cell. The following depicts a load cell and a cross section of a load cell:

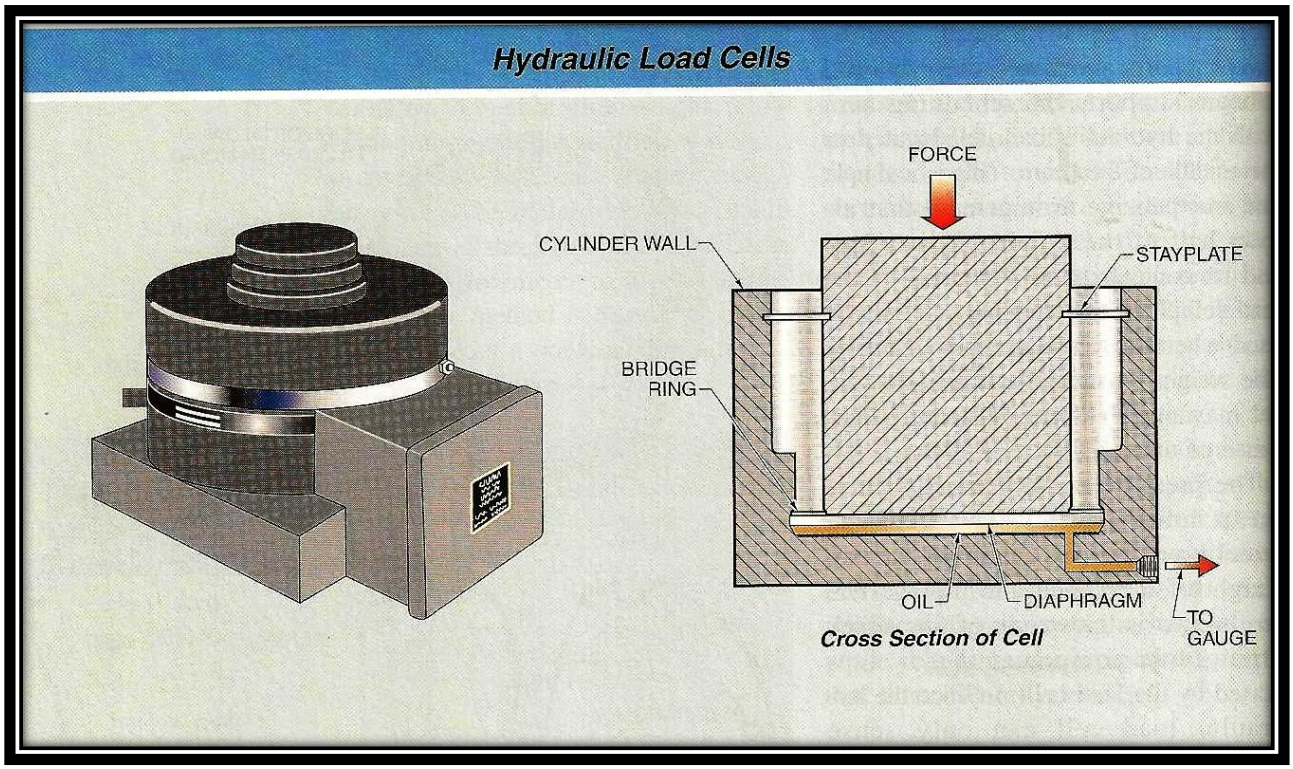


Figure 4-43 page 158

In practice, a single hydraulic load cell can only be used to measure a vessel and its content if there are additional pivoted supports to provide a stable base for the vessel. The most common arrangement is with a vessel that has four equally spaced supports.

Two of the supports are placed on pivots and a beam supports the remaining two supports, with the hydraulic load cell located on the middle of the beam. This is a simple and inexpensive arrangement that allows half of the weight of the vessel and its contents to be supported by the load cell. This configuration can only be used when the vessel contains a liquid material. The weight of a granular solid material in a vessel may not be evenly distributed because of uneven flow.

To improve overall accuracy, three or four hydraulic load cells can be used. The hydraulic pressures are then combined in a hydraulic summation unit. The problem with using more than one load cell is ensuring that the weight is equally distributed among all cells so that the load cells cannot bottom out. The following picture depicts a single Load Cell and a Four Load Cell Installation:

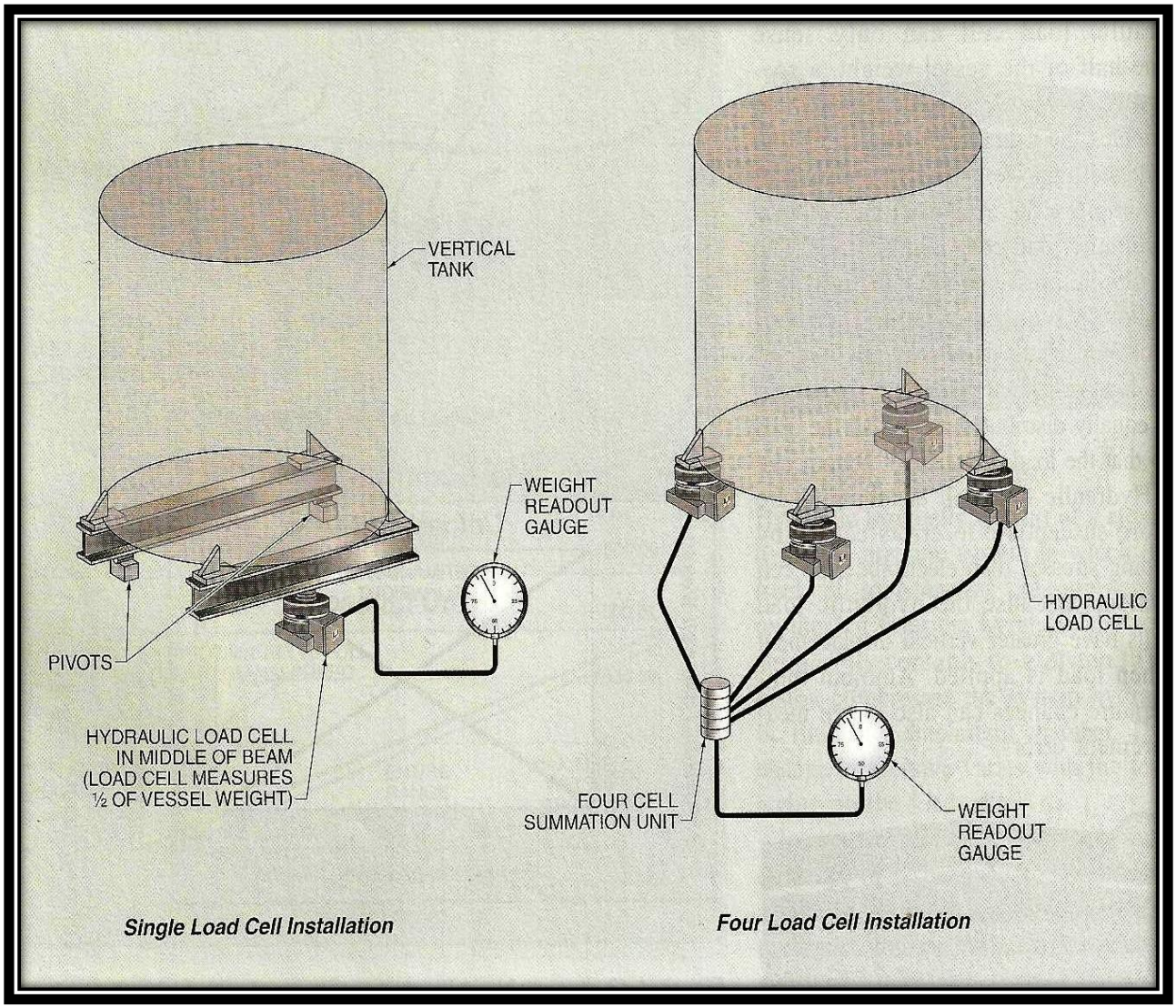


Figure 4-43 page 158

KEO 3. 28 **DESCRIBE TWO DISADVANTAGES** associated with **HYDRAULIC LOAD CELLS** that are not associated with **ELECTRONIC LOAD CELLS**.

There two disadvantages for Hydraulic Load Cells as they are more susceptible to errors caused by:

- Piping Stress because the hydraulic load cell have a greater vertical displacement than Electronic Cells when load is applied
- Ambient temperature can also create greater measurement errors over Electronic Cells.

SUMMARY

- Load Cells are either piston-cylinder devices that produce a hydraulic output pressure or strain gauge assemblies that provide an electrical output proportional to the applied load.
- The use of Load Cells for level measurement requires an accurate value for the density of the material being measured.
  Shear-Type Load Cells are used to measure the weight in vertical tanks/vessels are placed under the feet of a tank to measure the weight of the material in the tank.
- Compression-Type load cells are used for long horizontal vessels where one end of the vessel needs to be free floating to allow for dimensional changes with temperature changes and allow the tank to move slightly still providing a means to measure the weight of the material in the tank.
- Tension-Type load cells are used where the tank hangs from a ceiling or beam and the tank is actually attached to the load cells attached to the ceiling or beam.
- Load Cells can be calibrated using four methods:
  1. Load Cell Simulation Method
  2. Physical Weight Calibration Method
  3. Weighted-Liquid Calibration Method
  4. Portable Load Cell Calibration Method
- Hydraulic Load Cells are part of a closed hydraulic pressure system in which the load cell transfers the pressure acting on the cell from the weight of the vessel and its contents to a piston. The piston compresses the system hydraulic fluid in a diaphragm chamber. This change in pressure varies with the load acting on the load cell.
- In practice, a single hydraulic load cell can only be used to measure a vessel and its content if there are additional pivoted supports to provide a stable base for the vessel and can only be used to measure liquid due the fact that weight of a granular solid material in a vessel may not be evenly distributed because of uneven flow.
- To improve overall accuracy, three or four hydraulic load cells can be used. The hydraulic pressures are then combined in a hydraulic summation unit. The problem with using more than one load cell is ensuring that the weight is equally distributed among all cells so that the load cells cannot bottom out.
- Hydraulic Load Cells as they are more susceptible to errors caused by: Piping Stress because the hydraulic load cell have a greater vertical displacement than Electronic Cells when load is applied, and ambient temperature can also create greater measurement

errors over Electronic Cells.

KEO 3. 29 **DESCRIBE** difficult or complicated **SITUATIONS** associated with the measurement of level for **BULK SOLIDS IN SILOS AND TANKS** and what can be done to ensure safe and reliable operation of level sensors.

The flow and handling of **BULK SOLIDS** in tanks and silos is extremely complex. A Bulk Solid is a granular solid, such as gravel, sand, sugar, grain, wet cement, or other solid material that can be made to flow.

The top surface of a Bulk Solid in a silo may not b even across the top. The surface may be heaped up in the middle as the silo is being filled or it may be lowered in the middle as the bulk solid flows out of the bottom of the silo. Measurement of the level of Bulk Solids depends on the flow properties of the Bulk Solid.

Flow Properties of Bulk Solids make level measurement difficult. The two types of flow of bulk solids in silos are **Funnel Flow** and **Mass Flow.**

Funnel Flow is the flow of a bulk solid where the material empties out of the bottom of a silo and the main material is down the center of the silo, with stagnant areas at the sides and bottom of the silo.

Mass Flow is the flow of a bulk solid where all material in a silo flows down toward the bottom at the same rate. Mass Flow is the most desired flow regime, but it rarely exists in practice.

The undesirable effects of **Funnel** or **Mass Flow** include **Ratholing** and **Bridging.**

Ratholing is a condition arising in a silo when material in the center has flowed out the feeder at the bottom, leaving large areas of stagnant material on the side.

Bridging is a condition arising in a silo when the material has build up over the feeder, blocking all flow out of the silo.

Silo Design significantly affects the flow of bulk solids. Free-Flowing materials will flow out of bins with flat or nearly flat bottoms while cohesive materials need bins with steep sides to allow material to flow.

The following picture depicts Bulk Solid Flow and addresses Funnel Flow, Mass Flow, Ratholing and Bridging to show how these conditions can affect level determinations:

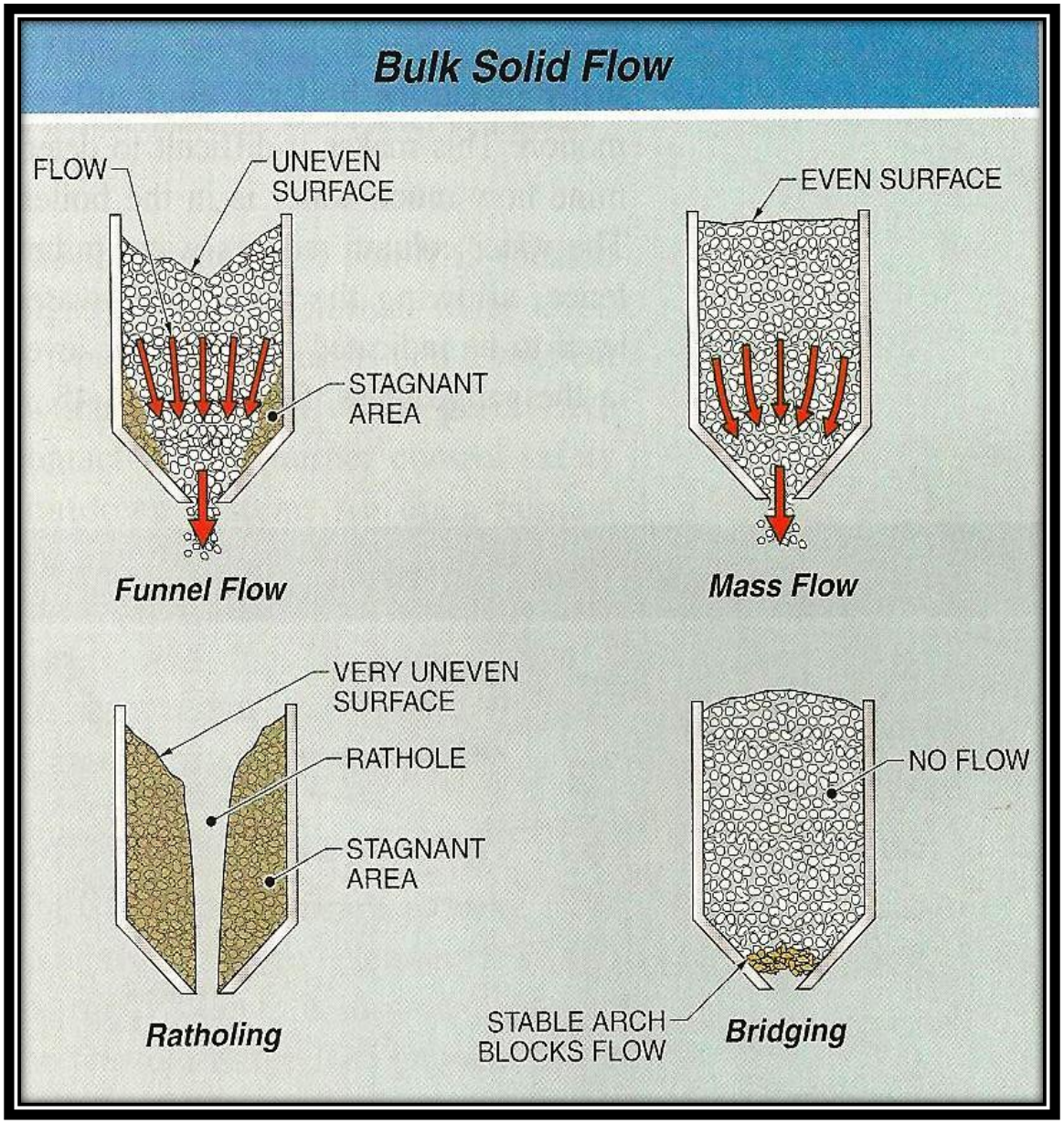


Figure 4-44 page 159

Many bulk solids are very dusty when dumped into a silo. The presence of dust in a silo can interfere with most types of level measuring instruments. Dust can block most light and laser instruments. Some radar instruments can penetrate the dust and get a reflection from the surface.

SUMMARY

- The flow and handling of Bulk Solids in tanks and silos is extremely complex.
- A Bulk Solid is a granular solid, such as gravel, sand, sugar, grain, wet cement, or other solid material that can be made to flow.
- Flow Properties of Bulk Solids make level measurement difficult. The two types of flow of bulk solids in silos are Funnel Flow and Mass Flow.
- Funnel Flow is the flow of a bulk solid where the material empties out of the bottom of a silo and the main material is down the center of the silo, with stagnant areas at the sides and bottom of the silo.
- Mass Flow is the flow of a bulk solid where all material in a silo flows down toward the bottom at the same rate. Mass Flow is the most desired flow regime, but it rarely exists in practice.
- Ratholing is a condition arising in a silo when material in the center has flowed out the feeder at the bottom, leaving large areas of stagnant material on the side.
- Bridging is a condition arising in a silo when the material has build up over the feeder, blocking all flow out of the silo.
- The presence of dust in a silo can interfere with most types of level measuring instruments. Dust can block most light and laser instruments. Some radar instruments can penetrate the dust and get a reflection from the surface.

KEO 3. 30 **DESCRIBE** difficult or complicated **SITUATIONS** associated with the measurement of level for **WATER LEVEL IN A BOILER** and what can be done to ensure safe and reliable operation of level sensors.

There are a number of general and facility specific procedures to monitor and control water level in boilers. Regulations require that all boilers have two means of measuring the water level for boilers. This could be either two gauge glasses, or one gauge glass and one remote level indicator, or one gauge glass and Try Cocks. Additionally all boilers must have two automatic burner shutdown devices for low water level conditions.

The reason water level must be monitored and shutdown devices are required is that a loss of water in a boiler can lead to the burning out of tubes and/or a boiler explosion.

A **Boiler Water Column** is a boiler fitting that reduces the movement of boiler water to provide an accurate water level in the gauge glass. When the boiler is producing steam, the water inside the boiler is constantly in motion making it difficult to determine how much water is actually in the boiler. The **Water Column** reduces water turbulence, allowing the true boiler water level to be indicated by the water level in the gauge glass. With the Water Column and the Gauge Glass

as one of two required methods of measuring water level, this true accurate reading is essential to boiler operations. The following picture depicts a **Boiler Water Column** system to include the **Gauge Glass**, and **Try Cocks**:

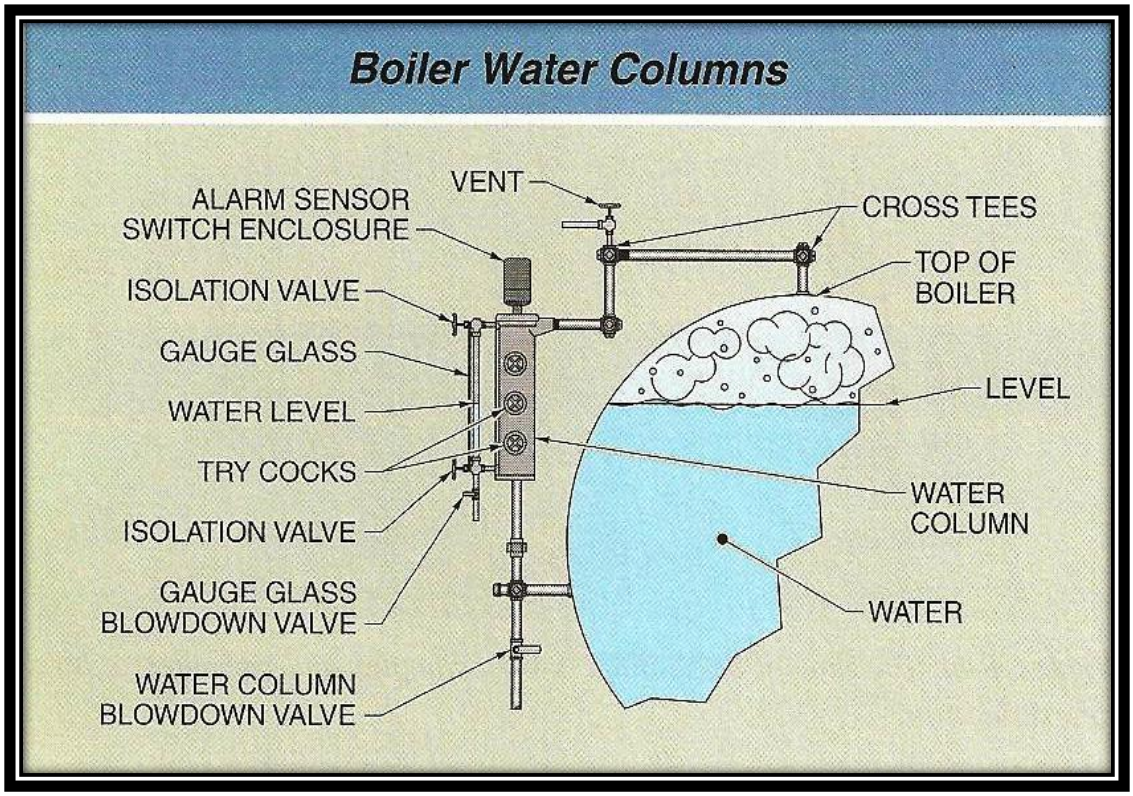


Figure 4-45 page 160

The Boiler Water Column is part of the instrumentation used to measure water level. This system includes a Vent, Alarm Sensor Switch Enclosure, Isolation Valves, Gauge Glass, Try Cocks, Gauge Glass Blowdown Valve, and Water Column Blowdown Valve.

A **TRY COCK** is a valve located on a water column used to determine the boiler water level if the gauge glass is not functional or able to be read. There are typically three **Try Cock** valves installed on a water column as depicted above.

The middle **Try Cock** is installed at the normal operating water level **(NOWL).** The top **Try Cock** is mounted at the highest acceptable water level. The bottom **Try Cock** is mounted at the lowest acceptable water level. If the boiler water is at the proper level, steam and water should be discharged from the middle **Try Cock**. Water discharged from the top **Try Cock** indicates a high water condition in the boiler.

Water discharged from the bottom **Try Cock** indicates a low water condition in the boiler. The following picture depicts how the Boiler Water Column System operates to include the three **Try Cocks** for Normal, High, or Low Water Conditions:

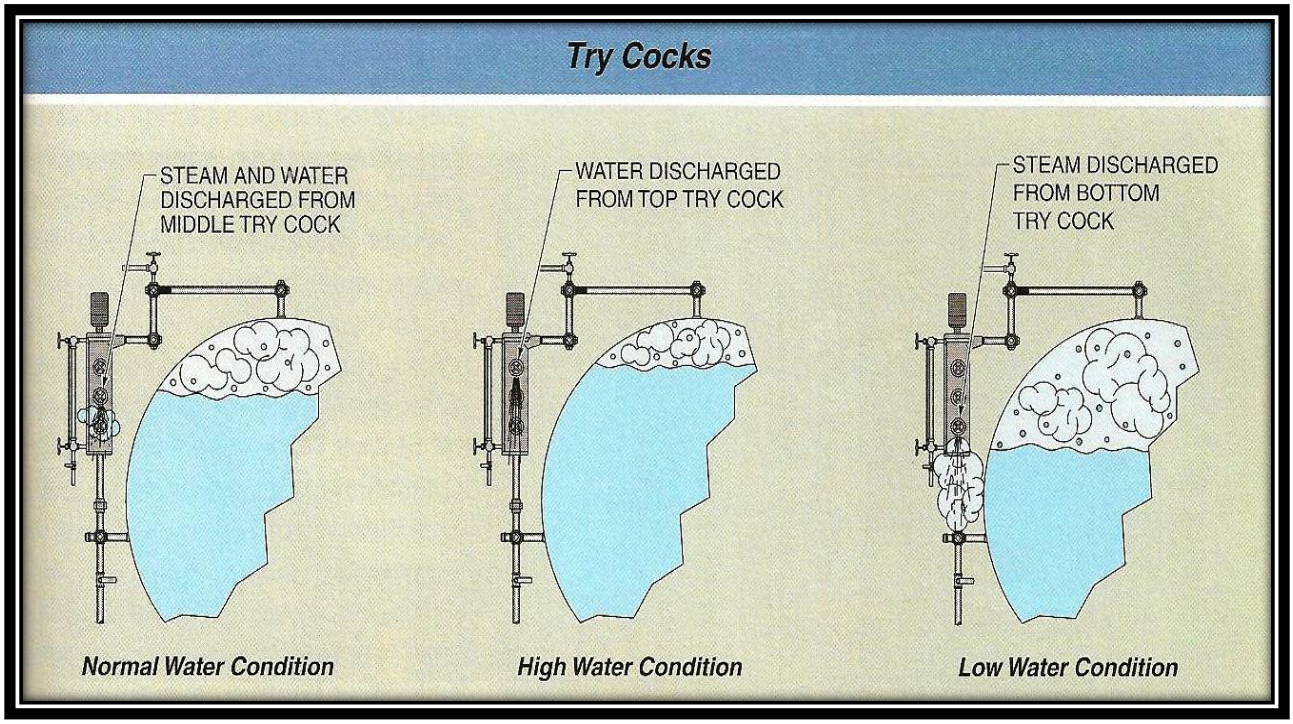


Figure 4-46 page 161

TRY COCK NOTE:

Try Cock are typically used for pressures up to 250 psi. Pressures above 250 psi are difficult to distinguish between the water and the flash steam that blows out of a Try Cock.

A condition in the Water Column System causes an accumulation of sludge and or sediment which must be periodically removed by a Blowdown Process. This process is accomplished opening the lowest valve on the Water Column System called the Water Column Blowdown valve. This valve is opened for about 5 to 10 seconds allowing water and any lludge or dediment to be discharged.

The Gauge Glass Blowdown Valve is also opened to perform a blow down of the gauge glass. Both the Water Column and Gauge Glass Blowdown procedure is typically performed each operational shift. Free flowing boiler water from the water column and gauge glass is crucial for providing an accurate boiler water level reading.

If at any time the accuracy of the gauge glass water level is suspect, the Try Cocks should be used to determine the actual boiler water level.

Low Water Fuel Cutoff is the safety feature required of boilers as a loss of water in a boiler not only damages the boiler, but it could cause an explosion. The **Low Water Fuel Cutoff** is a boiler fitting that shuts the burner **OFF** in the event of a low water condition as depicted below:

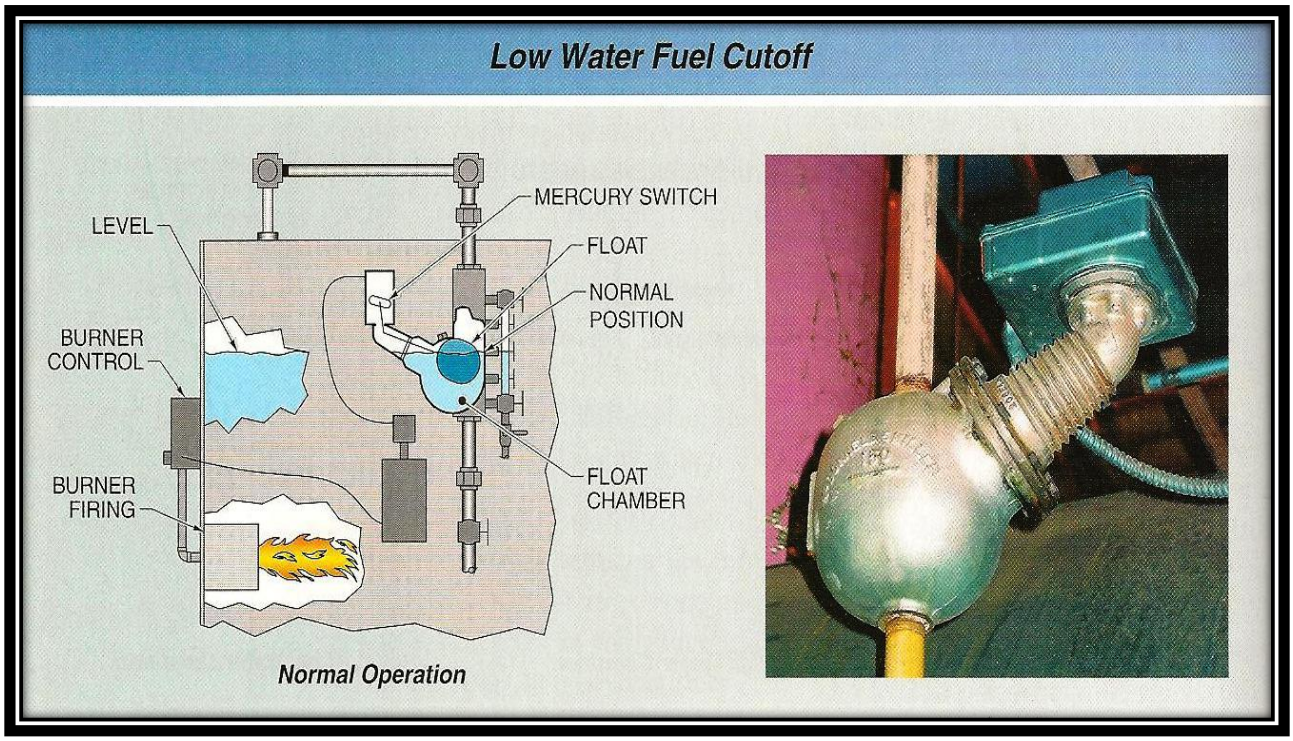


Figure 4-47 page 162

The Low Water Fuel Cutoff is located slightly below the normal water level. A typical cutoff consists of a level float switch that is part of the permissive contact of the burner control safety shutdown system. An open contact fails the burner permissive interlock system, de-energizes the burner safety system, and shuts down the burner if the water level drops below the safe operating level.

SUMMARY

- Regulations require that all boilers have two means of measuring the water level for boilers.
- All boilers must have two automatic burner shutdown devices for low water level conditions.
- The reason water level must be monitored and shutdown devices are required is that a loss of water in a boiler can lead to the burning out of tubes and/or a boiler explosion.
- A Boiler Water Column is a boiler fitting that reduces the movement of boiler water to provide an accurate water level in the gauge glass.
- The Water Column reduces water turbulence, allowing the true boiler water level to be indicated by the water level in the gauge glass.
- A Try Cock is a valve located on a water column used to determine the boiler water level if the gauge glass is not functional or able to be read.
- There are typically three Try Cock valves installed on a water column and the top is for a high level, the middle a normal level, and the bottom for a low level.
- The middle Try Cock is installed at the normal operating water level (NOWL). The top Try Cock is mounted at the highest acceptable water level. The bottom Try Cock is mounted at the lowest acceptable water level.
- If at any time the accuracy of the gauge glass water level is suspect, the Try Cocks should be used to determine the actual boiler water level.
- Low Water Fuel Cutoff is the safety feature required of boilers as a loss of water in a boiler not only damages the boiler, but it could cause an explosion.

KEO 3. 31 **DESCRIBE** difficult or complicated **SITUATIONS** associated with the measurement of level for **CORROSIVE FLUIDS** and what can be done to ensure safe and reliable operation of level sensors.

There are times when a pressure measurement is used to determine level and process liquid is incompatible with the instrumentation used to measure the **CORROSIVE FLUID**. The level sensing instrument must be protected from the **Corrosive Fluids**. A common way to respond to this condition is to use **Diaphragm Seals** with **Differential Pressure Cell Devices**. The choice of which Diaphragm Seals to use should be made with help from the manufacture to ensure chemical compatibility.

Typical Diaphragm Seals are attached to flanges on a tank or vessel with a diaphragm that senses head pressure generated by the corrosive liquid level. When using Differential Pressure Cell Devices, capillary tubing will be connected to the opposite side of the diaphragm device not making contact with the corrosive fluid and filled with a fluid having the same specific gravity weight of the corrosive solution being measured.

When using the Differential Pressure method, two diaphragm seals are attached to the vessel, one above the highest level on the tank and the other one to the lowest point possible on the tank. Then capillary tubing is attached to the diaphragm seals and the differential pressure device.

If the weight of the fluid in the capillary tubing is the same as the weight of the corrosive fluid the differential pressure device will provide an accurate level measurement of the corrosive fluid in the tank. If the capillary tubing fluid is not the same weight as the corrosive fluid, the differential pressure device will have to be adjusted to compensate for this difference. This compensation will be addressed in the next knowledge objective (KEO 3.32.).

Depicted below is a tank with SULFURIC ACID with two Diaphragm Seals attached to measure its differential pressure liquid level:

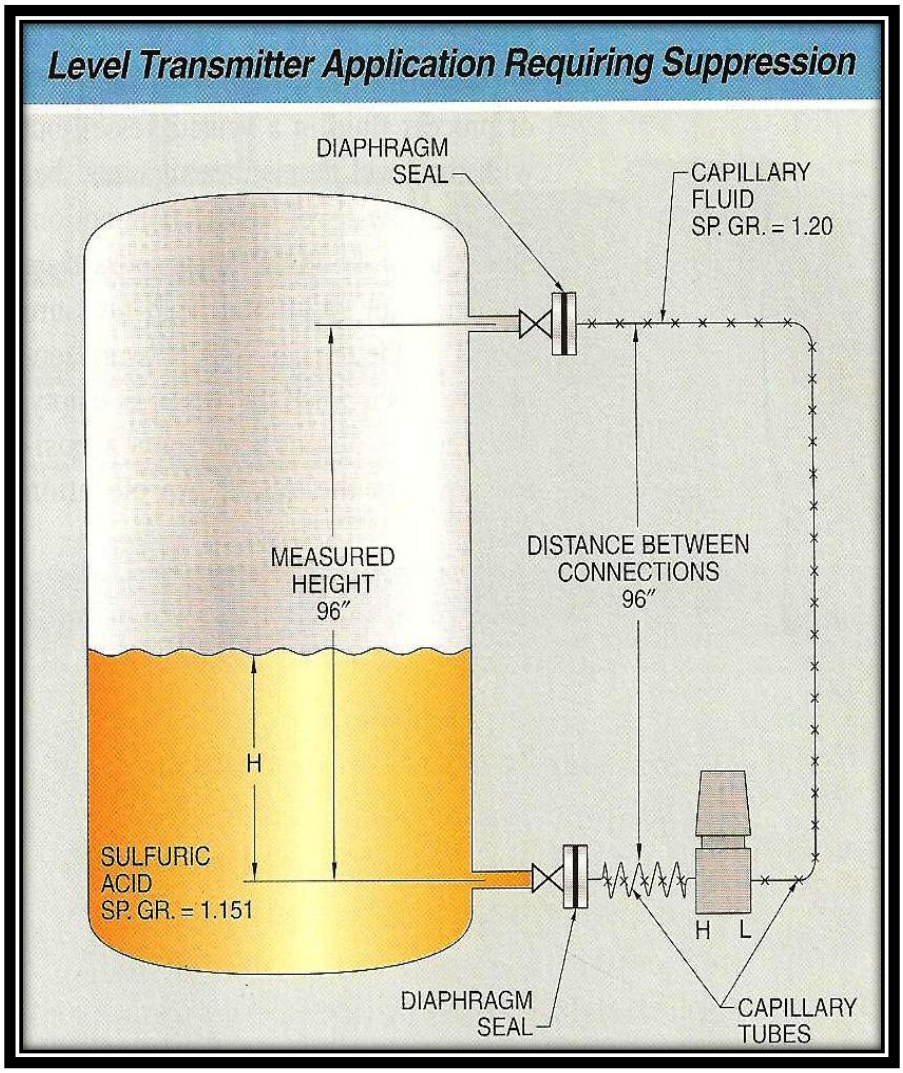


Figure 4-48 page 163

KEO 3. 32 **DESCRIBE** how to compensate for level measurement of a level transmitter using a **CAPILLARY FIELD SYSTEM REQUIRING SUPPRESSION**.

Common Capillary Filled Systems are filled with Glycerin and Silicone and other fluids having a specific gravity ranging from 0.85 to 1.85. When a tank in measuring differential pressure it requires the use of two diaphragm seal devices and may require the use of transmitter suppression.

The below picture depicts a tank having Sulfuric Acid with a Specific Gravity of 1.51 and the Capillary Tubing to the Differential Pressure has a fluid with a Specific Gravity of 1.20 requiring transmitter suppression:

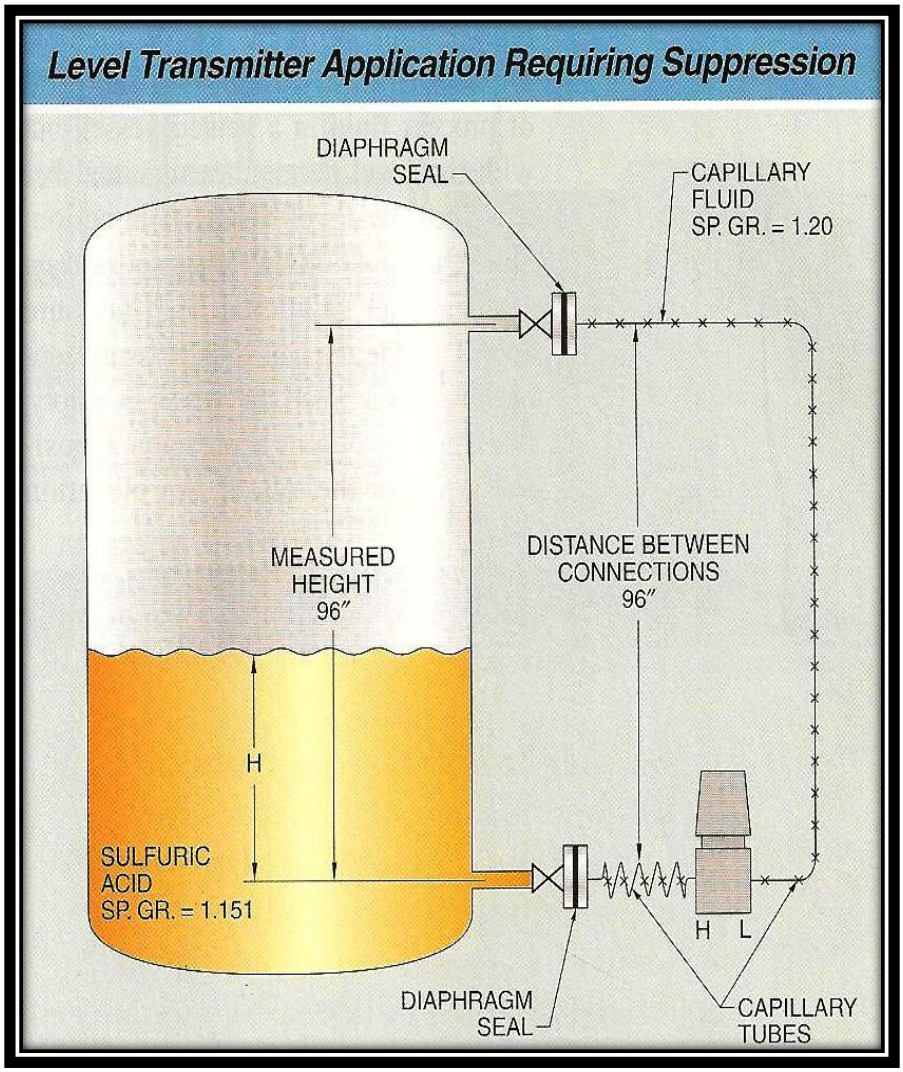


Figure 4-48 page 163

To provide the suppression necessary to read actual tank level, you multiply the distance between connections of 96 inches to the capillary fluid specific gravity of 1.20 which equals 115.20 inches. The zero setting for this transmitter needs to be set at 115.20 inches to be able to read the sulfuric acid fluid of 96 inches.

The transmitter will then read a range of 96 inches times the specific gravity of 1.151 equaling 110.496 inches. This 110.496 or 110.5 inches is the range or span in inches of water column the transmitter will be calibrated to above the 115.20 zero suppressed signal.

KEO 3. 33 **ESCRIBE** how to compensate for level measurement of level transmitter REQUIRING ELEVATION when the solution being measured is applied to the transmitter located below the tank being measured.

There are level measurement applications that may require the level transmitter to have an elevation incorporated into its calibration range. With this application, generally a differential pressure transmitter will be used and connected directly to the vessel or tank below the low-level vessel connection as depicted below:

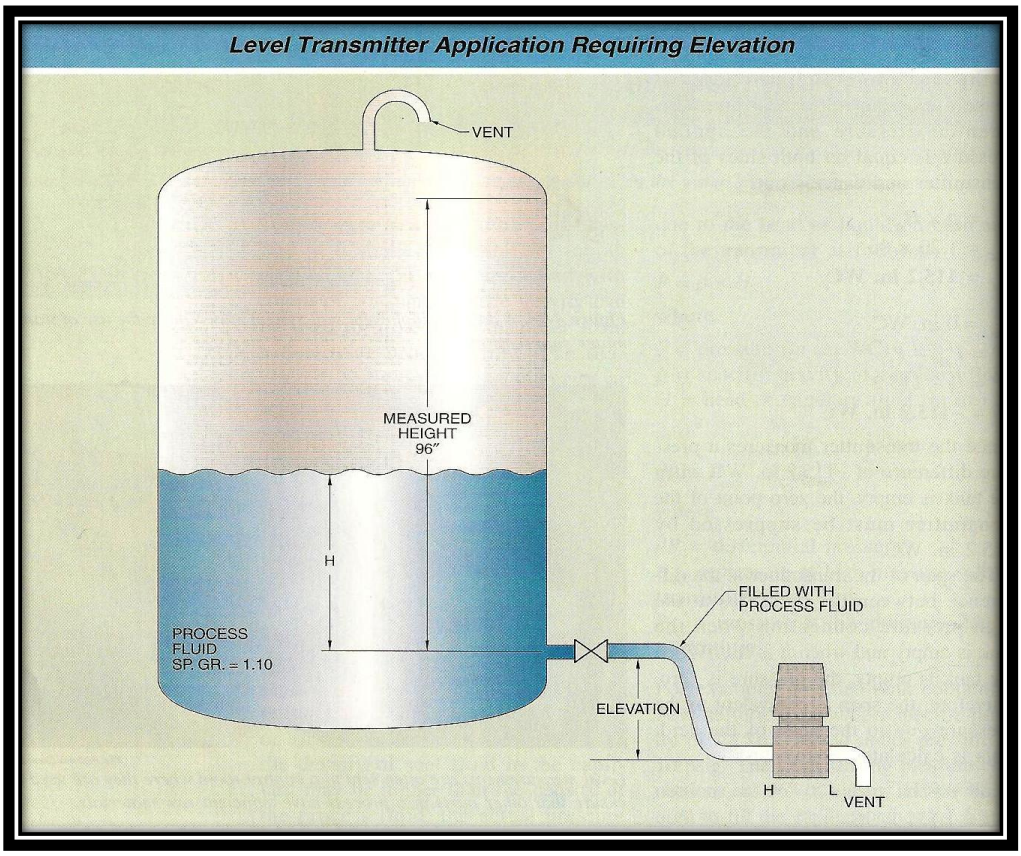


Figure 4-49 page 164

In the above example, there is always an additional head pressure applied to the Differential Pressure Transmitter (High Pressure side) and the low pressure side of the transmitter is vented to the atmosphere as is the top of the vessel. The amount of elevation (in inches) is added to the actual height of the tank measurement and this height is the zero setting for the calibration of this transmitter. If the distance was 20 inches, then the 20 inches is multiplied by the process fluid's specific gravity of 1.10 equaling 22 inches. This elevation of 22 inches would be the zero setting for the transmitter calibration.

Level Measurement Note:

Level measurement based on differential pressure is often referred to as Hydrostatic Tank Gauging (HTG).

SUMMARY

- When dealing with Corrosive Fluids, the level sensing instrument must be protected from the Corrosive Fluids. A common way to respond to this condition is to use Diaphragm Seals with Differential Pressure Cell Devices.
- Diaphragm Seals are attached to flanges on a tank or vessel with a diaphragm that senses head pressure generated by the corrosive liquid level.
- When using the Differential Pressure method, two diaphragm seals are attached to the vessel, one above the highest level on the tank and the other one to the lowest point possible on the tank. Then capillary tubing is attached to the diaphragm seals and the differential pressure device.
- The capillary tubing if filled with a special fluid that is used to provide the differential pressure signal to the transmitter.
- If the capillary tubing fluid is not the same weight as the corrosive fluid, the differential pressure device will have to be adjusted to compensate for this difference.
- Common Capillary Filled Systems are filled with Glycerin and Silicone and other fluids having a specific gravity ranging from 0.85 to 1.85.
- When a transmitter is located below the lowest point of the vessel, there is always an additional elevated head pressure applied to the Differential Pressure Transmitter (High Pressure side).
- The amount of elevation (in inches) is added to the actual height of the tank measurement and this height is the zero setting for the calibration of a transmitter.
- Level measurement based on differential pressure is often referred to as Hydrostatic Tank Gauging (HTG).

STEP TWO

Level Measurement Course

Skill/Performance Objectives

Skill Knowledge Introduction:

Below are the skill knowledge objectives. How these objectives are performed depend on equipment and laboratory resources available. With each skill objective it is assumed that a set of standard test equipment and tools be provided.

For example, to be able to perform temperature calibration tasks, the following tools and equipment will be required:

1. **A set of weights or pressure sources to include psi, inches of water column, or inches of mercury, load cell calibrators or simulators, etc.**
2. **A calibration standard to measure the applied pressure or weight**
3. **Equipment capable of measuring level such as gauge, transducer, transmitter, switches, etc.**
4. **A measuring device capable of measuring / indicating the output signal such as meter or smart calibrator**
5. **An appropriate power supply to power the equipment being calibrated**

Skill Terminal Objective (STO)

STO 3. 1 **Given a Level Measurement Task Checklist, under the direction of an instructor, complete a series of tasks using calibration equipment, level indicating devices, and level transmitting devices to demonstrate mastery of both knowledge and skill objectives associated with the measurement of level.**

Skill Enabling Objectives (SEO)

SEO 3. 1 Calibrate a Fisher Leveltrol Displacement System



SEO 3. 2 **Calibrate** a Level Loop System



SEO 3. 3 **Calibrate** a Pneumatic Level Controller

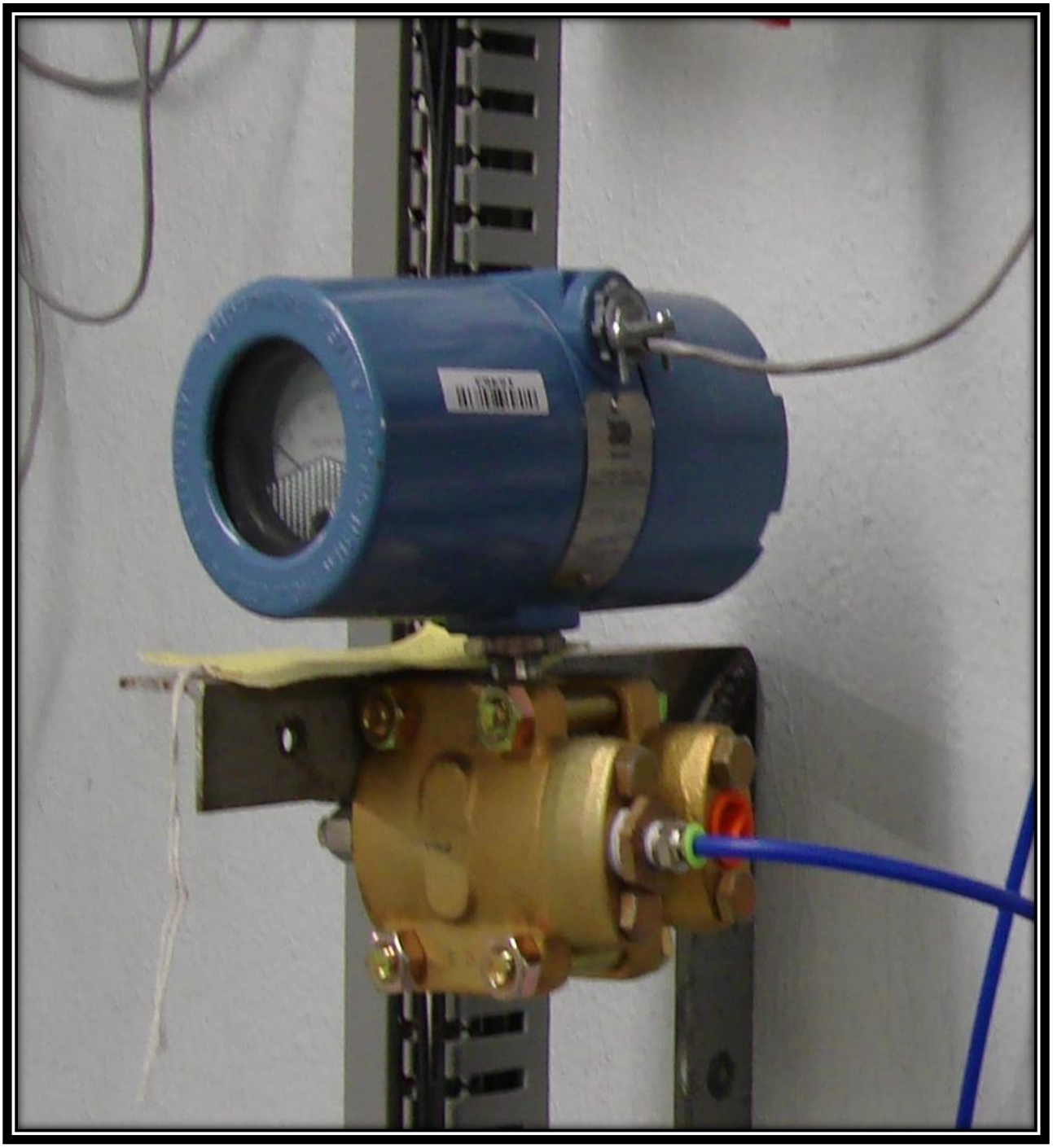
SEO 3. 4 **Calibrate** a Bubbler Loop System



SEO 3. 5 **Calibrate** an Ultrasonic Level Transmitter



SEO 3. 6 **Calibrate** a Differential Pressure Transmitter



SEO 3. 7 **Calibrate** a Capacitance Level Switch Transmitter

